Health, Retirement, and the Fiscal Returns to Social Security Reform*

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Abstract

Social Security faces a financing gap, and raising the retirement age is the usual response. We compare it with reforms that steepen the reward for delayed claiming, using a life-cycle model of participation, hours, saving, and claiming under health, mortality, wage, and medical-expense risk, estimated on the 1920–1940 PSID cohorts and MEPS medical spending. Raising the normal retirement age to 66–68 lowers benefits at every claiming age, lowers welfare, and reduces net government spending; reversing it would deliver 0.16–0.18 dollars of welfare per dollar of net spending. Raising the delayed retirement credit from 4.5 to 8–10 percent, or steepening the adjustment around the normal retirement age, raises welfare, lowers net government spending, and raises participation after the normal retirement age. Under the same accounting, first-bracket benefit increases have marginal values of public funds of 0.52–1.14, Medicare expansions about 0.15, old-age income-tax cuts 0.23–0.26, and saving subsidies at least 0.84–0.96.

JEL codes: D15, H31, H51, H55, I13, J14, J26.

Keywords: Social Security; retirement; delayed retirement credit; normal retirement age; health; disability insurance; Medicare; marginal value of public funds; life-cycle model; method of simulated moments.

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1 Introduction

1.1 The financing problem and two ways to change the claiming schedule

The Old-Age, Survivors, and Disability Insurance (OASDI) program is not financed over the long run at its scheduled benefits and tax rates: the 2024 Trustees Report puts its 75-year actuarial deficit at 3.50 percent of taxable payroll (Board of Trustees, OASDI 2024). The response most often legislated or proposed is a higher retirement age. The 1983 amendments raise the normal retirement age (NRA), the age at which a retiree receives the full benefit, from 65 to 67 for cohorts born in 1960 and later, and the Congressional Budget Office has analyzed an option that raises it to 70 for cohorts born in 1981 and later (Congressional Budget Office 2024).

A higher NRA is one of two ways to change the schedule that relates a retiree's benefit to the age at which it is first claimed. Under the rules faced by the cohorts studied here, benefits can be claimed from age 62; the benefit falls by 6.67 percentage points of the full amount for each year of claiming before the NRA of 65 and rises by a delayed retirement credit (DRC) of 4.5 percentage points for each year of claiming after it, up to age 70. Raising the NRA shifts this schedule: the benefit at every claiming age falls, whether or not anyone changes the age at which they claim. The second way leaves the benefit at the NRA unchanged and changes the slope of the schedule around it, through a higher DRC or a steeper adjustment on both sides of the NRA—a steeper actuarial adjustment of the benefit to the claiming age. Benefits then change only for individuals who claim away from the NRA: those who delay receive more, and those who claim early receive the same or less. The first instrument makes the retirement age more rigid; the second rewards a flexible claiming age without moving the statutory one.

If no one changed behavior, a higher NRA would save money, because every benefit falls; a higher DRC would cost money, because every late claimant receives more; and a symmetric adjustment would save or cost depending on how many claim before and how many after the NRA. Which reform saves more per unit of welfare lost, or whether one of them saves money while raising welfare, depends on who changes their claiming age and their labor supply in response, and on what happens to the taxes they pay and the other benefits they receive. That is an empirical question about the behavior of a population that differs in health, wages, assets, and expected longevity, and this paper answers it in an estimated life-cycle model.

1.2 What we do

We estimate a life-cycle model of labor force participation, hours, saving, and Social Security claiming in which individuals face health, mortality, wage, and medical-expense risk, in the tradition of French (2005) and French and Jones (2011). Health takes three states, good,

poor, and disability; it governs mortality, out-of-pocket medical spending, and wages, and it changes the fixed time cost of participation, which is estimated separately for healthy and for unhealthy workers. Exit from the labor force is reversible at an estimated reentry cost, and claiming is irreversible. The model includes the claiming-age schedule, the benefit formula, the retirement earnings test, disability insurance, Medicare, defined-benefit pensions, a progressive income tax, the payroll tax, and a consumption floor. It is estimated in two steps on the 1920–1940 birth cohorts of the Panel Study of Income Dynamics (PSID), with medical spending from the Medical Expenditure Panel Survey (MEPS). The first step estimates health transitions, survival, medical spending, and wage profiles by health; the second estimates seven preference parameters by the method of simulated moments on life-cycle profiles of participation and hours by health, assets, and reentry. Health reaches the reform results through the first-step profiles—mortality, medical spending, and wages by health—and through the health-dependent participation cost; the reform tables report participation by age group, and Section 7.3 states what the profiles imply, in the model, about who moves.

We then use the model to evaluate seven families of reforms with one welfare measure and one fiscal accounting. Four change Social Security: a higher NRA; a higher DRC; a symmetric steepening of the schedule around the NRA; and increases in the three bracket rates of the benefit formula, one at a time or together. Three change other programs: expansions of Medicare coverage after age 65; reductions in income taxes after age 62; and subsidies to retirement saving, read as changes in the tax treatment that retirement saving already receives through individual retirement accounts and 401(k) plans. Welfare is the consumption-equivalent change in the ex ante expected utility of the cohort at age 30. The fiscal accounting separates the change in the outlays of the program being reformed from the change in every other tax and transfer the cohort pays or receives, and combines them into the change in the government's net spending on the cohort, all as per-capita present values at age 62. The marginal value of public funds (MVPF) of each reform is the ratio of the cohort's willingness to pay to the change in net government spending (Hendren and Sprung-Keyser 2020). Because the model solves the individual's problem under each set of rules, the willingness to pay is exact for discrete reforms with fixed participation costs and irreversible claiming rather than a first-order approximation, and the fiscal response of every program is computed jointly with it. Section 5 maps each column of the reform tables into the MVPF and relates the structural MVPF to the consumption-smoothing benefit and the fiscal multiplier of the sufficient-statistics literature.

The exercise takes the perspective of one cohort. Reforms are permanent changes in the rules it faces from age 30; fiscal savings are valued at face value, with a unit welfare weight on their recipients outside the cohort; no transition is modeled; and the MVPF is reported as a

shadow price to which any intergenerational weight can be applied. A reform contributes to closing the Trustees' gap if it lowers the government's net spending on the cohort.

1.3 Results

Raising the NRA to 66, 67, or 68 lowers the cohort's willingness to pay by 0.38, 0.74, and 1.09 percent of consumption, lowers the present value of its retirement benefits by 6.13, 11.82, and 16.75 percent, and lowers net government spending by 0.87, 1.75, and 2.38 percent. The ratio of the welfare loss to the fiscal saving is the MVPF of reversing the reform: lowering the NRA from 66, 67, or 68 back to 65 would deliver 0.17, 0.16, or 0.18 dollars of welfare per dollar of net spending. Raising the DRC from the cohort's 4.5 percent to 8–10 percent per year raises the willingness to pay by 0.16 to 0.3 percent of consumption and lowers retirement-benefit outlays by 2.46 to 5.05 percent; the government's other net outlays fall as well, by 0.04 to 0.10 percent, and net spending falls by 0.44 to 0.78 percent. Setting equal early-claiming penalties and delayed credits of 7 to 15 percent per year raises the willingness to pay by 0.10 to 0.80 percent of consumption and lowers net spending by 0.12 to 1.57 percent. These reforms raise ex ante welfare and generate a fiscal surplus, so their MVPF is infinite, and participation at ages 65–69 rises under every one of them, by 0.22 to 2.61 percentage points. Two pairs hold the fiscal saving approximately fixed. An NRA of 66 and a symmetric adjustment of 10 percent per year lower net spending by 0.87 and 0.86 percent; the willingness to pay is -0.38 percent of consumption under the first and $+0.29$ under the second. An NRA of 67 and a symmetric adjustment of 15 percent lower it by 1.75 and 1.57 percent, with willingness to pay of -0.74 and $+0.80$. A benefit increase for late claimants lowers net spending; a benefit cut for every claimant lowers it by a similar amount and lowers welfare. Participation responses by age locate the difference. Under a steeper adjustment participation rises after the NRA under every reform and, beyond the smallest steps, falls at ages 62–64: the reform moves work and claiming from before the NRA to after it, and the steepest adjustments shift the incidence toward those who claim early.

The quantitative analysis opens by removing each policy instrument in turn. Eliminating Social Security benefits lowers welfare by 5.89 percent of consumption; the loss from incomplete Medicare coverage after age 65 is 4.28 percent; raising the net return on saving to the gross return raises welfare by 3.60 percent and eliminating the income tax by 3.16 percent; eliminating disability benefits lowers welfare by 0.34 percent. Social Security, the instrument with the most welfare at stake, is taken up first.

Within reforms that raise the generosity of retirement benefits, the first bracket of the benefit formula is where a dollar of net spending buys the most welfare. Increases in the first bracket rate have MVPFs from a lower bound of 0.52, for the smallest step, which is approximately self-financing, to 1.14; proportional increases in all three rates have MVPFs of

0.30 to 0.61, and increases in the third bracket 0.29 to 0.30. Expansions of Medicare coverage after age 65 by 20 percentage points, for everyone or for nonparticipants only, have MVPFs of 0.146 and 0.155. Reductions in income taxes after age 62 raise participation at every age and have MVPFs of 0.23 to 0.26. Subsidies to retirement saving of 0.2, 0.4, and 0.8 percentage points of return have MVPFs whose lower bounds, computed against the subsidy alone, are 0.96, 0.92, and 0.84. Across the menu, the reforms that lower net spending are the claiming-schedule reforms and the smallest first-bracket increase; every other reform raises welfare and net spending together, and the MVPF orders them.

1.4 Relation to the literature

The model belongs to the class of estimated life-cycle models of retirement with health and medical-expense risk. Rust and Phelan (1997) study Social Security and Medicare under incomplete markets; French (2005) estimates the effects of health, wealth, and wages on labor supply and retirement; French and Jones (2011) add employer-provided and public health insurance. Other members of the class study retiree health insurance (Blau and Gilleskie 2008), Social Security and the saving of low-income households (van der Klaauw and Wolpin 2008), medical expenses and the lifetime costs of bad health (De Nardi et al. 2010, 2025), disability insurance (Low and Pistaferri 2015), and medical spending and the evolution of health over the life cycle (Pashchenko and Porapakkarm 2019; Hosseini et al. 2022). Haan and Prowse (2014), Laun et al. (2019), Li (2018), Iskhakov and Keane (2021), Jones and Li (2023), Bairoliya and McKiernan (2023), Borella et al. (2023), and Yu (2024) use models of this class to evaluate pension and Social Security reforms, among them increases in the retirement age and changes in the claiming-age adjustment and the benefit formula; Blundell et al. (2016) survey the literature. Relative to French (2005) and French and Jones (2011), the model here lets the fixed cost of participation depend on health, models disability insurance explicitly, allows work at every age until death, and is estimated on the general population rather than on male household heads (Section 3.7).

Social Security reform has been studied in general-equilibrium overlapping-generations models (Nishiyama and Smetters 2007; Imrohoroğlu and Kitao 2012; Kitao 2014) and in life-cycle models of labor supply and social insurance (Huggett and Parra 2010; Erosa et al. 2012); Imrohoroğlu and Kitao (2012) and Kitao (2014) include an increase in the normal retirement age among the reforms they evaluate. An empirical literature studies the US and Swiss increases in the statutory age (Mastrobuoni 2009; Behaghel and Blau 2012; Deshpande et al. 2024; Lalive et al. 2023), the response of retirement to financial incentives and future entitlements (Coile and Gruber 2007; Manoli and Weber 2016), the gain from delaying claiming and the means to finance a delay (Shoven and Slavov 2014; Goda et al. 2018), how claiming decisions are made and framed (Shoven et al. 2017; Brown et al. 2016; Liebman and Luttmer

2015; Seibold 2021), and old-age support and the earnings test (Fetter and Lockwood 2018; Gelber et al. 2022). Gruber and Wise (1998) and the volumes they edited (1999; 2004; 2007) compare countries; Coile (2019) documents the US trend toward working longer in a later volume of the same series. This paper evaluates the two instruments for one cohort, at fixed prices, with the same welfare measure and the same fiscal accounting for both, and reports who moves by age group.

The accounting draws on the marginal value of public funds of Hendren (2016), Hendren and Sprung-Keyser (2020), and Finkelstein and Hendren (2020), whose antecedents are the excess-burden and marginal-cost-of-funds measures of Mayshar (1990), Slemrod and Yitzhaki (1996), and Kleven and Kreiner (2006), and which Finkelstein and McKnight (2008) and Finkelstein et al. (2019) apply to Medicare and to health-insurance subsidies. The sufficient-statistics approach to social insurance (Baily 1978; Chetty 2006, 2008, 2009; Chetty and Finkelstein 2013; Kleven 2021) values a program by the consumption-smoothing benefit of its transfers against their mechanical and behavioral fiscal cost; Landais and Spinnewijn (2021) and Kolsrud et al. (2024) apply it to unemployment insurance and to pension design, and Haller (2022) to pension reforms, summarizing the cost side by a fiscal multiplier. Section 5 writes the structural MVPF as the ratio of one plus a consumption-smoothing term to a fiscal multiplier and states which of these objects the reform tables identify.

The companion paper (Ndiaye and Yu 2026) characterizes the constrained-efficient allocation in the same economy and decomposes retirement behavior into the contributions of participation costs, health-related productivity differences, and mortality, in the dynamic optimal-taxation tradition of Diamond and Mirrlees (1978), Golosov and Tsyvinski (2006), Farhi and Werning (2013), Golosov et al. (2016), Stantcheva (2017), and Shourideh and Troshkin (2015), and of the work on retirement and taxation of Cremer et al. (2004), Grochulski and Kocherlakota (2010), Michau (2014), Choné and Laroque (2014), and Hosseini and Shourideh (2019). This paper uses the estimated model as a laboratory for reforms of the statutory rules; it does not use the constrained-efficient allocation as a benchmark, and it makes no claim that any reform in the menu approximates it.

1.5 Scope and roadmap

The analysis is partial equilibrium for one cohort at fixed factor prices. Section 3.7 lists the further restrictions—an individual decision-maker, accidental bequests, health observed by the government, time-separable preferences, and statutory ages that act only through incentives—and Section 11 states how each bears on the results and gives a reduced-form expression for how factor-price responses would enter the fiscal accounting.

Section 2 describes the rules faced by the 1920–1940 cohorts, the two ways of changing the claiming schedule, the reform menu, and the cohort perspective. Section 3 presents the model

Table 1: Social Security Benefit as a Percentage of the Primary Insurance Amount by Claiming Age, 1920–1940 Cohorts (NRA 65, DRC 4.5 Percent)

		Benefit, as a percentage of PIA, payable at age								
NRA	DRC	62	63	64	65	66	67	68	69	70
65	4.5	80	86.67	93.33	100	104.5	109	113.5	118	122.5

Notes: Benefit payable at ages 62–70 as a percentage of the PIA, from the rules of the Social Security Administration; the reduction before the NRA is 6.67 percentage points per year and the delayed retirement credit after the NRA is 4.5 percentage points per year. NRA, normal retirement age; DRC, delayed retirement credit, in percent per year. Source: Social Security Administration.

and Section 4 the data, the estimation, and the fit, with the differences by health on which the reforms act. Section 5 defines the welfare measure, the fiscal columns, and the MVPF, and relates them to the sufficient-statistics objects. Section 6 removes each instrument in turn. Section 7 evaluates the claiming-schedule reforms, Section 8 the benefit-formula reforms, and Section 9 the Medicare, income-tax, and saving-subsidy reforms. Section 10 places every reform side by side. Section 11 discusses scope and sensitivity, and Section 12 concludes. Appendix A to Appendix D contain the data and first-step estimation, the solution method and the fit of assets and reentry, the construction of the reforms, and the derivations.

2 Social Security Rules, the Financing Problem, and the Reform Menu

2.1 Rules faced by the 1920–1940 cohorts

Individuals born between 1920 and 1940 could first claim Old-Age, Survivors, and Disability Insurance (OASDI) retirement benefits at the early retirement age (ERA) of 62 and faced, on average over birth years, a normal retirement age (NRA) of 65. The benefit is the primary insurance amount (PIA) multiplied by a claiming-age factor that falls by 6.67 percentage points for each year of claiming before the NRA and rises by the delayed retirement credit (DRC) for each year of claiming after it, up to age 70. The DRC ranged from 3 to 6 percent per year by year of birth, and the paper uses the cohort average of 4.5 percent. Table 1 reports the schedule: 80 percent of the PIA at age 62, 100 percent at 65, and 122.5 percent at 70.

The PIA applies factors of 0.9, 0.32, and 0.15 to average indexed monthly earnings (AIME), the average of the highest 35 years of indexed earnings, below \$3,720, between \$3,720 and \$22,392, and above \$22,392, with AIME capped at \$43,800 (1987 dollars, as are the other amounts in this subsection). Under the retirement earnings test (RET), a beneficiary loses one dollar of benefits for every two dollars of earnings above \$6,000 before the NRA and one for every three dollars above \$8,186 from the NRA to age 69, with no test from age 70; withheld benefits are credited to later benefits (French and Jones 2011). Disability insurance (DI) pays

a PIA-based benefit, without the early-claiming reduction, to a worker with a disability who is older than 50, younger than the NRA, and earns at most $3,600 a year. A payroll tax of 3 to 7.65 percent on earnings up to $14,459–$64,974 finances OASDI and Hospital Insurance. Table A1 collects these parameters; Section 3 states how each enters the individual's problem.

2.2 A higher normal retirement age or a steeper actuarial adjustment

The schedule of Table 1 is the factor $\psi(a^c)$ that multiplies the PIA in (9). A reform can change it in two ways.

The first way moves the NRA. Reforms AN1–AN3 raise it to 66, 67, and 68, and the whole schedule moves one year later for each year added: the factor at age 62 falls from 80 to 75, 70, and 65 percent of the PIA, and the factor at age 70 from 122.5 to 118, 113.5, and 109. Every claimant receives a lower benefit at a given claiming age, whether or not the claiming age changes. This is the instrument of the 1983 amendments, which raise the NRA to 67 for cohorts born in 1960 and later, and of the proposal, analyzed by Congressional Budget Office (2024), to raise it to 70 for cohorts born in 1981 and later.

The second way leaves the factor at the NRA at 100 percent and changes the slope of the schedule around it. Reforms AD1–AD5 raise the DRC from the cohort's 4.5 percent to 8, 8.5, 9, 9.5, and 10 percent per year and leave the reduction before the NRA unchanged, so the factor at age 70 rises from 122.5 to 140 under AD1 and 150 under AD5 while the factors at ages 62–65 are those of the status quo. Reforms AL1–AL9 set the early-claiming reduction and the DRC equal in absolute value, at ±7 to ±15 percent per year in steps of one percentage point; the factor at age 62 falls from 79 under AL1 to 55 under AL9, and the factor at age 70 rises from 135 to 175. Under both the AD and the AL reforms a claimant at the NRA receives the status-quo benefit; what changes is the price of claiming earlier or later. AN1 and AD1 applied together, an NRA of 66 and a DRC of 8 percent, are the rules that the cohort born in the 1950s faced in Yu (2024); here each is applied alone to the cohort that faced an NRA of 65 and a DRC of 4.5 percent, and the combination was not simulated.

2.3 The reform menu

Table 2 lists every reform evaluated in Section 7 to Section 9.[1] The benefit-formula reforms raise the first, second, or third PIA factor in steps of one percent of its initial value (P1.1–P1.9, P2.0–P2.9, P3.0–P3.9) or all three factors proportionally (L0–L9); L9 and P3.9 set the same factors. Two Medicare reforms raise the share of total medical spending covered after age 65 by 20 percentage points, for everyone (Part A) or for nonparticipants only (Part B), because

[1] There are 66 reform rows in seven families; L9 and P3.9 are the same policy, and D3 and K3 are the same experiment.

Table 2: The Reform Menu

Family	IDs	Instrument changed	Grid	Ages affected
Normal retirement age	AN1–AN3	NRA; the claiming schedule of Table 1 moves one year later per step	NRA 66, 67, 68 (status quo 65)	Every claiming age 62–70
Delayed retirement credit	AD1–AD5	DRC, the credit per year of claiming after the NRA; the reduction before the NRA is unchanged	8, 8.5, 9, 9.5, 10 percent per year (status quo 4.5)	Claiming ages 66–70
Symmetric adjustment	AL1–AL9	Early-claiming reduction and DRC set equal in absolute value	±7 to ±15 percent per year, in 1 pp steps	Claiming ages 62–64 and 66–70
Benefit formula: progressivity	P1.1–P1.9	First PIA factor	0.91 to 0.99, 1 pp per step (status quo 0.90)	All beneficiaries, from the claiming age
	P2.0–P2.9	Second PIA factor, with the first factor at 0.99	0.3232 to 0.352, 0.32 pp per step (status quo 0.32)	All beneficiaries, from the claiming age
	P3.0–P3.9	Third PIA factor, with the first and second factors at 0.99 and 0.352	0.1515 to 0.165, 0.15 pp per step (status quo 0.15)	All beneficiaries, from the claiming age
Benefit formula: level	L0–L9	All three PIA factors	+1 percent of each factor per step, from (0.909, 0.3232, 0.1515) to (0.99, 0.352, 0.165); L9 coincides with P3.9[a]	All beneficiaries, from the claiming age
Medicare	Part A	Share of total medical spending covered, everyone aged 65 and older	+20 percentage points	65 and older
	Part B	Share of total medical spending covered, nonparticipants aged 65 and older	+20 percentage points	65 and older, nonparticipants
Income tax	I1–I5	Income tax after age 62; after-tax income $y_t - T_t(y_t)$ rises	+1 to +5 percent, in 1 pp steps	62 and older
Saving subsidy	K1–K3	Net return on assets r	+0.2, +0.4, +0.8 pp (status quo 3.2 percent); K3 coincides with D3[b]	All ages

Notes: NRA, normal retirement age; DRC, delayed retirement credit; PIA, primary insurance amount; pp, percentage points. The status-quo values are those of the 1920–1940 cohorts (Table 1 and Table A1). The rule change behind each family is stated in Appendix C.1; the Medicare rows are in percentage points of the coverage share of total medical spending.
[a] L9 and P3.9 set the same three factors, (0.99, 0.352, 0.165), and are the same policy; Table C1 reports identical outcomes for the two rows.
[b] K3 raises the net return from 3.2 to 4 percent, the gross return; it is the experiment reported as D3 in Table 5.

employer-provided plans already cover physician and outpatient services for most workers. Income-tax reforms I1–I5 lower income taxes after age 62 so that after-tax income rises by 1 to 5 percent. Saving-subsidy reforms K1–K3 raise the net return on assets by 0.2, 0.4, and 0.8 percentage points; they stand for an expansion of the tax preferences that retirement saving already receives through individual retirement accounts and 401(k) plans, and K3, which brings the net return of 3.2 percent to the gross return of 4 percent, is the experiment reported as D3 in Section 6. Appendix C gives the implementation of each reform.

2.4 The cohort perspective and the financing benchmark

Each reform is a permanent change in the rules faced by one cohort from age 30. Welfare is the ex ante average of expected utility at age 30 (Section 5). Fiscal flows are the cohort's own—benefits, Medicare payments, and transfers received, less payroll and income taxes paid—as per-capita net present values at age 62 at the net return of 3.2 percent; a reform changes the cohort's net position by the change in net government spending that the Net column of every reform table measures (Appendix D). Fiscal savings are valued at face value: a dollar the cohort does not receive is available to taxpayers or beneficiaries outside it, on whom we place a unit welfare weight. Incidence within the cohort is valued by the ex ante average, not by the distribution of gains and losses across realized histories. Incidence across cohorts is not weighted here: the marginal value of public funds is a shadow price, the welfare delivered to the cohort per dollar of net spending on it (Hendren and Sprung-Keyser 2020), to which any intergenerational weight can be applied through the Net column. No transition is modeled, and nothing is said about the welfare of other cohorts.

The benchmark on the fiscal side is the long-run financing gap: the 2024 Trustees Report puts the 75-year actuarial deficit of OASDI at 3.50 percent of taxable payroll (Board of Trustees, OASDI 2024). A reform contributes to closing it if its net fiscal effect on the cohort is negative. The Net column is in percent of the cohort's baseline net spending; expressing it in percent of taxable payroll would require the present value of the cohort's payroll, which the reported baseline aggregates do not include, so the comparison with the Trustees' figure is one of sign.

3 A Life-Cycle Model of Health, Work, Saving, and Claiming

This section sets out the model in which the reforms are evaluated: a cohort of individuals who choose participation, hours, saving, and the claiming age under health, mortality, wage, and medical-expense risk, facing the Social Security rules of Section 2, disability insurance, pensions, taxes, and a consumption floor. Section 4 estimates it, Section 5 defines the welfare and fiscal accounting applied to it, and Section 11 discusses the assumptions collected in Section 3.7.

3.1 Environment and timing

Individuals enter the model at age 30 and live at most to age $T = 85$; a period is one year, so $t = 30, \ldots, 85$ indexes 56 annual periods. They enter the labor market at age 25, so that at age 30 they carry five years of earnings, imputed as 2,000 hours per year at the age-30 wage. Health h_t takes three values: 0 for good health, 1 for poor health, and 2 for a disability;

throughout the paper "unhealthy" refers to poor health and disability pooled. In each period the individual chooses consumption c_t, hours of work $n_t \geq 0$, and, when eligible, whether to claim Social Security retirement benefits, $b_t \in \{0, 1\}$. Participation is the discrete part of the hours choice, $P_t \equiv \mathbb{1}\{n_t > 0\}$, and savings follow from the budget constraint.

The life cycle has three stages. From ages 30 to 61 individuals choose consumption and labor supply and are not eligible for pension or Social Security retirement benefits. From ages 62 to 69 they also decide whether to claim retirement benefits; claiming is a one-time, irreversible decision. From ages 70 to 85 everyone has claimed and continues to choose consumption and labor supply; claiming by age 70 is mandatory in the model, so $b_t = 1$ for all $t \geq 70$.

3.2 Preferences and participation costs

Period utility is Cobb–Douglas in consumption and leisure and of the constant-relative-risk-aversion form in the composite,

$$u(c_t, n_t; h_t) = \frac{1}{1-\nu} \left(c_t^{\gamma} \, l_t^{1-\gamma} \right)^{1-\nu}, \tag{1}$$

where $\gamma \in [0, 1]$ is the weight on consumption and ν governs risk aversion and intertemporal substitution. Preferences are nonseparable between consumption and leisure within the period and separable across periods.[2] Leisure is the time endowment L net of hours and of two fixed time costs,

$$l_t = L - n_t - \phi_n(h_t)\, \mathbb{1}\{n_t > 0\} - \phi_{RE}\, RE_t, \qquad P_t \equiv \mathbb{1}\{n_t > 0\}, \quad RE_t \equiv P_t\,(1 - P_{t-1}). \tag{2}$$

The participation cost $\phi_n(h_t)$ is paid in every period with positive hours and depends on health; the reentry cost ϕ_{RE} is paid in a period of participation that follows a period of nonparticipation. Because the sample contains few individuals with a disability, poor health and disability share one participation cost, so the estimated costs are $\phi_n(h = 0)$ and $\phi_n(h = 1, 2)$. The fixed cost makes participation a discrete choice: an individual either works enough hours to cover $\phi_n(h_t)$ or does not work. The reentry cost makes exit from the labor force reversible but costly. Future utility is discounted at the factor β and by survival. The preference parameters $(\gamma, \nu, \phi_n(h = 0), \phi_n(h = 1, 2), \phi_{RE}, L, \beta)$ are estimated in Section 4.

3.3 Health, mortality, medical spending, and wages

Health follows a Markov chain on $\{0, 1, 2\}$ with age-dependent transition probabilities,

$$\pi_{j,i,t+1} = \Pr\left(h_{t+1} = j \mid h_t = i,\, t + 1\right), \qquad i, j \in \{0, 1, 2\}. \tag{3}$$

[2] Grochulski and Kocherlakota (2010) study preferences that are separable in consumption and leisure and nonseparable over time; Section 11 discusses time separability.

The distribution of health at age 30 is drawn from the data (Appendix A). Survival to age $t+1$ depends on age and on health at age t, and no one survives beyond T,

$$s_{t+1}(h_t) = S(h_t, t+1), \qquad s_{T+1} = 0, \quad T = 85. \tag{4}$$

The hourly wage has a deterministic component that depends on health and age and a persistent idiosyncratic component θ_t, which we call ability,

$$\ln w_t(\theta_t, h_t) = W(h_t, t) + \ln \theta_t, \qquad \ln \theta_t = \rho \ln \theta_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2). \tag{5}$$

The profiles $W(h,t)$ and the parameters $(\rho, \sigma_\varepsilon^2)$ are estimated in the first step of Section 4; through $W(h,t)$, a deterioration in health lowers the wage at every ability level. Out-of-pocket medical spending depends on health, age, and participation,

$$m_t = M(h_t, t, P_t), \tag{6}$$

and equals total medical spending net of the coverage provided by private insurance before age 65 and by Medicare from age 65.[3] Spending rises with age and with worse health, and the dependence on participation reflects differences in coverage between workers and nonworkers.

3.4 Social Security, disability insurance, and pensions

This subsection states the rules as they enter the individual's problem; Section 2 and Table A1 give the statutory values for the 1920–1940 cohorts. Let $e_t \equiv w_t n_t$ denote labor earnings.

Lifetime earnings. Retirement benefits are based on the average indexed monthly earnings, which average the highest 35 years of earnings. The model carries an annualized version, $aime_t$, with law of motion

$$aime_{t+1} = \begin{cases} \min\left\{ aime_t + \dfrac{e_t}{35},\ aime_{\max} \right\} & \text{if } t < 60, \\ \min\left\{ aime_t + \max\left\{ 0, \dfrac{e_t - aime_t}{35} \right\},\ aime_{\max} \right\} & \text{if } t \geq 60, \end{cases} \tag{7}$$

[3] Blau and Gilleskie (2008), Imrohoroğlu and Kitao (2012), and French and Jones (2011) model employer-provided health insurance explicitly, and Ozkan (2023) models preventive medical spending. The model abstracts from heterogeneity in health insurance within a health state and age.

where $aime_{\max}$ is the statutory cap. Before age 60 every year of earnings raises the average. From age 60 only a year whose earnings exceed the current average raises it, which limits the incentive to keep working once 35 years of earnings have accumulated.[4]

The primary insurance amount. The base benefit is a piecewise-linear function of $aime_t$ with bend points $aime_0 < aime_1$,

$$\begin{aligned} PIA_t = 0.9\ \min\{aime_t, aime_0\} + 0.32\ \min\{\max\{aime_t - aime_0, 0\},\ aime_1 - aime_0\} \\ + 0.15\ \max\{aime_t - aime_1, 0\}. \end{aligned} \tag{8}$$

The replacement rate falls from 0.9 in the first bracket to 0.15 in the third, so the formula is progressive. The bend points are \$3,720 and \$22,392 and the cap is \$43,800, in 1987 dollars.

Claiming and the claiming-age adjustment. The indicator $b_t \in \{0, 1\}$ equals one once the individual has claimed retirement benefits. Claiming is possible from the early retirement age of 62, is irreversible ($b_t \geq b_{t-1}$), and is mandatory by age 70 ($b_t = 1$ for $t \geq 70$). Let a^c denote the claiming age, the first t with $b_t = 1$. The benefit before the earnings test is the primary insurance amount scaled by a factor that depends on the claiming age and, through the crediting rule stated below, on the years in which benefits were withheld under the earnings test,

$$ssb_t = \psi(a^c)\, PIA_t. \tag{9}$$

Under the status quo, $\psi(\cdot)$ is the schedule of Table 1 divided by 100. It equals one at the normal retirement age of 65, falls by 6.67 percentage points for each year of claiming before 65, and rises by the delayed retirement credit of 4.5 percentage points for each year of claiming after 65, from 0.80 at age 62 to 1.225 at age 70. The reforms of Section 7 replace $\psi(\cdot)$ by the schedules of Table 6 and leave every other rule unchanged; in particular, the normal retirement age at which the earnings-test parameters of (10) change and the age condition of (11) remain those of the 1920–1940 cohorts under the reforms that raise the normal retirement age (Appendix C).

The retirement earnings test. A beneficiary under age 70 who earns more than a threshold y_{ret} loses τ_{ret} dollars of benefits for each dollar of earnings above the threshold, until the benefit is exhausted:

$$ss_t = \begin{cases} \max\{0,\ ssb_t - \tau_{ret} \max\{0,\ e_t - y_{ret}\}\} & \text{if } t < 70, \\ ssb_t & \text{if } t \geq 70. \end{cases} \tag{10}$$

[4]Earnings are indexed to the personal consumption expenditure price index, which stands in for the Social Security Administration's indexation of earnings to average wage growth up to two years before eligibility.

The pair (τ_{ret}, y_{ret}) is 50 percent and \$6,000 before the normal retirement age and 33 percent and \$8,186 from the normal retirement age to age 69, in 1987 dollars; there is no test from age 70. Benefits withheld under the test are not lost: following French and Jones (2011), a year of withheld benefits raises future benefits by the early-claiming reduction factor if it occurs before the normal retirement age and by the delayed retirement credit if it occurs after, so that a year of withholding raises the factor ψ in (9) as a year of later claiming would.

Disability insurance. Disability insurance pays the primary insurance amount, without the early-claiming reduction, to an individual with a disability whose earnings do not exceed a threshold and who is older than 50 and younger than the normal retirement age:

$$db_t = \pi^{db}\, PIA_t\, \mathbb{1}\{h_t = 2\}\, \mathbb{1}\{e_t \leq y_{db}\}\, \mathbb{1}\{50 < t < NRA\}. \tag{11}$$

The threshold y_{db} is \$3,600 in 1987 dollars, the annual equivalent of the Social Security Administration's \$300 monthly substantial gainful activity threshold for nonblind individuals with a disability. The application decision is not modeled; benefits are granted to everyone who meets the health, earnings, and age conditions, and the factor π^{db}, the probability of receiving benefits for older applicants with severe work limitations estimated by Low and Pistaferri (2015), scales the payment to the average amount received by individuals with a disability. The disability state and the disability benefit follow Yu (2024).

Pensions. Defined-benefit pension income depends on the work history through the primary insurance amount and is illiquid before age 62:

$$pb_t = [\, pb(PIA_t) + \zeta_t \,]\, \mathbb{1}\{t \geq 62\}. \tag{12}$$

The function $pb(\cdot)$ is a regressive function of the primary insurance amount with the parameters of Gustman and Steinmeier (1999), and the age-specific term ζ_t captures the accrual pattern of defined-benefit plans documented by Gustman et al. (2000) and Ippolito (1997), which creates incentives to leave the labor force at particular ages; the imputation follows French (2005).

3.5 Taxes, transfers, and the budget constraint

Taxable income is $y_t \equiv ra_t + e_t + pb_t$: the return on assets, labor earnings, and pension income. Social Security, disability, and transfer income are not taxed. The income tax is the parametric function of Bénabou (2002) and Heathcote et al. (2017),

$$T_t(y_t) = y_t - \lambda_t\, y_t^{1-\tau_t}, \tag{13}$$

where τ_t measures progressivity and λ_t the level of taxation; both depend on age, with profiles estimated from PSID data as in Borella et al. (2023), and individuals anticipate the age variation. A proportional payroll tax at rate τ_t^p applies to earnings up to a cap $\bar{y}_t^p$,

$$T_t^p = \tau_t^p \, \min\{e_t, \; \bar{y}_t^p\}. \tag{14}$$

The rate and the cap follow the employee share of the combined Old-Age, Survivors, and Disability Insurance and Hospital Insurance payroll tax, which has ranged from 3 to 7.65 percent since 1960, with caps from \$14,459 to \$64,974 in 1987 dollars.

Assets a_t earn the net return r. The budget constraint is

$$\begin{aligned} a_{t+1} &= a_t + Y_t + b_t \, ss_t + db_t + tr_t - m_t - c_t, \\ y_t &= r a_t + e_t + pb_t, \qquad Y_t = y_t - T_t(y_t) - T_t^p, \\ a_{t+1} &\geq 0, \qquad r = (1 - \tau_k) \, r_f, \end{aligned} \tag{15}$$

where Y_t is after-tax income, ss_t is received only once benefits have been claimed, and tr_t is a government transfer. The borrowing limit is zero: individuals cannot borrow against future Social Security benefits and can rarely borrow against pension wealth, and because the lower bound of the wage distribution is zero, the natural borrowing limit is also zero.[5] The net return is the gross return r_f less a proportional reduction τ_k that stands for corporate and other capital taxes not modeled explicitly, as in Stantcheva (2017); with r_f equal to 4 percent (French 2005) and τ_k equal to 20 percent, the net return is 3.2 percent.[6] Transfers guarantee a consumption floor $\underline{c}$, as in Hubbard et al. (1995) and De Nardi et al. (2010), and stand for programs such as the Supplemental Nutrition Assistance Program and Supplemental Security Income:

$$tr_t = \max\left\{0, \; \underline{c} + m_t - (a_t + Y_t + b_t \, ss_t + db_t)\right\}. \tag{16}$$

The floor is \$3,500 in 2016 dollars, the value of French and Jones (2011).[7] Bequests are accidental: there is no bequest motive, and assets at the terminal age satisfy $a_{T+1} = 0$. The absence of a bequest motive bears on the fit of assets after the mid-50s (Section 4) and on the saving-subsidy reforms (Section 9).

3.6 Recursive problem

The state at age t is $X_t = (a_t, w_t, h_t, b_{t-1}, P_{t-1}, aime_t)$: assets, the wage, health, whether benefits have been claimed, whether the individual worked in the previous period, and aver-

[5] The assumption is standard (e.g., Hubbard et al. 1995; Zeldes 1989).

[6] Capital income is also taxed through y_t in (13); the distinction between r_f and r captures taxes outside that function. The reduction of 20 percent follows Stantcheva (2017).

[7] Braun et al. (2017) model the safety-net programs explicitly.

age earnings. Medical spending, pension income, and retirement and disability benefits are functions of the state and the choices and are not carried separately. The value function solves

$$V_t(X_t) = \max_{c_t, n_t, b_t} \Big\{ u(c_t, n_t; h_t) + \beta\, s_{t+1}(h_t) \int V_{t+1}(X_{t+1})\, dF_t(X_{t+1} \mid X_t, c_t, n_t, b_t) \Big\},$$
$$\text{subject to (2)–(16)}, \quad b_t = 0 \text{ for } t < 62, \quad b_t \geq b_{t-1}, \quad b_t = 1 \text{ for } t \geq 70, \quad a_{T+1} = 0, \tag{17}$$

where $F_t(\cdot \mid \cdot)$ is the age-t distribution of next period's state implied by the health chain, the survival and wage processes, and the budget constraint, and $V_{T+1} \equiv 0$ because survival beyond T is zero. The problem is solved by backward induction from age T. Its solution is a set of decision rules for consumption, hours, and claiming as functions of the state; the participation rule is the indicator that the hours rule is positive, and next period's assets follow from the budget constraint. Simulated life histories are generated by drawing initial states from the data and applying the decision rules forward with simulated shocks. The discretization of the state space, the quadrature for the wage innovation, and the interpolation of the value function are described in Appendix B.

3.7 Scope and relation to earlier models

The model is a partial-equilibrium model of one cohort. The cohort is small relative to the economy: it faces a world interest rate r_f and age–health wage profiles $W(h, t)$ that do not respond to its choices. With constant returns and mobile capital the wage is pinned down by r_f, so fixed r and fixed wages are one assumption, not two; this is the environment of Farhi and Werning (2013) and Golosov et al. (2016) and of the companion paper (Ndiaye and Yu 2026). Section 11 states how factor-price responses would enter the fiscal accounting.

Five further restrictions are maintained. First, the decision-maker is an individual; the estimation targets are profiles for household heads and spouses in the general population, and the joint retirement of couples and spousal benefits are outside the model (Borella et al. 2023). Second, bequests are accidental. Third, health is observed by the government: disability insurance pays on the disability state and the earnings condition, and Medicare coverage after age 65 conditions on participation; no reform considered below conditions on health. Fourth, preferences are time separable. Fifth, the statutory ages act only through the budget constraint and the benefit rules; there are no reference-point or default effects at the early or normal retirement age (Seibold 2021). Section 11 discusses each restriction and the direction in which it bears on the reform results.

Relative to French (2005) and French and Jones (2011), the model differs in five respects. Work is allowed at every age until death rather than ending at age 70. The estimation targets are profiles for the general population rather than for male household heads. The disability

state and disability insurance are modeled explicitly. The fixed time cost of participation depends on health, whereas French and Jones (2011) specify a health time cost that does not depend on participation and a participation cost that does not depend on health, so that here the nonconvexity in labor supply varies with the health state. Exit from the labor force is reversible subject to an estimated reentry cost; French (2005) allow reversible exit without a reentry cost, and French and Jones (2011) include a reentry cost. At the estimated reentry cost the model generates no reentry in equilibrium (Section 4), and claiming is irreversible by construction; the reform simulations of Section 6 to Section 9 use this estimated model without further restriction on exit or reentry.

4 Data, Estimation, and Fit

The model is estimated in two steps, as in Cagetti (2003) and Gourinchas and Parker (2002). The first step estimates from the data, or takes from program rules and the literature, every input that does not require solving the model: health transitions, survival, medical spending, wage profiles and the ability process, the initial distribution at age 30, and the institutional parameters of Section 2. The second step estimates the seven preference parameters by the method of simulated moments (MSM), taking the first-step inputs as given. Section 4.6 collects the differences by health in the data profiles on which the reforms of Section 6 to Section 9 act; Appendix A and Appendix B give the details.

4.1 Data and samples

The primary data source is the Panel Study of Income Dynamics (PSID), the longitudinal survey of US families conducted by the Survey Research Center at the University of Michigan since 1968, with interviews every year until 1997 and every two years thereafter.[8] We use the 1968–2015 waves and the Survey Research Center component of the original sample, and we keep household heads and spouses born between 1920 and 1940, a cohort that faced similar Social Security rules and whose outcomes the PSID records across the life cycle. The sample has 2,248 individuals and 44,404 person-year observations (Table A2). It provides the health transitions, survival probabilities, wage profiles and wage dynamics, the initial distribution of state variables, and the labor supply and asset profiles that the model is estimated to match. Medical spending and insurance coverage come from the Medical Expenditure Panel Survey (MEPS) of the Agency for Healthcare Research and Quality, in which spending reported by individuals is cross-validated with medical providers and insurers.[9] We use the 1999–2012

[8] Panel Study of Income Dynamics, public use dataset, produced and distributed by the Survey Research Center, Institute for Social Research, University of Michigan, https://psidonline.isr.umich.edu/.

[9] Medical Expenditure Panel Survey, Agency for Healthcare Research and Quality, https://meps.ahrq.gov/.

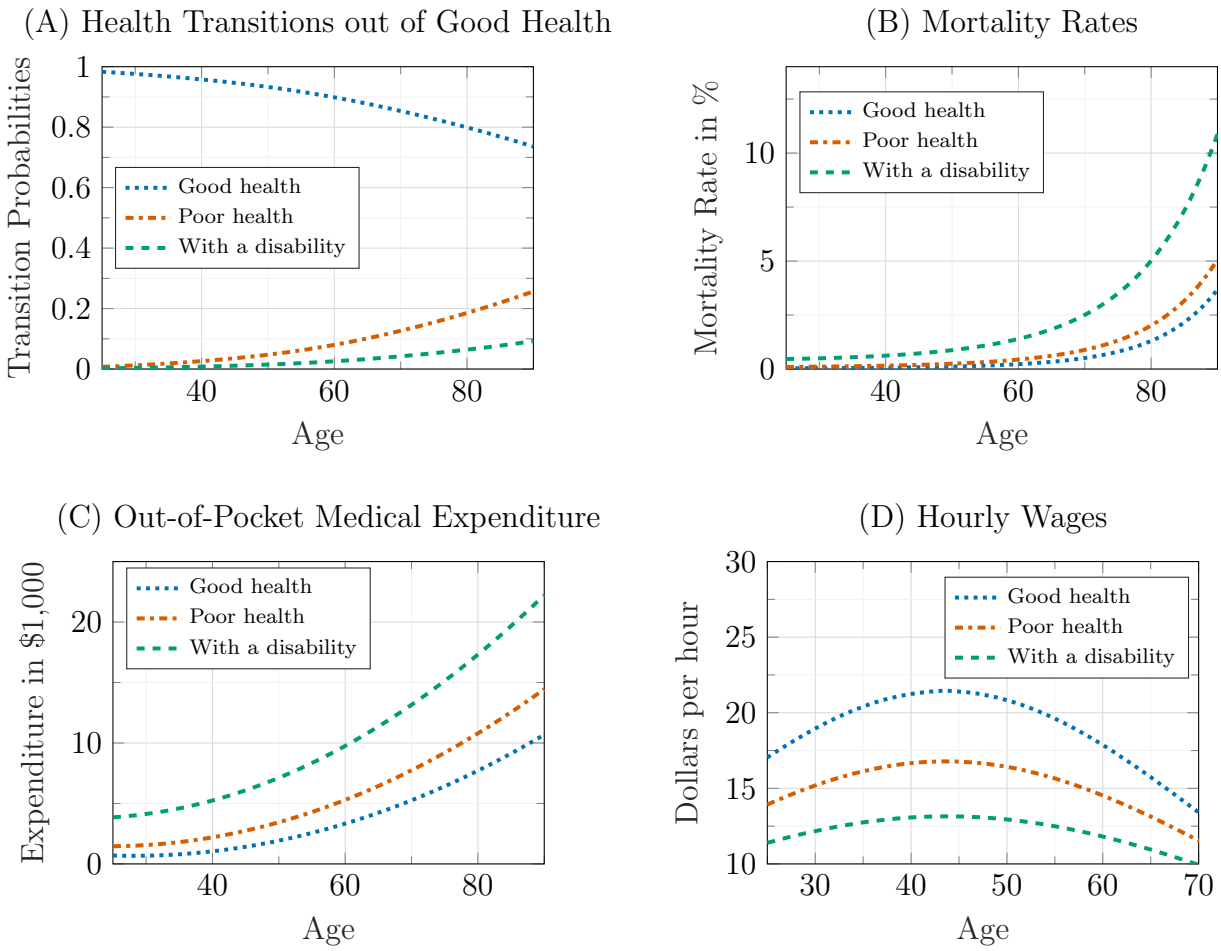
waves and keep household heads aged 20 and older with nonmissing age, medical spending, insurance coverage during working ages, and health status: 120,731 individuals and 211,709 person-year observations (Table A3). All monetary values in the paper are in 2016 dollars, except the statutory Social Security parameters of Table A1, which are in 1987 dollars.

Health is measured in the PSID from three work-limitation questions asked in every wave, following Low and Pistaferri (2015). Good health ($h = 0$) is no condition that limits work, and poor health ($h = 1$) is a condition that limits work somewhat or just a little. A disability ($h = 2$) is a condition that keeps the respondent from doing any type of work and limits the amount of work a lot, the criterion of the Social Security Administration for disability insurance. In the MEPS, health is classified from self-rated health in the three interviews of a year, following Pashchenko and Porapakkarm (2019): a disability is a rating of poor in any interview, good health a rating of good or better in all three, and poor health the remainder. The two measures give similar shares with a disability by age band—9.3 percent in the PSID and 10 percent in the MEPS at ages 50–60, 12.9 and 11.7 percent at ages 60–70—while the MEPS classifies more individuals as being in poor health (Table A4). Appendix A reproduces the survey items.

4.2 First step: profiles by health

Health transitions are probit regressions of the indicator of next year's health state on age dummies, estimated on the subsample in each current state, with the predicted probabilities interpolated linearly across ages. Survival is a logit of an indicator of survival on a quadratic in age, lagged health, and their interaction, estimated on 132,070 PSID person-years (Table A5). Out-of-pocket medical spending is total spending times one minus the coverage share, both estimated on age and year dummies by health group in the MEPS following Pashchenko and Porapakkarm (2019), with coverage from private insurance before age 65 and from Medicare afterward. The profiles are multiplied by 1.60 before age 65 and by 1.90 from age 65 to match the National Health Expenditure Accounts, following De Nardi et al. (2025), and smoothed on a quadratic in age. Hourly wages are annual earnings divided by annual hours in the PSID, trimmed below \$6.50 and above \$250, and corrected for selection into work with the two-stage procedure of Heckman (1976) as applied by Guner et al. (2012), in which log wages are regressed on age polynomials, their interactions with health, and the inverse Mills ratio from a participation probit, on 64,941 observations (Table A6). The persistence and innovation variance of ability in (5) are estimated by maximum likelihood with the Kalman filter on the wage residuals at ages 25–65, dropping the top 0.5 percent of residuals; the estimates are $(\rho, \sigma_\varepsilon^2) = (0.981,\ 0.016)$. The initial distribution at age 30 is the empirical joint distribution of wages, health, assets, and participation of sample members aged 28–32, with the mean of log wages adjusted to the estimated profile within each health group, and with average indexed

Figure 1: First-Step Profiles by Health Status



Notes: (A) Probabilities of transition from good health at age $t-1$ to good health, poor health, and a disability at age t; (B) annual mortality rate in percent; (C) out-of-pocket medical spending in thousands of 2016 dollars; (D) hourly wages in 2016 dollars, corrected for selection into work. Blue dotted: good health; red dash-dotted: poor health; green dashed: with a disability. Sources: PSID 1968–2015 (A, B, D); MEPS 1999–2012 (C).

earnings imputed from five years of work at 2,000 hours a year at the age-30 wage, following French (2005) (Table A7).

Figure 1 shows the four first-step profiles by health. The probability of remaining in good health falls with age, and the decline is absorbed mainly by transitions into poor health. Mortality rises with age for every health state and fastest, at older ages, for individuals with a disability. Out-of-pocket medical spending rises with age and is ordered by health at every age, highest for individuals with a disability. Hourly wages are hump-shaped in age for every health state and are ordered the other way: healthy workers earn more than workers in poor health, who earn more than workers with a disability. Section 4.6 reports the values at the ages that matter for the reforms.

4.3 Second step: method of simulated moments

The second step estimates the seven preference parameters, $\mathcal{P} = (\gamma, \nu, \phi_n(h{=}0), \phi_n(h{=}1,2), \phi_{RE}, L, \beta)$. The targets are life-cycle profiles for ages 30 to 69 constructed from the PSID with the fixed-effects regression of French (2005),

$$\begin{aligned} Z_{it} = f_i + \sum_k B_{gk}\, \mathbb{1}\{age_{it} = k\}\, \mathbb{1}\{h_{it} = 0\} + \sum_k B_{bk}\, \mathbb{1}\{age_{it} = k\}\, \mathbb{1}\{h_{it} \neq 0\} \\ + \sum_f B_f\, \mathbb{1}\{famsize_{it} = f\} + B_u\, U_t + u_{it}, \end{aligned} \tag{18}$$

where Z_{it} is participation or hours of individual i at age t, f_i is an individual effect, the age dummies are interacted with good health ($h = 0$) and with poor health or disability ($h \neq 0$), $famsize_{it}$ is family size, and U_t is the unemployment rate. The profiles are evaluated at a family size of 3, an unemployment rate of 6.5 percent, and the mean individual effect of individuals aged 50, in average health at 50, and born in 1935, so that family size, the business cycle, and cohort effects do not enter them. Participation is annual hours above 800; hours are conditional on participation; assets are nonpension net worth, divided equally between partners following Borella et al. (2018); the asset and reentry profiles are not conditioned on health (Appendix A). Seven blocks of 40 age-specific moments—participation of healthy and of unhealthy individuals, hours of healthy and of unhealthy workers, mean assets, median assets, and the mean reentry rate—give 280 moments for seven parameters. The estimator is

$$\hat{\mathcal{P}} = \underset{\mathcal{P}}{\operatorname{argmin}} \; \frac{I}{1+\varsigma}\, \hat{\varphi}(\mathcal{P};\chi)'\, \hat{\mathbf{W}}_I\, \hat{\varphi}(\mathcal{P};\chi), \tag{19}$$

where $\hat{\varphi}(\mathcal{P};\chi)$ stacks the differences between the data moments and the moments of profiles simulated from the model at $\mathcal{P}$ and at the first-step inputs χ, I is the number of individuals in the data, ς is the ratio of the number of data observations to the number of simulated observations, and $\hat{\mathbf{W}}_I$ is the inverse of the diagonal matrix formed from the diagonal of the data variance–covariance matrix. Identification follows French (2005) under two assumptions: work decisions do not affect health or wages, and age changes work and saving incentives but not preferences. Appendix B writes out the moment vector and the estimation steps.

4.4 Estimates and identification

Table 3 reports the estimates. The discount factor is 0.98, identified by intertemporal substitution and therefore by the labor supply and asset profiles, in particular at the ages at which wealth is accumulated for retirement. The time endowment is 4,491 hours a year, and the fixed cost of participation is 791 hours for healthy workers and 868 hours for workers in poor health or with a disability. These three parameters are identified by participation and

Table 3: Estimated Preference Parameters

Parameter	Definition	Estimates	S.E.
γ	Consumption weight	0.5354	0.0047
ν	CRRA for flow utility	1.7029	0.0681
β	Time discount factor	0.9797	0.0023
L	Time endowment	4490.5	21.65
ϕ_{RE}	Labor market reentry cost	3311.5	82.02
$\phi_n(h=0)$	Fixed cost of work, healthy	791.07	15.65
$\phi_n(h=1,2)$	Fixed cost of work, unhealthy	867.54	22.68
χ^2 statistic (degrees of freedom = 273)		1463	

Notes: Method of simulated moments on 280 moments (ages 30–69); standard errors in the last column; χ^2 statistic with 273 degrees of freedom. L, ϕ_{RE}, $\phi_n(h{=}0)$, and $\phi_n(h{=}1,2)$ are in hours per year. Point estimates are those of the May 2025 version of the companion paper (Ndiaye and Yu); standard errors and the χ^2 statistic were recomputed in 2026. The May 2025 manuscript reported standard errors of 0.0042, 0.0587, 0.0020, 24.71, 5.1666, 20.003, and 30.613 and a χ^2 statistic of 1386.8 for the same point estimates.

hours by health: the fixed cost generates a reservation level of hours below which working does not pay, and the endowment scales hours; the estimates lie within the range of French and Jones (2011) and Borella et al. (2023). Yu (2024), whose specification adds a time cost of being unhealthy that is paid whether or not the individual works, estimates participation costs of 936 hours for healthy and 755 hours for unhealthy workers on the 1920–1935 PSID cohort; the specification here has no separate health time cost, and the participation cost is higher for unhealthy workers. The reentry cost is 3,312 hours, identified by reentry rates that stay below 5 percent at every age in the data; at this value the model generates no reentry (Section 4.5). The curvature parameter ν is 1.70 and the consumption weight γ is 0.54, which imply a coefficient of relative risk aversion with respect to consumption, $-c_t\, u_{cc}/u_c$, of about 1.38; they are identified by the asset and labor supply profiles, since higher risk aversion raises precautionary saving and work early in life. The χ^2 statistic of 1463 with 273 degrees of freedom rejects the overidentifying restrictions, as the corresponding statistics do for the estimated models of French (2005) and French and Jones (2011).

4.5 Fit

Figure 2 compares the model's participation and hours profiles by health with the PSID profiles and their 95 percent confidence bands; Figure B1 in Appendix B does the same for mean and median assets and for reentry.

Healthy participation is above the data at ages 30–38 (93.25 percent against 88.34 at age 30), within a few percentage points of the data from age 39 through the mid-50s, and below the data at every age from 58 to 69: 59.19 against 64.34 at age 60 and 27.30 against 28.71 at age 65. Unhealthy participation is above the data at ages 33–34 and from age 36 to age 50, and above the 95 percent confidence band at age 39 and at ages 43–49 (84.97 against

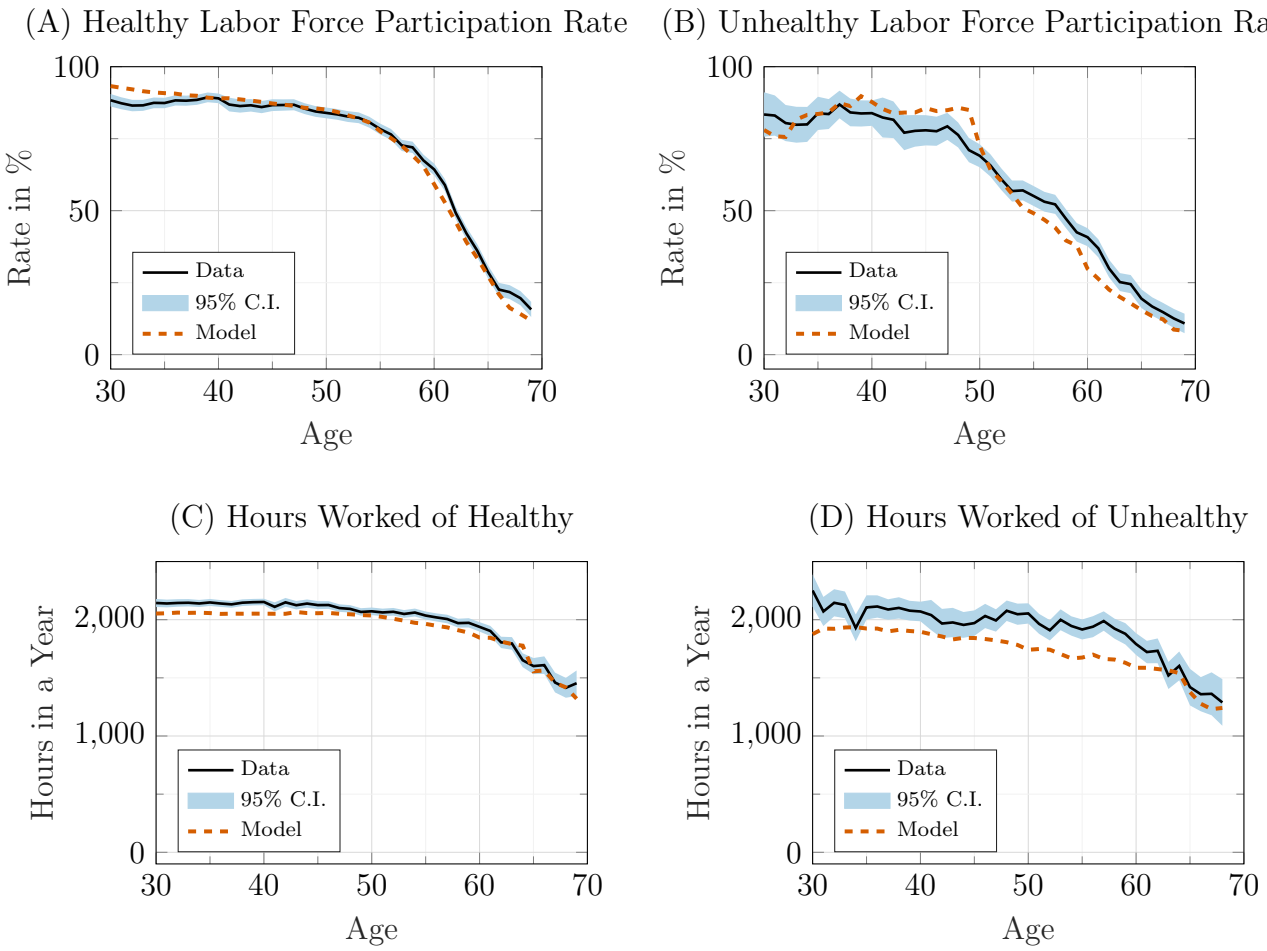
71.03 at age 49); it is below the data at every age from 51 to 69, with the under-prediction concentrated at ages 54–66 (29.83 against 40.79 at age 60 and 15.46 against 19.45 at age 65), and the model's decline through the 50s is steeper than the data's. Hours conditional on participation are below the data at most ages for both groups: 2,053 against 2,144 at age 30 and 1,556 against 1,601 at age 65 for healthy workers, and 1,876 against 2,251 at age 30 for unhealthy workers. Mean assets are matched through the 40s; from the early 50s the mean, and from the mid-50s the median, fall below the data, and the gap widens with age: at age 60 the model has 1.642 against 3.218 for the mean and 1.183 against 2.252 for the median, in hundreds of thousands of 2016 dollars; the median is above the data from the 30s through the early 50s (1.833 against 1.471 at age 50). The model has no bequest motive, so it does not generate the saving in old age that a motive for planned bequests produces (De Nardi et al. 2010; Lockwood 2018); the motive is omitted so that the environment remains comparable with the life-cycle taxation literature with exogenous retirement (Farhi and Werning 2013; Golosov et al. 2016; Stantcheva 2017). Reentry is zero in the model at every age, against a data profile that stays below 5 percent and peaks at 4.47 percent at age 63.

Each misfit bears on a part of the reform analysis. Healthy participation is under-predicted at ages 58–69 and unhealthy participation at ages 54–66. The claiming-schedule reforms of Section 7, whose participation effects are reported at ages 62–64 and 65–69, and the disability-insurance elimination of Section 6 are therefore evaluated from a baseline with fewer workers at those ages than the data. The over-prediction of unhealthy participation at 43–49 falls at ages at which the reforms change incentives only through the accrual of future benefits, the 30–49 column of the reform tables. The under-prediction of hours lowers the earnings base on which payroll taxes, the income tax after age 62, and the earnings test are computed, which bears on the fiscal columns of every family and in particular on the income-tax reforms of Section 9. The under-prediction of assets after the mid-50s bears on the saving-subsidy reforms, whose fiscal cost is proportional to the asset base; the direction of the resulting bias is not determined by the reported output. The absence of reentry means that the participation responses to every reform operate through the timing of exit from the labor force and not through returns to it.

4.6 The health gradient in retirement

The reforms of Section 6 to Section 9 act on a cohort whose labor supply, mortality, medical spending, and wages differ by health, and the estimated participation cost adds a difference in the fixed cost of working. In the PSID profiles (black lines of Figure 2; Figure A1 in Appendix A), participation of healthy and unhealthy individuals is 83.94 and 69.06 percent at age 50, 64.34 and 40.79 percent at age 60, and 28.71 and 19.45 percent at age 65: the gap is about 5 percentage points at age 40, widens from the mid-40s, more than a decade before the early retirement age, and exceeds 20 percentage points at every age from 52 to 61. Disability

Figure 2: Model Fit: Labor Force Participation and Hours by Health Status



Notes: Black: PSID profiles from (18); shaded: 95 percent confidence bands; red dashed: model. Ages 30–69 (panel D: ages 30–68). (A) and (B) labor force participation in percent, defined as annual hours above 800; (C) and (D) annual hours conditional on participation. "Unhealthy" pools poor health and disability. Model series recomputed in 2026; data profiles and confidence bands are unchanged from the May 2025 version. Source: PSID 1968–2015.

insurance is a route out of the labor force before the normal retirement age for workers with a disability (Maestas et al. 2013; French and Song 2014; Low and Pistaferri 2015; Li 2018); in the model it enters through (11). In the first-step profiles (Figure 1), the probability of remaining in good health from one year to the next is 0.8986 at age 60; annual mortality at age 70 is 0.51 percent in good health, 0.89 percent in poor health, and 2.51 percent with a disability; out-of-pocket medical spending at age 70 is \$5,260, \$7,741, and \$13,150; and the hourly wage at age 60 is \$17.87, \$14.53, and \$11.81. At the ages at which the claiming schedule applies, a worker in poor health or with a disability therefore faces a lower wage, a higher participation cost, a shorter expected horizon over which a delayed benefit is paid, and higher medical spending than a healthy worker of the same age. These profiles, with the participation cost of Table 3,

determine in the model who exits before the normal retirement age and who continues past it, and therefore who is affected by a change in the early-claiming reduction and who by a change in the delayed retirement credit (Section 7). The decomposition of retirement behavior into these channels is in the companion paper (Ndiaye and Yu 2026). The reform tables that follow report participation by age group, and Section 7.3 uses these profiles to say who, in the model, moves under each reform.

5 Welfare and Fiscal Accounting

Every reform in Sections 6–10 is evaluated with one welfare measure and one fiscal accounting. This section defines both, maps each column of the reform tables to a formula, and relates the marginal value of public funds to the objects of the sufficient-statistics literature; Appendix D collects the algebra. Superscript 0 denotes the status quo, ρ a reform, and Δ the difference.

5.1 Welfare measures and willingness to pay

Let V_{30}^i be the value function (17) of simulated individual i at age 30, the entry age of the model. Cohort welfare is the average of V_{30}^i over the I^s simulated individuals, V^0 under the status quo and V^ρ under reform ρ: an ex ante measure, taken before health, mortality, wage, and medical-expense shocks are realized. The consumption-equivalent (CE) welfare change x_ρ^{CE} is the log change in status-quo consumption at every age and state, with leisure held fixed, that makes the cohort as well off as under the reform; $e^{x_\rho^{\mathrm{CE}}} - 1$ is the corresponding proportional change, which x_ρ^{CE} approximates to first order. Flow utility (1) is homogeneous of degree $\gamma(1-\nu)$ in consumption, so x_ρ^{CE} solves $V^\rho = e^{\gamma x_\rho^{\mathrm{CE}}(1-\nu)} V^0$:

$$x_\rho^{\mathrm{CE}} = \frac{\ln(V^\rho/V^0)}{\gamma(1-\nu)}, \qquad V^0 \equiv \frac{1}{I^s}\sum_{i=1}^{I^s} V_{30}^{i,0}, \qquad V^\rho \equiv \frac{1}{I^s}\sum_{i=1}^{I^s} V_{30}^{i,\rho}. \tag{20}$$

With $\nu > 1$ (Table 3) both averages are negative, so the ratio V^ρ/V^0 is positive; a welfare-improving reform makes V^ρ less negative, the ratio lies below one, and dividing its negative logarithm by $\gamma(1-\nu) < 0$ gives $x_\rho^{\mathrm{CE}} > 0$. The WTP column of every reform table reports $100\, x_\rho^{\mathrm{CE}}$, the CE change in percent, the cohort's willingness to pay (WTP) for the reform as a share of consumption.

The leisure-equivalent (LE) change scales leisure instead of consumption:

$$x_\rho^{\mathrm{LE}} = \frac{\ln(V^\rho/V^0)}{(1-\gamma)(1-\nu)} = \frac{\gamma}{1-\gamma}\, x_\rho^{\mathrm{CE}}. \tag{21}$$

At the estimate $\gamma = 0.5354$ the factor $\gamma/(1-\gamma)$ is 1.15, so the LE column of Table 5 is a rescaling of the CE column and appears nowhere else.

The dollar value of a reform converts x_ρ^{CE} with baseline consumption:

$$\mathcal{W}_\rho = x_\rho^{\mathrm{CE}}\,\bar{C}_s, \qquad \bar{C}_s \equiv \sum_{t=s}^{85}(1+r)^{-(t-62)}\,\mathbb{E}\big[c_t^0\big]\,, \qquad s \in \{62, 30\}, \tag{22}$$

where $\mathbb{E}[\cdot]$ averages over the simulated cohort, r is the net return of Section 3, and $\bar{C}_s$ is the per-capita net present value (NPV) at age 62 of status-quo consumption from age s onward. The window is $s = 62$ for the Social Security, Medicare, and income-tax families, whose reforms change flows received from age 62 onward, and $s = 30$ for the saving subsidies, which change the return on assets at every age; for $t < 62$ the factor compounds forward. The CE change scales consumption at all ages while $\bar{C}_{62}$ counts consumption from age 62, so $\mathcal{W}_\rho$ is the dollar value of the CE change applied to consumption after 62; because $\bar{C}_{30}$ and $\bar{C}_{62}$ differ, MVPF levels of the saving-subsidy family are not compared with those of the other families.

5.2 Direct, indirect, and net fiscal effects

Three aggregates are computed for each family, all per capita and discounted to age 62 at the net return $r = 3.2$ percent with the factor $(1+r)^{-(t-62)}$: B, the outlay of the program being reformed; O, the other taxes and transfers listed in Table 4; and G, government spending on the cohort, whose baseline value G^0 is the base of the Net column and whose change satisfies $\Delta G = \Delta B + \Delta O$ when every flow the reform changes is counted in B or in O. Table 4 states what B and O contain in each family. The three columns are

$$\mathrm{Direct}_\rho = 100\,\frac{\Delta B}{\mathrm{base}_f}, \qquad \mathrm{Indirect}_\rho = 100\,\frac{\Delta O}{O^0}, \qquad \mathrm{Net}_\rho = 100\,\frac{\Delta G}{G^0}, \qquad \Delta G = \Delta B + \Delta O, \tag{23}$$

where $\mathrm{base}_f = B^0$ (baseline retirement benefits) for the Social Security families, $T^{p,0}$ (baseline payroll-tax collections) for Medicare, $|T^0|$ (baseline income-tax collections) for the income-tax reforms, for which B is revenue forgone, and G^0 for the saving subsidies, for which B is the subsidy. The Direct bases are positive by construction, so a negative Direct entry is a reduction in the program's own outlay, and a negative Net entry is a reduction in government spending on the cohort; the Indirect column is read with the same convention, a negative entry being a reduction in the government's other net outlays.

Because the three bases differ, the Net column is not the sum of the other two: for I1 the Direct and Indirect entries are 0.70 and −0.81 while Net is 0.69; for P1.1 they are 0.57 and −0.10 while Net is −0.03. The relation that holds is

$$\text{Net}_\rho = \omega_D \,\text{Direct}_\rho + \omega_O \,\text{Indirect}_\rho, \qquad \omega_D \equiv \frac{\text{base}_f}{G^0}, \qquad \omega_O \equiv \frac{O^0}{G^0}, \tag{24}$$

which follows from (23) whenever $\Delta G = \Delta B + \Delta O$. The weights are ratios of baseline aggregates and are not printed; they sum to one only if $G^0 = B^0 + O^0$, which is not assumed. For the saving subsidies both equal one (the subsidy is zero at the baseline and the Direct base is G^0), and Net equals Direct plus Indirect in every K row.

5.3 The marginal value of public funds and the table columns

The marginal value of public funds (MVPF) of a reform is the ratio of the beneficiaries' willingness to pay to the net cost to the government, inclusive of behavioral responses (Hendren 2016; Hendren and Sprung-Keyser 2020; Finkelstein and Hendren 2020). Here both terms are per capita for the cohort:

$$\text{MVPF}_\rho = \frac{\mathcal{W}_\rho}{\Delta G_\rho}. \tag{25}$$

Substituting (22) and (23),

$$\text{MVPF}_\rho = \kappa_f \,\frac{\text{WTP}_\rho}{\text{Net}_\rho}, \qquad \kappa_f \equiv \frac{\bar{C}_s}{G^0}, \tag{26}$$

so within a family the MVPF column is proportional to WTP/Net. For P1.2, the ratio of WTP to Net is 0.22/0.09; the MVPF of 0.97 converts it to dollars of welfare per dollar of net spending.

When $\mathcal{W}_\rho > 0$ and $\Delta G_\rho > 0$ the MVPF is a positive number. When $\mathcal{W}_\rho > 0 > \Delta G_\rho$ the reform raises welfare and reduces net spending: it is self-financing, the MVPF is infinite, and the tables print ∞; when in addition $\Delta B_\rho > 0$ they report a lower bound that counts only the program's own outlay,

$$\text{MVPF}^{\text{LB}}_\rho = \frac{\mathcal{W}_\rho}{\Delta B_\rho} = \frac{\bar{C}_s}{\text{base}_f}\,\frac{\text{WTP}_\rho}{\text{Direct}_\rho}, \tag{27}$$

printed as $[\text{MVPF}^{\text{LB}}_\rho, +\infty)$; P1.1 is the one such row, with a lower bound of 0.52. When $\Delta B_\rho < 0$ as well as $\Delta G_\rho < 0$, as in every AD and AL row, no finite bound exists. When $\mathcal{W}_\rho < 0$ and $\Delta G_\rho < 0$, as in the AN rows, the positive ratio is the welfare loss per dollar of net savings and, since x^{CE} and ΔG both change sign when a reform is reversed, equals the MVPF of the reverse reform except for the difference between the consumption aggregates $\bar{C}_s$

Table 4: Accounting Conventions by Reform Family

Family	Program outlay B	Base of Direct (base$_f$)	Base of Indirect (O^0)	Base of Net (G^0)	Window for $\mathcal{W}$ ($\bar{C}_s$)	MVPF cell
Social Security: benefit formula (P, L); claiming schedule (AN, AD, AL)	NPV of retirement benefits paid by the program, $b_t\, ss_t$, from age 62	B^0: baseline retirement benefits	All other taxes and transfers: disability benefits and transfers, net of income and payroll taxes	Baseline government spending	From age 62	Number if $\Delta G > 0$; $[\mathcal{W}/\Delta B, +\infty)$ if $\Delta G < 0 < \Delta B$; ∞ if $\Delta G < 0$ and $\Delta B < 0$; for $\mathcal{W} < 0$, $\Delta G < 0$ (AN) the number is the value of the reverse reform
Medicare Part A; Part B	NPV of medical payments by the program to the targeted beneficiaries after age 65	$T^{p,0}$: baseline payroll-tax collections	All taxes and transfers: Social Security benefits and transfers, net of income and payroll taxes	Baseline government spending	From age 62	Number
Income tax (I1–I5)	NPV of income-tax revenue forgone after age 62, $-\Delta T$	$\lvert T^0\rvert$: baseline income-tax collections	All other taxes and transfers: Social Security benefits and transfers, net of payroll taxes	Baseline government spending	From age 62	Number
Saving subsidy (K1–K3)	NPV of the subsidy *Sub* paid on assets at all ages	G^0: baseline government spending	Government spending excluding the subsidy	Baseline spending including the subsidy	From age 30, compounded to 62	$[\mathcal{W}/Sub,\ \mathcal{W}/\Delta G^{\text{full}}]$

Notes: All fiscal aggregates are per-capita net present values (NPVs) discounted to age 62 at the net return r with the factor $(1+r)^{-(t-62)}$; the columns are defined in (20), (22)–(24), and (25)–(28), and the three fiscal columns are related by (24), not by addition. Participation columns report changes in labor force participation at ages 30–49, 62–64, and 65–69 in percentage points. The definitions used in computing the tables are the following. *Social Security:* "the direct fiscal effect is calculated as the percentage change in the net present value (NPV) of total retirement benefits paid out by the program, discounted to age 62 by a factor of $(1+r)^{-1}$. The indirect fiscal effects are computed as the percentage change in the NPV of all other taxes and transfers, including payments to disability benefits and government transfers net of income and payroll tax collections. The net fiscal effect combines the direct and indirect effects." The factor $(1+r)^{-1}$ in this definition is applied for each year between t and age 62, that is, $(1+r)^{-(t-62)}$. *Medicare:* "the direct fiscal effect is calculated as the changes in the NPV of medical payments made by the program to the targeted beneficiaries after age 65 (Part A for everyone; Part B for labor market nonparticipants), expressed as a share of payroll tax collections at the baseline. The indirect fiscal effects are computed as the percentage change in the NPV of all taxes and transfers, including payments to all Social Security benefits and government transfers net of income and payroll tax collections. The net fiscal effect is measured as the sum of the NPV of the change in government spending and the total medical payments to the targeted beneficiaries, divided by the NPV of government spending at the baseline." *Income tax:* "the direct fiscal effect is calculated as the percentage change in the NPV of negative total income tax collection. The indirect fiscal effect is computed as the percentage change in the NPV of all other taxes and transfers, including payments to all Social Security benefits and government transfers net of payroll tax collections. The net fiscal effect combines the direct and indirect effects." *Saving subsidy:* "the direct fiscal effect is calculated as the NPV of the subsidy as a share of government spending at the baseline. The indirect fiscal effect is computed as the percentage change in the NPV of government spending, excluding subsidies. The net fiscal effect is measure[d] as the percentage change in the NPV of the government spending, including subsidies." The welfare NPV for this family starts at age 30 and "the lower bound uses only the subsidy in the denominator."

of (22) under the two rules; Section 7 reads the AN rows in this way. For the saving subsidies the tables print an interval,

$$\text{MVPF}_\rho \in \left[\frac{\mathcal{W}_\rho}{Sub_\rho},\ \frac{\mathcal{W}_\rho}{\Delta G_\rho^{\text{full}}}\right], \tag{28}$$

whose upper bound uses the net cost $\Delta G_\rho^{\text{full}}$ computed as if all capital gains were realized and taxed in each period.

Hendren and Sprung-Keyser (2020) write that "the creation of MVPFs requires various judgment calls." Ours are the following. Welfare is the equal-weighted ex ante average of expected utility at age 30. Fiscal flows are per-capita NPVs at age 62 at the net return r and enter at face value: no marginal cost of public funds multiplies the Net column, unlike in the marginal-cost-of-funds tradition of Mayshar (1990), Slemrod and Yitzhaki (1996), and Kleven and Kreiner (2006); the reform's own revenue response is inside ΔG, and the cost of raising the funds enters by comparing MVPFs across policies. The accounting is for one cohort, with no transition (Section 2 states the stance on incidence outside the cohort), and at fixed prices (Section 11 sets out how factor-price responses would enter $\mathcal{W}$ and ΔG). The participation columns report changes in labor force participation at ages 30–49, 62–64, and 65–69 in percentage points (pp); they are the behavioral responses behind the fiscal effects and enter no formula directly.

5.4 Relation to sufficient statistics: consumption smoothing and the fiscal multiplier

In the sufficient-statistics approach to social insurance (Baily 1978; Chetty 2006, 2008, 2009; Chetty and Finkelstein 2013), a marginal expansion of a program raises welfare when the consumption-smoothing value of the transfer exceeds its fiscal cost, the mechanical outlay plus the fiscal externality from behavioral responses. Haller (2022) extends the formula to a dynamic retirement setting and evaluates increases in the early retirement age and reductions in benefit levels with a fiscal multiplier, the total fiscal effect per mechanical dollar, estimated from Austrian administrative data; the multiplier is the object to which the fiscal columns are mapped below. Split the change in program outlays into a mechanical and a behavioral part,

$$\Delta B = \Delta B^M + \Delta B^B, \tag{29}$$

where ΔB^M applies the reformed benefit, coverage, tax, or return rule to the status-quo histories with every policy function held at its status-quo value, and ΔB^B is the remainder; ΔO is behavioral except for mechanical interactions through the consumption floor (16) and the income-tax base in (15). Define

$$\mathrm{FM} \equiv \frac{\Delta G}{\Delta B^M} = 1 + \frac{\Delta B^B + \Delta O}{\Delta B^M}, \qquad \mathrm{FM}_B \equiv \frac{\Delta B}{\Delta B^M} = 1 + \frac{\Delta B^B}{\Delta B^M}, \tag{30}$$

the fiscal multiplier and its within-program part, and

$$\hat{\pi} \equiv \frac{\mathcal{W}}{\Delta B^M} - 1, \qquad \mathrm{MVPF} = \frac{1+\hat{\pi}}{\mathrm{FM}}, \qquad \mathrm{MVPF}^{\mathrm{LB}} = \frac{1+\hat{\pi}}{\mathrm{FM}_B}. \tag{31}$$

Both equalities in (31) are identities: $\mathcal{W} = (1+\hat{\pi})\,\Delta B^M$ by the definition of $\hat{\pi}$ and $\Delta G = \text{FM}\ \Delta B^M$ by (30), so their ratio is (25), and replacing ΔG by ΔB gives (27); both require only $\Delta B^M \neq 0$ and hold exactly for discrete reforms. An MVPF of one is the condition $1+\hat{\pi} = \text{FM}$, the Baily–Chetty condition when a dollar of public funds is valued at one, that is, relative to a financing instrument whose own MVPF is one; an infinite MVPF with $\Delta B^M > 0$ means $\text{FM} < 0$, behavioral fiscal gains exceeding the mechanical outlay.

For a marginal change in a transfer rule, the envelope theorem applied to (17) gives $\mathcal{W} \approx \sum_t (1+r)^{-(t-62)}\,\mathbb{E}[\omega_t\,\delta_t^M]$, where δ_t^M is the mechanical change in the individual's resources at age t and ω_t is the marginal utility of consumption at t, discounted at β and compounded to age 62 at r, relative to the average marginal utility implicit in the conversion of x^{CE} to dollars (Appendix D). Then $\pi \approx \mathbb{E}[\omega\,\delta^M]/\mathbb{E}[\delta^M] - 1$ is the transfer-weighted average of relative marginal utilities minus one, the consumption-smoothing value of the transfer: the term of Baily (1978) and Chetty (2006), positive when the transfer accrues in states of high marginal utility such as poor health, low assets, and survival to old age (see also Landais and Spinnewijn 2021; Kolsrud et al. 2024). With nonseparable utility and $\nu > 1$, marginal utility of consumption falls with leisure, so a dollar is valued less after labor-force exit than while working at the same consumption. Two cautions apply. First, the structural $\mathcal{W}$ is exact for the discrete reforms considered, whereas the envelope expression is first-order in the size of the reform and omits the utility changes of individuals who switch participation or claiming state, discrete choices with fixed costs (Kleven 2021); $\hat{\pi}$ is defined by (31) so that the identities hold regardless. Second, the printed Direct column is ΔB, not ΔB^M: it combines the mechanical change with the within-program behavioral response, while the Indirect column holds the behavioral response outside the program. Separating ΔB^M from ΔB^B requires an accounting pass over saved status-quo histories with behavior held fixed, which is not part of the present output: the tables identify $(1+\hat{\pi})/\,\text{FM}$ and $(1+\hat{\pi})/\,\text{FM}_B$ but neither FM nor $\hat{\pi}$ separately, the fiscal multiplier cannot be read from the Direct column, and the comparison with Haller (2022) is one of objects, not of magnitudes.

Two sign inferences follow from the printed cells and the schedules of Table 6, and no others. For AD1–AD5 every factor after the NRA rises and every factor before it is unchanged, so with claiming held at its status-quo age outlays rise for everyone who claims after age 65 or whose benefits withheld under the earnings test after the NRA are credited at the DRC (Section 3.4), and fall for no one: $\Delta B^M \geq 0$. Direct and Net are negative in all five rows, so $\Delta B < 0$ and $\Delta G < 0$, hence $\text{FM}_B < 0$ and $\text{FM} < 0$ whenever $\Delta B^M > 0$: the behavioral change in outlays more than offsets the mechanical increase. For AN1–AN3 every factor falls, so $\Delta B^M < 0$ by construction, and WTP is negative; the finite ratio is the reversal value of Section 5.3. For the benefit-formula, Medicare, and income-tax families $\Delta B^M > 0$, but the

Table 5: Welfare Changes from Eliminating Each Policy Instrument

		CE Welfare Δ	LE Welfare Δ
(D1)	No Social Security Benefits	-5.89	-6.79
(D2)	Loss From Incomplete Medicare Coverage After Age 65	-4.28	-4.93
(D3)	Full Savings Subsidy	3.60	4.15
(D4)	No Income Tax	3.16	3.64
(D5)	No Disability Benefits	-0.34	-0.40

Notes: Consumption-equivalent (CE) and leisure-equivalent (LE) welfare changes, in percent, for each counterfactual simulation that eliminates a policy instrument, computed from (20) and (21). D1 sets Social Security benefits to zero. D2 sets out-of-pocket medical expenses after age 65 to zero for nonparticipants and participants; the entry is the welfare of the status quo relative to this counterfactual, the loss from incomplete Medicare coverage after age 65. D3 sets the net return to assets r to the gross interest rate r_f, from 3.2 to 4 percent; it is the same experiment as K3 in Table 9. D4 sets the income tax to zero. D5 sets disability benefits to zero. LE equals CE times $\gamma/(1-\gamma)$, 1.15 at the point estimates. The blue gradient indicates larger absolute values. Source: authors' simulations of the estimated model.

sign of ΔB^B is not identified, so whether FM exceeds one is undetermined; the AL rows move factors in both directions, so ΔB^M itself is unsigned.

6 What Each Instrument Is Worth: Elimination Experiments

Before pricing marginal reforms, we remove each policy instrument in turn and measure the welfare change with the consumption-equivalent (CE) measure (20). Table 5 reports five experiments. D1 sets Social Security benefits to zero. D2 sets out-of-pocket medical spending after age 65 to zero for participants and nonparticipants and reports the status quo relative to that counterfactual, that is, the loss from incomplete Medicare coverage after age 65. D3 raises the net return on assets from 3.2 to 4 percent, the gross return; it is the same experiment as the largest saving subsidy, K3, of Section 9. D4 sets the income tax to zero, and D5 sets disability benefits to zero.

Eliminating Social Security benefits lowers welfare by 5.89 percent of consumption, the largest effect among the five eliminations. The loss from incomplete Medicare coverage after age 65 is 4.28 percent: covering all out-of-pocket spending after 65 would raise welfare by less than the existing Social Security benefits do. Raising the net return to the gross return raises welfare by 3.60 percent and eliminating the income tax by 3.16 percent; eliminating disability benefits lowers welfare by 0.34 percent. The ranking—Social Security, Medicare coverage, the return on saving, the income tax, disability insurance—orders the instruments by the welfare at stake. The LE column is the CE column multiplied by $\gamma/(1-\gamma)$, 1.15 at the point estimates, by (21); it carries no information beyond the CE column and is not reported for the reforms.

The participation effects of the five eliminations are reported as age profiles by health status in Appendix C; no table summarizes them. The profiles show the largest change in participation under D1, from age 63 on, and they show that eliminating disability benefits

raises the participation of people with a disability after age 50, a group small enough that the welfare effect of D5 is the smallest in Table 5.

French and Jones (2011) emphasize the role of Medicare for retirement; in the earlier working-paper version, French and Jones (2004) find a larger effect of Social Security than of Medicare on labor supply. In the welfare metric used here, eliminating Social Security benefits has a larger effect than completing Medicare coverage after age 65, and the sections that follow show that reforms of the two instruments have different fiscal returns. These experiments remove whole instruments; the sections that follow price marginal changes to each, beginning with Social Security, the instrument with the most welfare at stake.

7 Reforms to the Claiming Schedule: A Higher Normal Retirement Age versus a Steeper Actuarial Adjustment

This section evaluates the two ways of changing the claiming schedule described in Section 2.2: raising the NRA (AN1–AN3), raising the DRC (AD1–AD5), and setting equal early-claiming reductions and delayed credits (AL1–AL9). Table 6 and Figure 3 show the schedules; Table 7 reports the outcomes, with the columns defined in Section 5 and Table 4.

7.1 How each reform moves the benefit schedule

Table 6 reports the factor $\psi(a^c)$ of (9) under the status quo (AN0) and under each reform, as a percentage of the PIA at claiming ages 62 to 70. Raising the NRA moves the whole schedule: under AN1–AN3 the factor at age 62 is 75, 70, and 65 instead of 80, and the factor at age 70 is 118, 113.5, and 109 instead of 122.5; every entry in the AN rows lies below the AN0 entry at the same age. Raising the DRC leaves the factors at ages 62–65 at their status-quo values of 80, 86.67, 93.33, and 100 and raises every factor after 65: at age 70, 140 under AD1 and 150 under AD5 instead of 122.5. The symmetric adjustments rotate the schedule around the NRA, where the factor stays at 100: the factor at age 62 is 79 under AL1 and 55 under AL9, and the factor at age 70 is 135 and 175. Figure 3 plots AN0, AN2, AD5, and AL4.

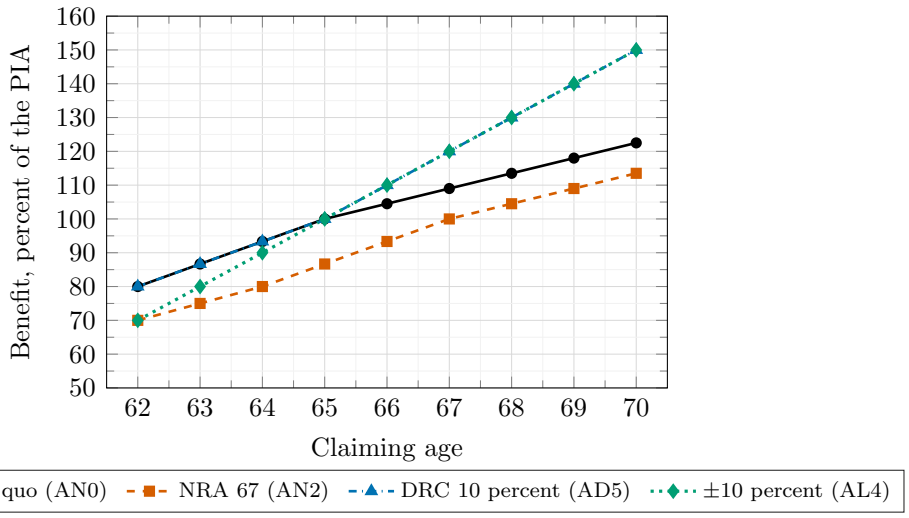
The two kinds of reform differ in who is affected without any change in behavior. Under a higher NRA every claimant receives a lower benefit at any claiming age, so the reform cuts benefits even if no one changes the claiming age. Under a higher DRC a claimant at or before the NRA receives the status-quo benefit and a claimant after it a higher one; under a symmetric adjustment a claimant at the NRA receives the status-quo benefit, an earlier claimant less, and a later claimant more. These reforms change the price of moving the claiming age away

Table 6: Benefit as a Percentage of the PIA by Claiming Age under a Higher Normal Retirement Age (AN), a Higher Delayed Retirement Credit (AD), and a Symmetric Adjustment (AL)

Reform		Benefit as a Percentage of the PIA, by Claiming Age								
		62	63	64	65	66	67	68	69	70
(AN0)	NRA at 65	80	86.67	93.33	100	104.5	109	113.5	118	122.5
(AN1)	NRA at 66	75	80	86.67	93.33	100	104.5	109	113.5	118
(AN2)	NRA at 67	70	75	80	86.67	93.33	100	104.5	109	113.5
(AN3)	NRA at 68	65	70	75	80	86.67	93.33	100	104.5	109
(AD1)	DRC at 8%	80	86.67	93.33	100	108	116	124	132	140
(AD2)	DRC at 8.5%	80	86.67	93.33	100	108.5	117	125.5	134	142.5
(AD3)	DRC at 9%	80	86.67	93.33	100	109	118	127	136	145
(AD4)	DRC at 9.5%	80	86.67	93.33	100	109.5	119	128.5	138	147.5
(AD5)	DRC at 10%	80	86.67	93.33	100	110	120	130	140	150
(AL1)	Linear ±7%	79	86	93	100	107	114	121	128	135
(AL2)	Linear ±8%	76	84	92	100	108	116	124	132	140
(AL3)	Linear ±9%	73	82	91	100	109	118	127	136	145
(AL4)	Linear ±10%	70	80	90	100	110	120	130	140	150
(AL5)	Linear ±11%	67	78	89	100	111	122	133	144	155
(AL6)	Linear ±12%	64	76	88	100	112	124	136	148	160
(AL7)	Linear ±13%	61	74	87	100	113	126	139	152	165
(AL8)	Linear ±14%	58	72	86	100	114	128	142	156	170
(AL9)	Linear ±15%	55	70	85	100	115	130	145	160	175

Notes: Social Security retirement benefit payable upon claiming at ages 62–70, as a percentage of the primary insurance amount (PIA), under the status quo of the 1920–1940 cohorts (AN0) and under each reform. AN1–AN3 raise the NRA to 66, 67, and 68. AD1–AD5 raise the DRC from the cohort's 4.5 percent to 8, 8.5, 9, 9.5, and 10 percent per year, leaving the reduction before the NRA unchanged. AL1–AL9 set equal early-claiming penalties and delayed credits of ±7 to ±15 percent per year. The blue gradient marks values above 100, the red gradient values below 100, and white marks 100 percent of the PIA. Sources: rules of the Social Security Administration (AN0) and the reform definitions of Table 2.

Figure 3: Benefit Schedules under the Status Quo, a Higher NRA (AN2), a Higher DRC (AD5), and a Symmetric ±10 Percent Adjustment (AL4)



Notes: Social Security retirement benefit payable at each claiming age, as a percentage of the primary insurance amount (PIA), from Table 6. The status quo is the schedule of the 1920–1940 cohorts, with a normal retirement age (NRA) of 65, a reduction of 6.67 percentage points per year of claiming before the NRA, and a delayed retirement credit (DRC) of 4.5 percentage points per year after it. AN2 raises the NRA to 67; AD5 raises the DRC to 10 percent per year and leaves the schedule through age 65 unchanged; AL4 sets the reduction and the credit equal to 10 percent per year, so the schedule coincides with AD5 after the NRA and lies below the status quo before it. AD5 and AL4 coincide at ages 65–70. Sources: rules of the Social Security Administration (status quo) and the reform definitions of Table 2.

Table 7: Claiming-Schedule Reforms: Marginal Value of Public Funds, Willingness to Pay, Fiscal Effects, and Participation

Policy Reforms		%Δ Welfare		%Δ Budget			Δ Participation		
		MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
(AN1)	NRA at 66	0.17	-0.38	-0.87	-6.13	-0.02	-0.04	0.38	0.60
(AN2)	NRA at 67	0.16	-0.74	-1.75	-11.82	-0.12	-0.07	-0.04	1.98
(AN3)	NRA at 68	0.18	-1.09	-2.38	-16.75	-0.08	-0.14	-0.43	3.27
(AD1)	DRC at 8%	∞	0.16	-0.45	-2.46	-0.10	0.11	0.15	0.74
(AD2)	DRC at 8.5%	∞	0.19	-0.44	-2.87	-0.04	0.10	0.18	0.57
(AD3)	DRC at 9%	∞	0.23	-0.66	-4.22	-0.07	0.09	-0.36	1.20
(AD4)	DRC at 9.5%	∞	0.26	-0.73	-4.56	-0.09	0.12	-0.49	1.37
(AD5)	DRC at 10%	∞	0.3	-0.78	-5.05	-0.08	0.14	-0.97	1.39
(AL1)	Linear ±7%	∞	0.10	-0.12	-0.74	-0.01	0.05	0.24	0.22
(AL2)	Linear ±8%	∞	0.15	-0.46	-2.81	-0.07	0.11	0.05	0.61
(AL3)	Linear ±9%	∞	0.21	-0.72	-4.74	-0.07	0.11	-0.31	1.21
(AL4)	Linear ±10%	∞	0.29	-0.86	-5.54	-0.09	0.14	-0.93	1.42
(AL5)	Linear ±11%	∞	0.37	-1.17	-7.02	-0.19	0.19	-1.15	2.01
(AL6)	Linear ±12%	∞	0.47	-1.37	-7.72	-0.28	0.30	-1.46	2.61
(AL7)	Linear ±13%	∞	0.57	-1.38	-8.21	-0.23	0.32	-2.90	2.50
(AL8)	Linear ±14%	∞	0.69	-1.47	-8.22	-0.31	0.34	-3.30	2.42
(AL9)	Linear ±15%	∞	0.80	-1.57	-8.45	-0.37	0.36	-3.49	2.26

Notes: Columns as defined in Table 4. MVPF, marginal value of public funds (25); WTP, consumption-equivalent welfare change in percent of consumption (20); Net, Direct, and Indir., percentage changes in the net present value at age 62 of the cohort's net government spending, of retirement benefits paid by Social Security, and of all other taxes and transfers (23); 30–49, 62–64, and 65–69, changes in labor force participation at those ages in percentage points. ∞ denotes a reform with positive willingness to pay and a negative net fiscal effect; AN rows report the MVPF of reversing the reform. AN1–AN3 raise the NRA to 66, 67, and 68; AD1–AD5 raise the DRC from 4.5 percent to 8, 8.5, 9, 9.5, and 10 percent per year; AL1–AL9 set equal early-claiming penalties and delayed credits of ±7 to ±15 percent per year (Table 6). Blue shading marks entries in the direction of higher welfare, higher participation, or lower spending, and red shading the opposite. Source: authors' simulations of the estimated model.

from the NRA, and their fiscal effect depends on who moves.[10] Three pairs share the schedule after the NRA and differ only before it: AL2 and AD1 (108 to 140 at ages 66–70), AL3 and AD3 (109 to 145), and AL4 and AD5 (110 to 150); AD2 and AD4 have no AL counterpart. Comparing the members of each pair isolates the effect of the steeper early-claiming reduction.

7.2 Welfare and fiscal returns

An entry of ∞ in the MVPF column of Table 7 means that the reform has a positive willingness to pay and a negative net fiscal effect, so that no finite ratio of the two exists; the AN rows report the MVPF of reversing the reform.

The central fact of the table is that a benefit increase for late claimants reduces program outlays. Raising the DRC lowers the net present value of retirement benefits paid to the cohort by 2.46 percent under AD1 and 5.05 percent under AD5; the symmetric adjustments lower it by 0.74 percent under AL1 and 8.45 percent under AL9. The government's other net outlays fall as well: the indirect effect lies between −0.10 and −0.04 in the AD rows and between −0.01

[10]The gain from delaying claiming under current US rules, and whether households can finance a delay from their own assets, are examined by Shoven and Slavov (2014) and Goda et al. (2018).

and −0.37 in the AL rows. Net spending therefore falls in every AD and AL row, by 0.44 to 0.78 percent for AD1–AD5 and by 0.12 to 1.57 percent for AL1–AL9, while the willingness to pay is positive in every row, 0.16 to 0.3 percent of consumption for AD1–AD5 and 0.10 to 0.80 percent for AL1–AL9. These reforms are welfare-improving and self-financing: they raise ex ante welfare and generate a fiscal surplus.[11] The MVPF is infinite whether the denominator is the net or the direct fiscal effect, because both are negative; by (27), no finite lower bound exists when $\Delta B < 0$ and $\Delta G < 0$. The AD rows admit the sign inference of Section 5.4: the mechanical change in outlays is weakly positive, since the factors after the NRA rise and the factors before it are unchanged, while the Direct column is negative, so the within-program behavioral response more than offsets the mechanical outlay and the fiscal multiplier (30) is negative.

Raising the NRA has the opposite welfare sign. The willingness to pay is −0.38, −0.74, and −1.09 percent of consumption under AN1, AN2, and AN3; retirement benefits fall by 6.13, 11.82, and 16.75 percent; the indirect effect is −0.02, −0.12, and −0.08; and net spending falls by 0.87, 1.75, and 2.38 percent. Here the mechanical effect is a cut: every factor falls, so the reform saves money before any behavioral response. The MVPF column reports 0.17, 0.16, and 0.18, the welfare loss per dollar of net saving; read as the MVPF of the reverse reform, lowering the NRA from 66, 67, or 68 back to 65 would deliver 0.17, 0.16, or 0.18 dollars of welfare per dollar of net spending. In Hendren and Sprung-Keyser's (2020) words, the MVPF "measures the shadow price of raising revenue from the beneficiaries of the policy by reducing spending on the policy."

Two comparisons across families hold net spending approximately fixed. AN1 and AL4 reduce net spending by 0.87 and 0.86 percent; the willingness to pay is −0.38 percent under AN1 and +0.29 percent under AL4. AN2 and AL9 reduce it by 1.75 and 1.57 percent; the willingness to pay is −0.74 and +0.80. At approximately equal fiscal saving, the symmetric adjustment raises welfare where the higher NRA lowers it. Within the second way of changing the schedule, the three pairs with a common post-NRA schedule isolate the early-claiming side. AD1 and AL2 reduce benefits by 2.46 and 2.81 percent with willingness to pay of 0.16 and 0.15; AD3 and AL3 by 4.22 and 4.74 percent with 0.23 and 0.21; AD5 and AL4 by 5.05 and 5.54 percent with 0.3 and 0.29. Adding the steeper early-claiming reduction to a higher DRC saves more in benefits and lowers the willingness to pay slightly; the participation changes at ages 62–64 of the two members of each pair are close (0.15 and 0.05; −0.36 and −0.31; −0.97 and −0.93).

[11] The welfare statistic is the average consumption-equivalent change at age 30, (20). Effects on individuals who claim early are heterogeneous, and the AL reforms raise the penalty on early claiming; the exhibits do not report the distribution of welfare changes across realized histories.

7.3 Participation by age group

The last three columns of Table 7 report the behavioral responses behind the fiscal effects. Under a higher NRA, participation at ages 30–49 falls by 0.04, 0.07, and 0.14 percentage points and participation at ages 65–69 rises by 0.60, 1.98, and 3.27 points. At ages 62–64 the response is not monotone in the size of the reform: participation rises by 0.38 points under AN1 and falls by 0.04 and 0.43 points under AN2 and AN3. One reading of these signs is that the lower factor at every claiming age is a loss of lifetime resources to which older workers respond by working more, while the lower benefit that a year of earnings buys lowers the return to work at younger ages.

Under a higher DRC and under a symmetric adjustment, participation rises at ages 30–49 in every row, by 0.05 to 0.36 points, and at ages 65–69 in every row, by 0.22 to 2.61 points. At ages 62–64 the sign depends on the steepness of the schedule. Participation rises under AD1, AD2, AL1, and AL2, by 0.15, 0.18, 0.24, and 0.05 points, that is, for a DRC up to 8.5 percent and a symmetric adjustment up to ±8 percent. It falls from AD3 and AL3 on: by 0.36, 0.49, and 0.97 points under AD3–AD5, and by 0.31 points under AL3 to 3.49 points under AL9. AD1, which raises the DRC from 4.5 to 8 percent, raises participation in all three age groups, lowers benefit outlays, and raises welfare. Figure C1 in Appendix C plots the participation changes at ages 62–64 and 65–69 against the direct fiscal effect for every row of Table 7: in every family, beyond the smallest steps, the rows with the larger benefit saving are the rows with the larger participation loss at 62–64; the participation gain after 65 rises with the saving in the AN rows and, except at AD2 and beyond AL6 (Section 7.4), in the AD and AL rows.

In the model, the workers for whom continued work is most costly are those in poor health or with a disability—higher participation cost, mortality and medical spending, lower wages (Figure 1)—and they are the ones who accept the steeper early penalty, while the steeper reward pulls the others past 65. The tables report participation by age group only; no exhibit splits the response to a priced reform by health, so this reading is not tested here. Studies of increases in the statutory retirement age in the United States and Switzerland document claiming responses concentrated at the statutory age itself (Mastrobuoni 2009; Behaghel and Blau 2012; Deshpande et al. 2024; Lalive et al. 2023), and Manoli and Weber (2016) estimate responses of retirement timing to financial incentives. The model has no reference-point effect at the NRA, so the AN predictions above are the incentive component of the response; Section 11 discusses the direction in which reference effects would move the comparison between the two kinds of reform.

7.4 Steeper adjustments shift incidence toward early claimants

Along the AL grid, every step raises welfare and lowers net spending: the willingness to pay rises monotonically from 0.10 under AL1 to 0.80 under AL9, and net spending falls

monotonically from −0.12 to −1.57 percent. What changes along the grid is who bears the adjustment. The participation gain at ages 65–69 peaks at AL6, 2.61 points, and is 2.50, 2.42, and 2.26 points at AL7–AL9, while the participation loss at ages 62–64 keeps growing, to 2.90, 3.30, and 3.49 points. The indirect effect is not monotone over the steep end of the grid, −0.19, −0.28, −0.23, −0.31, and −0.37 for AL5–AL9, while the direct effect continues to fall, from −7.02 to −8.45. The steepest adjustment and the largest NRA increase lower the age-62 factor to similar levels, 55 percent of the PIA under AL9 and 65 under AN3, and both lower participation at ages 62–64, by 3.49 and 0.43 points. The welfare signs differ: the willingness to pay is +0.80 under AL9, the largest positive value in the table, and −1.09 under AN3. The difference is that AL9 leaves the factor at the NRA at 100 and raises it to 175 at age 70, so the steeper early penalty is the price of a steeper reward for delay, whereas AN3 lowers the factor at every age. Within the grid considered, no AL step lowers welfare or raises net spending; what the steeper steps do is shift the adjustment toward those who claim early.

7.5 The reforms against the financing benchmark

Every row of Table 7 has a negative net fiscal effect, so every reform in this section reduces the cohort's net claim on the government and, in the sense of Section 2.4, contributes to closing the financing gap. The steeper adjustments do so at a welfare gain and a higher NRA at a welfare cost; AL4 and AL9 match the net savings of AN1 and AN2 (Section 7.2), and the AL1 saving, 0.12 percent of net spending, is small. Because the Net column is in percent of the cohort's baseline net spending rather than of its taxable payroll (Section 2.4), the comparison with the Trustees' deficit is one of sign.

The section's result is a difference in sign: at approximately equal fiscal saving, a steeper adjustment of benefits to the claiming age raises ex ante welfare and a higher normal retirement age lowers it, and the steeper the adjustment, the more of its saving is borne by those who claim early.[12]

8 Benefit Levels versus Progressivity

The reforms of Section 7 change the claiming-age factor $\psi(a^c)$ in (9) and leave the benefit formula unchanged. This section does the reverse: it raises the bracket rates of the primary insurance amount (8), one bracket at a time or all three together, and leaves the claiming-age schedule at its status-quo values. The accounting is that of Section 5: the program outlay B is retirement benefits, the Direct column is on the base B^0, and the Net column is on baseline net spending G^0 (Table 4).

[12] Ndiaye and Yu (2026) find that requiring this cohort to finance a lump-sum transfer to earlier generations leaves retirement ages almost unchanged, the adjustment falling on consumption and hours. That experiment changes the cohort's resources without changing the price of delaying claiming, so its retirement response is a wealth effect; the reforms here change that price, and the two results are not in tension.

8.1 Reform grid

Four sequences of reforms are considered; Table 2 lists them. Reforms P1.1–P1.9 raise the first bracket rate from 0.90 by one percentage point per step, to 0.91 under P1.1 and 0.99 under P1.9, and leave the second and third rates at 0.32 and 0.15. Reforms P2.0–P2.9 start from P1.9 and raise the second bracket rate from 0.32 by 0.32 percentage points per step, one percent of its initial value, to 0.352 under P2.9. Reforms P3.0–P3.9 start from P2.9 and raise the third bracket rate from 0.15 by 0.15 percentage points per step, again one percent of its initial value, to 0.165 under P3.9. Reforms L0–L9 raise all three rates proportionally by one percent per step, from (0.909, 0.3232, 0.1515) under L0 to (0.99, 0.352, 0.165) under L9. The P sequence and the L sequence therefore begin at the status quo and end at the same policy—L9 and P3.9 are identical—but follow different paths: the P sequence takes the first bracket to its final value before touching the second, and the second before the third, whereas the L sequence moves the three rates together. The bend points, the cap on average indexed earnings, the claiming-age schedule, the earnings test, and the payroll tax are unchanged.

By (8), a higher first-bracket rate raises the benefit of everyone with positive average indexed earnings: by a fixed dollar amount for individuals whose earnings exceed the first bend point and by a smaller amount below it, so that as a share of the benefit the increase is largest at low lifetime earnings. A higher second- or third-bracket rate raises benefits only for individuals whose average earnings exceed the first or the second bend point. A proportional increase raises every benefit by the same percentage. The three sequences thus place the same kind of instrument—a higher replacement rate—at different points of the lifetime-earnings distribution.

8.2 Welfare, fiscal cost, and MVPF by bracket

Table 8 reports ten of the 39 rows of the grid: the first, a middle, and the last step of the first-bracket and level sequences, and a middle and the last step of the second- and third-bracket sequences; Table C1 reports all of them. Every reform in the grid raises welfare. The WTP rises from 0.10 percent of consumption under P1.1 to 0.80 under P1.9 and 0.98 under P3.9, and along the level sequence from 0.16 under L0 to 1.02 under L8 and 0.98 under L9. The direct cost rises with generosity: the net present value of retirement benefits increases by 0.57 percent of its baseline value under P1.1, 4.12 under P1.9, 9.97 under P2.9, and 10.70 under P3.9 and L9. The indirect effect is a saving in every row: all other taxes and transfers fall by between 0.05 and 0.23 percent of their baseline value. Net government spending rises in every row except P1.1, for which it falls by 0.03 percent of its baseline value; P1.1 is approximately self-financing, and its MVPF is reported as the interval $[0.52, +\infty)$ of (27), where 0.52 is the welfare gain per dollar of additional retirement benefits.

Table 8: Social Security Progressivity and Level Reforms

	Policy Reforms	%Δ Welfare		%Δ Budget			Δ Participation		
	PIA Brackets	MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
(P1.1)	(0.91, 0.32, 0.15)	$[0.52, +\infty)$	0.10	-0.03	0.57	-0.10	0.08	0.20	0.06
(P1.5)	(0.95, 0.32, 0.15)	0.91	0.50	0.21	2.32	-0.09	0.16	-0.34	-0.31
(P1.9)	(0.99, 0.32, 0.15)	0.69	0.80	0.44	4.12	-0.11	0.25	-0.85	-0.60
(P2.4)	(0.99, 0.336, 0.15)	0.38	0.92	0.93	7.21	-0.05	0.24	-1.60	-1.63
(P2.9)	(0.99, 0.352, 0.15)	0.29	0.95	1.27	9.97	-0.09	0.28	-2.26	-2.05
(P3.4)	(0.99, 0.352, 0.1575)	0.29	0.96	1.27	10.34	-0.13	0.28	-2.23	-2.03
(P3.9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	-0.20	0.28	-2.37	-1.99
(L0)	(0.909, 0.3232, 0.1515)	0.51	0.16	0.12	1.34	-0.05	0.11	-0.05	0.01
(L4)	(0.945, 0.336, 0.1575)	0.59	0.74	0.49	5.28	-0.21	0.21	-0.48	-0.43
(L9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	-0.20	0.28	-2.37	-1.99

Notes: Ten of the 39 reforms of the grid in Section 8.1; the full grid is Table C1. The second column gives the three bracket rates of the primary insurance amount (8) under the reform; the status quo is (0.9, 0.32, 0.15). L9 and P3.9 are the same policy. Columns are defined in Table 4 and (20)–(27): WTP is the consumption-equivalent welfare change in percent of consumption; Direct is the change in the net present value (NPV) of retirement benefits in percent of its baseline value; Indir. is the change in the NPV of all other taxes and transfers (disability benefits and transfers net of income and payroll taxes) in percent of its baseline value; Net is the change in net government spending in percent of its baseline value, related to the other two by (24) and not by addition; MVPF is the dollar welfare gain per dollar of net spending, and the entry $[0.52, +\infty)$ reports, for the one row in which net spending falls, the lower bound that counts only the increase in retirement benefits. The last three columns are changes in labor force participation at ages 30–49, 62–64, and 65–69 in percentage points. All NPVs are per capita at age 62 at the net return of 3.2 percent. Shading is blue for a welfare gain, a fiscal saving, a higher MVPF, or a participation increase and red for the opposite, with intensity proportional to the entry. Source: simulations of the estimated model.

For the remaining rows the MVPF is finite and falls as the increase moves up the earnings distribution: it lies between 0.69 and 1.14 in the first bracket (P1.2–P1.9), falls from 0.56 under P2.0 to 0.29 under P2.9 in the second, and is 0.29 or 0.30 in every third-bracket row; along the level sequence it lies between 0.30 and 0.61. The MVPF is not monotone within a sequence—P1.6 exceeds both P1.5 and P1.7, and L3 exceeds L2—but the ordering across sequences holds at every level of direct cost: a first-bracket row has a higher MVPF than the level row of comparable cost, and a second- or third-bracket row has no higher one. P1.9 and L3 raise outlays by 4.12 and 4.25 percent and have MVPFs of 0.69 and 0.61; P2.4 and L6 raise them by 7.21 and 7.41 percent and have MVPFs of 0.38 and 0.47; P2.9 and L8 raise them by 9.97 and 9.80 percent and have MVPFs of 0.29 and 0.35.

Within this menu, raising the first bracket rate is the most efficient way to increase the generosity of retirement benefits. Between P1.9 and P3.9 the WTP rises from 0.80 to 0.98 percent of consumption while the direct cost rises from 4.12 to 10.70 percent of outlays, so the spending added in the second and third brackets buys little additional welfare. A uniform increase delivers less welfare per dollar of net spending than the same spending concentrated in the first bracket and more than the same spending placed in the upper brackets. The comparison is between increases in generosity under the same accounting, not between progressive and proportional benefit formulas in general. The pattern follows from where the transfer accrues. By (8) a first-bracket increase is, as a share of the benefit, largest for individuals with

low lifetime earnings, who in the model also hold few assets in retirement and whose marginal utility of a transferred dollar is high in the sense of Section 5.4; second- and third-bracket increases accrue only to individuals above the bend points, whose benefits are already larger. Changes in the level and progressivity of the benefit formula have been studied in general-equilibrium overlapping-generations models by Nishiyama and Smetters (2007), Imrohoroğlu and Kitao (2012), and Kitao (2014), in the life-cycle model of Huggett and Parra (2010), and in a life-cycle model with health and claiming by Yu (2024); the rows here are single-cohort, fixed-price experiments evaluated with the accounting of Section 5, and the comparison with those studies is qualitative.

8.3 Participation responses

Participation at ages 30–49 rises in every row of the grid, from 0.08 percentage points under P1.1 to 0.28 under P2.9, under every third-bracket reform, and under L9. Participation at ages 62–64 rises under P1.1 and P1.3, by 0.20 and 0.15 pp, and falls in every other row, by as much as 2.41 pp under P3.5. Participation at ages 65–69 is unchanged or higher under P1.1, P1.2, P1.3, and L0 (0.06, 0.00, 0.01, and 0.01 pp) and lower in every other row, by as much as 2.1 pp under P2.8. P1.1 and P1.3 are the only reforms that raise participation in all three age groups; from P1.4 onward in the P sequence and from L1 onward in the L sequence, participation falls at both older age groups in every row.

The two margins respond to different components of the reform. At working ages a higher bracket rate raises, for individuals whose average indexed earnings fall in the affected bracket, the increment to future benefits from an additional year of earnings through (7) and (8), a substitution effect toward work. At the claiming ages a higher benefit at every claiming age is an income effect toward earlier exit, and it grows with the size of the increase. The indirect column is the fiscal counterpart of these responses together with the induced changes in income taxes, payroll taxes, transfers, and disability benefits; it is a saving in every row of the grid, and its components are not reported. The participation columns also qualify the MVPF ordering of Section 8.2: the reforms with the highest MVPFs are those under which old-age participation falls least, and the upper-bracket and larger level reforms combine the smallest welfare gain per dollar with the largest declines in participation at ages 62–69.

9 Medicare, Old-Age Income Taxes, and Retirement-Saving Subsidies

The three remaining families change programs other than Social Security retirement benefits: the coverage of medical spending after age 65, the income tax after age 62, and the net return on saving. Each is evaluated with the welfare measure of Section 5.1 and the fiscal

Table 9: Medicare, Old-Age Income-Tax, and Retirement-Saving Reforms

Panel A. Medicare Part A and Part B

Policy Reforms		%Δ Welfare		%Δ Budget			Δ Participation		
		MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
Medicare Part A	+ 20 pp	0.146	1.36	3.53	45.17	0.10	0.22	-1.70	-0.38
Medicare Part B	+ 20 pp	0.155	1.30	3.19	41.61	0.03	0.21	-1.95	-3.44

Panel B. Income-tax reductions after age 62

Policy Reforms		%Δ Welfare		%Δ Budget			Δ Participation		
		MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
(I1)	$y_{t\geq 62}^{\text{post-tax}}$ + 1%	0.25	0.44	0.69	0.70	-0.81	0.06	0.76	1.98
(I2)	$y_{t\geq 62}^{\text{post-tax}}$ + 2%	0.26	0.87	1.28	1.31	-1.51	0.15	1.58	4.12
(I3)	$y_{t\geq 62}^{\text{post-tax}}$ + 3%	0.25	1.29	2.01	2.03	-2.18	0.18	2.76	6.07
(I4)	$y_{t\geq 62}^{\text{post-tax}}$ + 4%	0.23	1.69	2.80	2.80	-2.78	0.2	3.19	7.71
(I5)	$y_{t\geq 62}^{\text{post-tax}}$ + 5%	0.23	2.09	3.46	3.47	-3.58	0.26	4.11	8.92

Panel C. Subsidies to saving

Policy Reforms		%Δ Welfare		%Δ Budget			Δ Participation		
		MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
(K1)	$\Delta r = +0.2$ pp	[0.96, 20.37]	0.87	0.08	1.77	-1.69	0.15	-1.27	-0.85
(K2)	$\Delta r = +0.4$ pp	[0.92, 19.38]	1.75	0.18	3.77	-3.59	0.36	-2.58	-1.75
(K3)	$\Delta r = +0.8$ pp	[0.84, 5.29]	3.6	1.34	8.44	-7.10	0.38	-5.98	-4.11

Notes: Panel A: the Part A reform raises the Medicare coverage share of total medical spending by 20 percentage points for everyone aged 65 and older; the Part B reform raises it by 20 percentage points for nonparticipants aged 65 and older. Panel B: after-tax income $y_t - T_t(y_t)$ after age 62 rises by 1 to 5 percent. Panel C: the net return on assets rises by 0.2, 0.4, and 0.8 percentage points at every age; K3 is the elimination experiment D3 of Table 5. Direct effect: panel A, change in the net present value (NPV) of Medicare payments to the targeted beneficiaries after 65 as a share of baseline payroll-tax collections; panel B, income-tax revenue forgone after 62 as a share of baseline income-tax collections; panel C, the subsidy as a share of baseline net government spending. Indir.: panel A, change in the NPV of all taxes and transfers other than the Medicare payments, in percent of its baseline value; panel B, the same for all taxes and transfers other than the income tax; panel C, change in government spending excluding the subsidy, in percent. Net: change in net government spending, including the program outlay, in percent of its baseline value; it is related to the first two columns by (24) and equals their sum only in panel C. MVPF in panels A and B is the dollar welfare gain per dollar of net spending (25); in panel C it is reported as [lower, upper] per (28): willingness to pay over the subsidy, and over the net cost under full realization and taxation of capital gains. WTP is the consumption-equivalent welfare change in percent of consumption; the dollar conversion uses baseline consumption from age 62 in panels A and B and from age 30 in panel C. The last three columns are changes in labor force participation at ages 30–49, 62–64, and 65–69 in percentage points. All NPVs are per capita at age 62 at the net return of 3.2 percent. Shading is blue for a welfare gain, a fiscal saving, a higher MVPF, or a participation increase and red for the opposite. Source: simulations of the estimated model.

accounting of Section 5.2. The program outlay B and the base of the Direct column differ by family (Table 4): added Medicare payments to the targeted beneficiaries as a share of baseline payroll-tax collections, income-tax revenue forgone as a share of baseline income-tax collections, and the saving subsidy as a share of baseline government spending. Table 9 reports the three families in one table, and the Indirect entries, which are on family-specific bases, are not compared across its panels.

9.1 Medicare Part A and Part B

Out-of-pocket medical spending $m_t = M(h_t, t, P_t)$ in (6) is total spending net of coverage, and from age 65 the coverage is that of Medicare. Two expansions are considered. Most beneficiaries receive Part A, which covers hospital and inpatient services, without a premium on the basis of their payroll-tax contributions, so the Part A reform raises the Medicare coverage share of total medical spending for everyone aged 65 and older by 20 percentage points. Part B, which covers physician and outpatient services, requires a premium, and most employer-provided plans already cover those services for individuals who work; the model therefore assigns Part B to nonparticipants aged 65 and older only, and the Part B reform raises their coverage share by 20 percentage points. The experiments start from a re-simulated baseline: the average Medicare coverage rates of individuals aged 65 and older are computed from the MEPS by health and participation status, the estimated model is simulated with these rates, and the two reforms are applied to that simulation; Appendix C gives the rule in symbols.

Both expansions raise welfare: the WTP is 1.36 percent of consumption for Part A and 1.30 for Part B. The direct effect is large on the base used for this family: the added Medicare payments to the targeted beneficiaries after 65 amount to 45.17 percent of baseline payroll-tax collections under Part A and 41.61 percent under Part B. The indirect effect is small and positive, 0.10 and 0.03 percent of baseline other taxes and transfers, so the rest of the budget does not offset the added payments; net government spending rises by 3.53 and 3.19 percent of its baseline value. The MVPFs are 0.146 and 0.155. Participation rises slightly at ages 30–49, by 0.22 and 0.21 percentage points, and falls at ages 62–64, by 1.70 and 1.95 pp, and at ages 65–69, by 0.38 pp under Part A and 3.44 pp under Part B. The Part B response at 65–69 is larger by construction: the reform conditions coverage on not working, so for a participant aged 65 or older, leaving the labor force now also brings the higher coverage, which lowers the return to participation at those ages. The Part A reform covers participants and nonparticipants alike, and in the model its participation effects at the older ages are income effects of lower expected out-of-pocket spending.

The size of the direct effect relative to the welfare gain reflects what a proportional coverage increase pays for. The out-of-pocket profiles of Figure 1 rise with age and are highest for individuals with a disability, so an increase in coverage by a fixed share of spending applies to larger amounts at the oldest ages. The age profile of the added payments is not reported. Medical-expense risk in old age is a motive for saving and for continued work in models of this class (De Nardi et al. 2010, 2025), and French and Jones (2011) conclude that Medicare matters for retirement timing on a par with Social Security; the participation responses here, negative at 62–64 and 65–69 under both expansions, are consistent with that conclusion.

Finkelstein and McKnight (2008) and Hendren and Sprung-Keyser (2020) place the MVPF of the introduction of Medicare in the range $[0.52, 3.83]$. An expansion from the status quo is a different object: the marginal dollar of coverage insures spending that is already partly insured, and with diminishing returns to insurance its value per dollar is lower than that of the first dollar. The MVPFs reported here are for expansions of 20 percentage points from the status quo and are not comparable in level to those of the introduction.[13]

9.2 Income-tax reductions after age 62

Reforms I1–I5 lower the income tax (13) from age 62 so that after-tax income $y_t - T_t(y_t)$ rises by 1 to 5 percent, in steps of one percent, with no change before age 62. The change applies to all taxable income after 62—asset income, earnings, and pension income—and therefore to nonparticipants as well as participants; Appendix C gives the implementation in terms of the level parameter λ_t.

The WTP rises from 0.44 percent of consumption under I1 to 2.09 under I5. Participation rises at every age and in every row: by 0.06 to 0.26 percentage points at ages 30–49, 0.76 to 4.11 pp at 62–64, and 1.98 to 8.92 pp at 65–69. The revenue forgone is 0.70 to 3.47 percent of baseline income-tax collections. The indirect effect is a saving of 0.81 to 3.58 percent of all other taxes and transfers. The two columns are on different bases, so their entries are not comparable in dollars. In dollars the saving offsets only part of the revenue loss: net government spending rises by 0.69 to 3.46 percent of its baseline value, and the MVPF is 0.25, 0.26, 0.25, 0.23, and 0.23 for I1 through I5.

The income-tax reductions and the steeper claiming-age adjustments of Section 7 both raise participation after the normal retirement age; their fiscal outcomes differ in sign. The adjustments reward delayed claiming within the benefit formula and reduce outlays, whereas the tax reductions lower the tax on all income after 62, including the asset and pension income of individuals whose participation does not change, and the participation response does not recover the revenue. Estimates of the labor-supply response of older workers to income taxes and to the earnings test are in Alpert and Powell (2020) and Gelber et al. (2022); here the response is the model's, and it is large at 62–69 in every row of panel B of Table 9.[14]

[13] Finkelstein et al. (2019) find a willingness to pay for health insurance of about 30 percent of the cost it imposes on the insurer among low-income adults, with about 70 percent of their care uncompensated, and Hendren and Sprung-Keyser (2020), attributing uncompensated care to the government, obtains an upper bound of 0.8 for the MVPF of health-insurance subsidies to low-income adults. These figures concern Medicaid and are not comparable to the Medicare expansions here.

[14] The income-tax reductions apply after age 62 and the saving subsidies of Section 9.3 at every age, and the dollar conversion of the welfare gain uses baseline consumption from age 62 for the former and from age 30 for the latter (Section 5.1). The two families are therefore not ranked against each other by the level of their MVPFs.

9.3 Subsidies to retirement saving

Reforms K1–K3 raise the net return on assets r in (15) by 0.2, 0.4, and 0.8 percentage points at every age. With $r = (1 - \tau_k)r_f$, r_f equal to 4 percent, and τ_k equal to 20 percent, K3 raises the net return to the gross return and is the elimination experiment D3 of Section 6 under another name. The reforms are read as changes in the tax treatment of retirement saving that already exists—the deferral of tax in traditional individual retirement accounts and 401(k) plans and the exemption of returns in their Roth counterparts—which raise the after-tax return on part of household saving; the model has a single asset, so the subsidy applies to all saving, and a higher r is equivalently a smaller proportional reduction τ_k of the gross return. The tax treatment of saving in life-cycle models with retirement is studied by Conesa et al. (2009) and Paluszynski and Yu (2023); the question here is confined to the welfare and fiscal returns of marginal changes in the net return.

The WTP is 0.87, 1.75, and 3.6 percent of consumption. The direct effect is the subsidy itself, 1.77, 3.77, and 8.44 percent of baseline government spending. The indirect effect is a reduction of 1.69, 3.59, and 7.10 percent in government spending excluding the subsidy. In the model this reduction comes from the income taxes paid on the asset income of younger households that save more; asset income ra_t is taxed each period through y_t in (13), which amounts to assuming that all capital gains are realized and taxed as they accrue. Net government spending rises by 0.08, 0.18, and 1.34 percent of its baseline value; in this family Net equals Direct plus Indirect because both are on the base G^0 (24). The MVPF is reported as an interval (28): the lower bound, 0.96, 0.92, and 0.84, divides the dollar welfare gain by the subsidy alone, and the upper bound, 20.37, 19.38, and 5.29, divides it by the net cost under full realization and taxation of capital gains. The lower bound is the value used in the text: it does not depend on the realization assumption and treats the indirect saving as absent. All three upper bounds exceed one; they are the values obtained if the induced saving is taxed in full as its returns accrue, and they fall with the size of the subsidy because the net cost rises faster than the welfare gain.

Participation rises at ages 30–49, by 0.15, 0.36, and 0.38 percentage points, and falls at ages 62–64, by 1.27, 2.58, and 5.98 pp, and at ages 65–69, by 0.85, 1.75, and 4.11 pp. Saving subsidies and the claiming-schedule reforms of Section 7 thus move old-age participation in opposite directions: a higher return makes retirement easier to finance, whereas a steeper adjustment makes it more costly to begin early. Two features of the model bear on these rows. First, the consumption-equivalent gain is converted to dollars with baseline consumption from age 30 rather than from age 62, because the reform changes the budget constraint at every age (Section 5.1); the MVPF levels in panel C of Table 9 are therefore not compared with those of the other families. Second, mean assets are matched through the 40s and under-predicted

from the early 50s, and median assets are over-predicted through the early 50s and under-predicted from the mid-50s (Section 4), and the model has no bequest motive (De Nardi et al. 2010; Lockwood 2018); the response to a return subsidy depends on the asset base, and the direction in which these misfits bias the WTP and the fiscal columns is not determined by the reported output. Across the three families the signs are the same: the Medicare expansions, the income-tax reductions, and the saving subsidies all raise welfare and all raise net spending; they differ in what they do to work after age 62, which the tax reductions raise and the other two lower.

10 The Menu as a Whole

Sections 7–9 evaluate the seven reform families one at a time; this section places every row side by side. Three columns are comparable across families: the MVPF, in dollars of welfare per dollar of net spending; the Net column, in percent of baseline government spending on the cohort; and the WTP column, in percent of consumption. The Indirect column is on a base that differs by family (Table 4) and is not compared across families, and the MVPF intervals of the saving subsidies, whose dollar conversion uses consumption from age 30 rather than from age 62, are kept out of the cross-family ordering (Section 5.1).

10.1 Ranking within the menu

The sign of the Net column splits the menu in two. Net spending falls under AN1–AN3, AD1–AD5, AL1–AL9, and P1.1, and rises under every other row: P1.2–P3.9, L0–L9, both Medicare expansions, I1–I5, and K1–K3. With the sign of the WTP column this gives three groups, the three populated quadrants of Figure 4. AD1–AD5 and AL1–AL9 raise welfare and lower net spending: they are self-financing, their MVPF is infinite, and no finite lower bound exists because the program's own outlays fall as well (Section 7.2). P1.1 belongs to this group approximately: its net saving is 0.03 percent of baseline government spending, and its MVPF is the interval $[0.52, +\infty)$ of (27), whose lower end counts only the added retirement benefits. AN1–AN3 lower both welfare and net spending; their MVPF entries, 0.17, 0.16, and 0.18, are the values of reversing them. Every remaining row raises both and has a finite MVPF.

Within this last group the MVPF orders the families as follows. First-bracket increases P1.2–P1.9 return 0.69 to 1.14 dollars of welfare per dollar of net spending; uniform increases L0–L9 return 0.30 to 0.61; second- and third-bracket increases P2.0–P3.9 return 0.29 to 0.56; income-tax reductions I1–I5 return 0.23 to 0.26. The Medicare expansions return 0.146 and 0.155, which with the reversal values of the NRA increases are the lowest finite values in the menu. The saving subsidies have lower bounds of 0.96, 0.92, and 0.84, which divide the dollar welfare gain by the subsidy alone, and upper bounds of 20.37, 19.38, and 5.29, which divide it by the net cost under full taxation of the induced asset income. The lower bounds are the

values used in the text, and because of the consumption window they are not ranked against the other families. In Figure 4, by (26), the MVPF within a family is proportional to the slope of the ray from the origin to a marker; the benefit-formula markers bend to the right from P2.5 on, which is the falling MVPF of Section 8.2. The largest welfare gains in the menu, 3.6 percent of consumption under K3 and 2.09 under I5, come with increases in net spending of 1.34 and 3.46 percent, the latter close to the largest in the menu, 3.53 percent under Medicare Part A, whereas the gains of the claiming-schedule reforms, up to 0.80 percent under AL9, come with net savings.

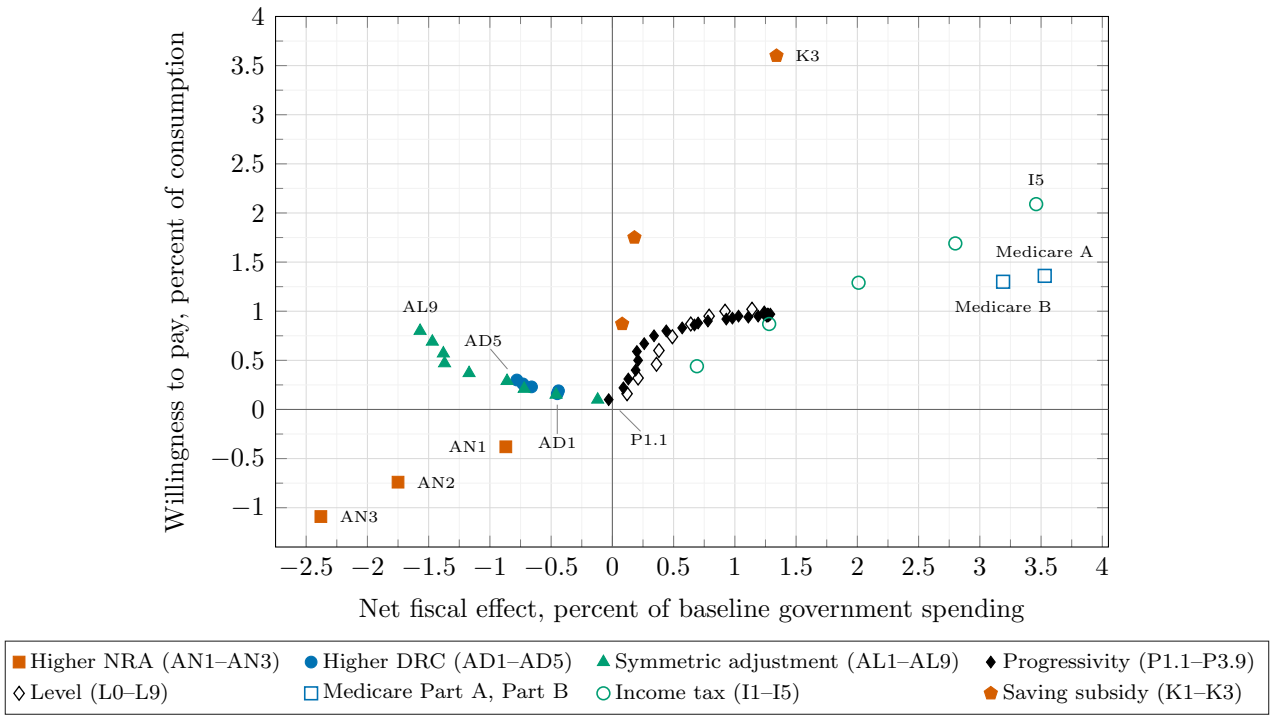
The ranking is a comparison within the menu under one accounting. It does not select a best reform, it says nothing about reforms outside the menu, and the MVPF enters it as a shadow price: a reform with positive net spending is worth undertaking when its MVPF exceeds the value placed on a dollar of public funds outside the cohort, and a self-financing reform at any such value (Hendren and Sprung-Keyser 2020). Appendix D.4 states this formally. In the sense of Section 2.4, the rows with negative Net contribute to closing the financing gap.

10.2 What the participation columns say about the fiscal externality

By (29) and (30), the fiscal effects beyond the mechanical outlay—the within-program behavioral change ΔB^B and the change ΔO in other taxes and transfers—are generated by behavioral responses: the fiscal externality of Chetty (2009) and the fiscal multiplier of Haller (2022). The participation columns report these responses on the extensive margin for three age groups; hours and saving responses are not reported. The signs sort the families. At ages 30–49 participation rises under every reform except the NRA increases, under which it falls. At ages 65–69 it rises under every claiming-schedule reform and every income-tax reduction, and falls under the Medicare expansions, the saving subsidies, and every benefit-formula row beyond P1.1–P1.3 and L0. At ages 62–64 the sign depends on the size of the step within the claiming-schedule and benefit-formula families: it is positive under AN1, AD1, AD2, AL1, AL2, P1.1, and P1.3 and negative in every other row of those families (Section 7.3, Section 8.3); it is positive in every income-tax row and negative under Medicare and the saving subsidies. The ranges are those of Table 7, Table C1, and Table 9.

The Indirect column is the fiscal counterpart of these responses within each family, and the correspondence is not one-to-one. Where participation after the NRA rises, under AN, AD, AL, and I, other taxes and transfers fall: by 0.01 to 0.37 percent of their base in the claiming-schedule rows and by 0.81 to 3.58 percent in the tax-reduction rows. Where it falls under Medicare, they rise, by 0.10 and 0.03 percent. Where it falls under the benefit-formula reforms they nevertheless fall, by 0.05 to 0.23 percent (Section 8.3), and under the saving

Figure 4: Willingness to Pay and Net Fiscal Effect across Reform Families



Notes: Each marker is one reform row of Table 7, Table 9, or Table C1. The vertical axis is the WTP column, the consumption-equivalent welfare change in percent of consumption (20); the horizontal axis is the Net column, the change in government spending on the cohort in percent of its baseline value, the Net base defined in Table 4. Rows with positive willingness to pay and a negative net fiscal effect (upper-left quadrant) are the rows with an infinite MVPF in the tables; the AN rows (lower-left quadrant) have negative willingness to pay and a negative net fiscal effect. Within a family the MVPF is proportional to the ratio WTP/Net by (26), with a constant that differs across families, so rays from the origin order the MVPF within a family only. L9 and P3.9 are the same policy and plot at the same point; AD1 and AL2 nearly coincide. Source: authors' simulations of the estimated model.

subsidies they fall by 1.69 to 7.10 percent because the reform taxes the asset income it induces (Section 9.3). The participation columns locate the externality by age but do not measure it, and the Indirect column measures it on a family-specific base; the fiscal multiplier itself can be signed from the printed cells for AD1–AD5 (negative) and AN1–AN3 (positive) only (Section 5.4). Read together, the tables separate the menu by the sign of the net fiscal effect, and within the rows that raise net spending the MVPF is highest where the added benefits accrue through the first bracket of the benefit formula, to low lifetime earners, and lowest for the Medicare expansions and the income-tax reductions.

11 Scope and Sensitivity

The reform tables of Section 6 to Section 10 are computed in the model of Section 3 at the estimates of Table 3. This section states the restrictions under which they hold and, for each, the direction in which relaxing it would move the results, in the order of Section 3.7.

11.1 Statutory ages as reference points

In the model the statutory ages act only through the budget constraint and the program rules. Age 62 opens eligibility for retirement benefits and makes pension income liquid. The normal retirement age is the kink of the claiming-age schedule in (9), the age at which the earnings-test rate and threshold change in (10), and the age at which eligibility for disability insurance ends in (11). Age 65 brings Medicare coverage into (6), and age 70 ends the earnings test and makes claiming mandatory. There are no default, norm, or reference-point effects: an individual who claims at 65 does so because the schedule, the earnings test, and the health and wage profiles make it worthwhile, not because 65 is the normal retirement age.

The evidence indicates that such effects exist. Seibold (2021) documents, from discontinuities in German pension rules, bunching of retirement at statutory ages that financial incentives alone do not account for. Mastrobuoni (2009), Behaghel and Blau (2012), and Deshpande et al. (2024) study the US cohorts whose normal retirement age was raised by the 1983 amendments and find that the spike in claiming and retirement at the normal retirement age moved with it; Lalive et al. (2023) find a similar response to a statutory-age increase in Switzerland. Liebman and Luttmer (2015) find that information about the program's rules raised the labor force participation of older workers but did not shift their claiming, and Brown et al. (2016) and Shoven et al. (2017) find that the framing of claiming information and the reasons individuals give for their claiming ages go beyond the actuarial gain from delay.

The paper takes one stance on what this implies. The predictions of Section 7 for a higher normal retirement age are the incentive component of the response. A reference effect at the new normal retirement age would move claiming later for individuals who, in the model, keep claiming at 62–64 after the reform. It would raise the government's saving, since delayed claims are paid for fewer years at a factor closer to 100; it would lower the benefit cut borne by those who move; and it would add a cost for those who delay for reasons the model does not price. For a steeper actuarial adjustment the effect works the other way. The normal retirement age stays at 65, so individuals anchored to it respond less to the steeper reward and the steeper penalty than the model predicts, and the participation gains at ages 65–69 and the benefit savings in the AD and AL rows of Table 7 would be overstated. Reference effects therefore move the contrast of Section 7 toward the normal-retirement-age reform: a higher statutory age moves a focal point, whereas a steeper adjustment changes the schedule without moving it. The welfare sign of the contrast does not change, because the benefit cut under a higher normal retirement age, every factor of Table 6 lower, does not depend on behavior. The size of either correction depends on the distribution of claiming ages in the model and in the data; claiming ages are not among the estimation targets and are not reported, so the discussion is qualitative.

11.2 Factor prices: partial equilibrium and a reduced-form adjustment

Every column of the reform tables is computed at the status-quo prices: the gross return r_f and the age–health wage profiles $W(h,t)$ of Section 3 do not respond to the cohort's choices. Section 3.7 states the assumption as that of a small open economy, in which fixed r and fixed wages are one assumption rather than two (Appendix D.3). The reforms nevertheless change what the cohort supplies to factor markets: the participation columns of Tables 7–9 report changes in labor supply by age group, and the saving subsidies change asset holdings (Section 9.3). In a closed economy, or if the reform applied to every cohort of a stationary economy, these changes would move the capital–labor ratio and with it the wage and the return.

Appendix D.3 derives a reduced-form adjustment that requires no new solution of the model. With a constant-returns technology and competitive factor markets, the proportional change in the wage is the capital share divided by the elasticity of substitution, times the change in the capital–labor ratio, and the proportional change in the return gross of depreciation has the opposite sign and the labor share in place of the capital share, as in (48). At fixed behavior, the price changes alter each individual's earnings, net asset income, and, through the earnings record, retirement benefits. Valued at the relative marginal utilities of Section 5.4, they add a term to the dollar willingness to pay, (49), and, through the income tax, the payroll tax, and the benefit base, a term to the change in net spending, (50); the adjusted MVPF is the ratio of the adjusted numerator to the adjusted denominator, (51). The two terms partition the cohort's change in factor income between its members and the government.

The direction of the adjustment follows from the participation columns and is collected in Table D1. Reforms that raise labor supply—the income-tax reductions at every age, and the claiming-schedule reforms after the normal retirement age—lower the wage and raise the return in a closed economy: workers lose part of the gain the tables record, asset holders gain, part of the indirect tax saving is returned, and the benefit base falls. Reforms that lower old-age labor supply—the larger benefit-formula increases and the Medicare expansions—raise the wage and lower the return, with the opposite incidence. The saving subsidies raise the capital stock and lower old-age participation, so they raise the wage and lower the return: the higher return they pay is partly undone, while labor-tax revenue and the benefit base rise. For the claiming-schedule reforms the sign of the aggregate labor-supply change depends on the weights, in efficiency units of labor (Appendix D.3), of age groups whose participation moves in opposite directions.

No adjusted MVPF is reported. The participation changes in the tables are at most a few percentage points within age groups that are a minority of the cohort's labor in efficiency units, and the adjustment scales them by a factor share and divides by the substitution elasticity; the paragraph states the direction of the adjustment and not its size. A number would require the capital share, the elasticity of substitution, and the depreciation rate; the reform-induced changes in the cohort's labor in efficiency units and in its assets, computable from the simulated panels behind the tables; the age profiles of the labor-tax, asset-tax, and benefit bases in (50); and the elasticity of benefits to the wage profile. The companion paper lists factor-price feedback among the questions it leaves open (Ndiaye and Yu 2026); the fixed-price environment used here is that of Farhi and Werning (2013) and Golosov et al. (2016).

11.3 Couples, bequests, health information, time separability, income risk, and reentry

Couples. The decision-maker is an individual, and the estimation targets are profiles for household heads and spouses in the general population (Table A2). Spousal and survivor benefits, which are computed from the worker's own benefit, and the coordination of claiming and labor supply within couples are outside the model; Borella et al. (2023) estimate a model of couples in which marriage-related Social Security rules shape the labor supply of both spouses. In such a model a reform of the claiming-age schedule also changes the base of the spousal and survivor benefits, and the claiming age of each spouse responds to the other's; neither margin is in the reform tables.

Bequests. Bequests are accidental and assets at the terminal age are zero (Section 3.5). Mean assets are matched through the 40s and fall below the data from the early 50s; median assets are above the data from the 30s through the early 50s and below it from the mid-50s; at age 60 the model has 1.642 against 3.218 for the mean, in hundreds of thousands of 2016 dollars (Figure B1). De Nardi et al. (2010) and Lockwood (2018) show that a bequest motive and the demand for self-insurance against late-life risks account for saving in old age of the kind the model under-predicts. The under-prediction bears on the saving-subsidy reforms, whose cost and induced income taxes are proportional to the asset base (Section 9.3).

Health information. Health is observed by the government: disability insurance pays on the disability state and the earnings condition in (11), the Medicare Part B expansion conditions coverage on nonparticipation, and no reform in the menu conditions on health. If health were private information, disability insurance and Medicare would become screening problems in which earnings conditions, asset tests, and waiting periods separate applicants by health, as in Golosov and Tsyvinski (2006) and Low and Pistaferri (2015), and the fiscal columns

would carry the cost of false claims. The private-information results of the companion paper (Ndiaye and Yu 2026) concern ability, not health.

Time separability. Preferences are nonseparable between consumption and leisure within the period and separable across periods (Section 3.2). Separability over time is a substantive restriction where consumption at the time of exit from the labor force affects the value of consumption afterward: Grochulski and Kocherlakota (2010) show that the design of a social security system changes when preferences are nonseparable over time, and Kolsrud et al. (2024) use changes in consumption at retirement to evaluate pension design. The welfare valuation of the claiming-schedule reforms, which move the exit age, would differ from (20) under such preferences, in a direction that requires a specification.

Income risk. Ability follows the AR(1) process of (5) with persistence 0.981 and innovation variance 0.016. Processes with non-Gaussian innovations, heterogeneous profiles, or age-varying variances (Storesletten et al. 2004; Heathcote et al. 2010; Guvenen et al. 2021) are outside the model. The dispersion of lifetime earnings determines the distribution of average indexed earnings across the brackets of (8), and therefore the incidence of the benefit-formula reforms of Section 8; the persistence of shocks governs the precautionary motive behind the saving and participation responses of every family.

Reentry and irreversibility. Exit from the labor force is reversible at the estimated reentry cost of 3,312 hours, at which the model generates no reentry; claiming is irreversible and mandatory by age 70. Every participation response in the reform tables therefore operates through the timing of exit: an individual who leaves under a steeper early-claiming penalty or a higher normal retirement age does not return, and the 62–64 and 65–69 columns are changes in the stock of individuals who have not yet exited. In the data, reentry rates stay below 5 percent at every age (Figure B1); a model that generated them would let part of the response occur through returns to work after exit.

11.4 Preference specification and what was not varied

Appendix B.4 reports a re-estimation with preferences that are logarithmic in consumption, separable in hours with a Frisch elasticity fixed at 0.5 following Chetty (2012), and with participation and reentry costs in units of utility. The estimated discount factor is 0.97 and the participation costs are 0.53 for healthy and 0.59 for unhealthy workers (Table B1); as in the baseline, the cost of working is higher for unhealthy than for healthy workers. The specification matches assets in old age more closely than the baseline, 3.542 against 3.218 in hundreds of thousands of 2016 dollars at age 60, and participation less closely, 96.86 against 83.94 percent for healthy individuals at age 50 (Figure B2). The reform tables were not re-

run under this specification, so the paper does not report how the welfare and fiscal columns depend on the curvature of utility or on the form of the participation cost.

The remaining inputs were not varied. Every reform table is computed at one parameter vector, the point estimates of Table 3, and reports point values, without propagating the sampling uncertainty of that table. The net present values are discounted at one rate, the net return of 3.2 percent; the welfare measure is taken at one age, 30, over one horizon, to age 85; and the dollar conversion uses one consumption window per family (Section 5.1). The sensitivities the tables contain are two: the choice of fiscal denominator, net or direct, which gives the lower bounds of Section 5.3 and coincides in sign with the net reading in every row; and the pair of bounds for the saving subsidies, which differ in whether the induced asset income is taxed as it accrues (Section 9.3). Neither is a sensitivity to preferences.

12 Conclusion

Social Security's financing gap can be addressed through the claiming-age schedule in two ways, and the estimated life-cycle model of this paper assigns them different welfare signs. A higher normal retirement age lowers the benefit at every claiming age; it reduces the cohort's benefit outlays and net government spending before anyone changes behavior, and it lowers ex ante welfare, so the welfare delivered per dollar of net spending by reversing it is small. A steeper reward for delayed claiming, through a higher delayed retirement credit or a symmetric adjustment around the normal retirement age, leaves the benefit at the normal retirement age unchanged and changes the price of moving the claiming age. In the model it raises ex ante welfare and lowers both benefit outlays and net government spending, so its marginal value of public funds is infinite, and it moves participation from the years before the normal retirement age to the years after it. At approximately equal fiscal savings the two reforms differ in the sign of their welfare effect. The steepest adjustments shift the burden toward individuals who claim early, and the effects on them are not described by the average welfare measure.

The same accounting ranks the other instruments. Among increases in the generosity of retirement benefits, raising the first bracket of the benefit formula delivers the most welfare per dollar of net spending, and the first step is approximately self-financing; proportional increases deliver less, and increases concentrated in the upper brackets the least, while lowering participation at older ages the most. Expansions of Medicare coverage after age 65 raise welfare and lower participation at ages 62–69, reductions in income taxes after age 62 raise welfare and participation at every age, and both return little per dollar of net spending. Subsidies to retirement saving return at least 0.84–0.96 dollars of welfare per dollar of subsidy, with an upper bound that depends on whether the induced asset income is taxed as it accrues.

Across the menu, the sign of the net fiscal effect, not the size of the welfare gain, separates the reforms that contribute to closing the financing gap from those that widen it.

The exercise has defined limits: a partial-equilibrium evaluation for one cohort at fixed factor prices, with no transition and no weights on other cohorts, with statutory ages acting only through incentives, an individual decision-maker, accidental bequests, and one parameter vector and one preference specification; Section 11 states the direction in which each bears on the results, and the marginal value of public funds is reported so that other weights on public funds can be applied. This paper evaluates reforms within the rules as they are written; the optimum against which those rules could be judged is characterized in the companion paper (Ndiaye and Yu 2026).

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Appendix

A Data and First-Step Estimation

This appendix collects the statutory parameters of Section 2, describes the PSID and MEPS samples and the health measures of Section 4.1, details the first-step procedures of Section 4.2, documents the construction of the target profiles of Section 4.3, and lists the parameters that are fixed outside the method of simulated moments. Monetary values are in 2016 dollars unless a table note states otherwise.

A.1 Social Security parameters

Table A1 lists the Social Security parameters that enter the model through (7)–(11) and (14), in the notation of Section 3. Individuals born between 1920 and 1940 faced an early retirement age of 62 and, on average over birth years, a normal retirement age of 65; their delayed retirement credit ranged from 3 to 6 percent per year of delay by year of birth, and the model uses the cohort average of 4.5 percent. The bend points of the primary insurance amount are \$3,720 and \$22,392 and the cap on average indexed earnings is \$43,800; the earnings test withholds 50 percent of earnings above \$6,000 before the normal retirement age and 33 percent of earnings above \$8,186 from the normal retirement age to age 69. The payroll tax is the employee share of the combined Old-Age, Survivors, and Disability Insurance and Hospital Insurance tax, 3 to 7.65 percent since 1960, on earnings up to \$14,459–\$64,974. The disability earnings threshold of \$3,600 a year is the annual equivalent of the monthly substantial gainful activity threshold of \$300 for nonblind individuals with a disability.

A.2 Samples

Panel Study of Income Dynamics. The PSID is a longitudinal survey of a representative sample of the US population that records labor market outcomes, income, health, and wealth.[15] It began in 1968 with interviews of approximately 4,800 families and has followed the original respondents and their descendants since, with interviews every year until 1997 and every two years thereafter. The original sample has two components: 1,872 low-income families drawn from the Survey of Economic Opportunity (SEO) and 2,930 families selected by the Survey Research Center (SRC) at the University of Michigan to represent the general population. The sample used here consists of individuals in the SRC component interviewed between 1968 and 2015; the SEO component is dropped because most of its families were dropped from the survey after 1997. Within the SRC component we keep household heads and, when

[15] Panel Study of Income Dynamics, public use dataset, produced and distributed by the Survey Research Center, Institute for Social Research, University of Michigan, https://psidonline.isr.umich.edu/.

Table A1: Social Security Parameters

Parameter	Description	Value		Source
τ_t^p	Payroll tax rate	3%–7.65%		SSA
$\bar{y}_t^p$	Threshold, payroll tax	\$14,459–\$64,974		SSA
y_{db}	Disability benefits income threshold	\$3,600		SSA
$aime_0$	1st bend point in PIA formula	\$3,720		SSA
$aime_1$	2nd bend point in PIA formula	\$22,392		SSA
$aime_{max}$	Maximum Social Security wealth	\$43,800		SSA
ERA	Early retirement age	62		SSA
NRA	Normal retirement age	65		SSA
DRC	Delayed retirement credit	4.5%		SSA
RET	Retirement earnings test	Under NRA	NRA-69	
τ_{ret}	Tax rate	50%	33%	SSA
y_{ret}	Threshold	\$6,000	\$8,186	SSA

Notes: Parameters of the Social Security rules faced by the 1920–1940 cohorts, as used in (7)–(11) and (14). τ_t^p and $\bar{y}_t^p$ are the payroll tax rate and cap (ranges over the years the cohort worked); y_{db} is the disability earnings threshold; $aime_0$, $aime_1$, and $aime_{\max}$ are the bend points and the cap of the primary insurance amount; ERA, NRA, and DRC are the early retirement age, the normal retirement age, and the delayed retirement credit per year of delay; τ_{ret} and y_{ret} are the withholding rate and the threshold of the retirement earnings test below the NRA and from the NRA to age 69. Monetary values in 1987 dollars; all other monetary values in the paper are 2016 dollars. Source: Social Security Administration (SSA).

present, their spouses, aged 20 to 90, with nonmissing labor and health information, born between 1920 and 1940. Table A2 reports each step of the selection; the final sample has 2,248 individuals and 44,404 person-year observations. The PSID data are used to estimate health transitions, survival probabilities, wage profiles and wage dynamics, the initial distribution of state variables, and the target profiles.

Table A2: PSID Sample Selection

Selection	Individuals	Observations
Initial sample	76,880	3,075,200
Non-SEO sample	36,430	1,457,200
Household heads and spouse if present	21,225	355,314
Age 20 to 90	21,031	350,617
No missing labor or health information	19,433	222,319
Born between 1920 and 1940	2,248	44,404

Notes: Number of individuals and of person-year observations remaining after each selection step. Non-SEO: the Survey Research Center component of the original sample. The last row is the estimation sample. Source: PSID 1968–2015.

Medical Expenditure Panel Survey. The MEPS is a nationally representative survey of families, individuals, their medical providers, and employers.[16] It covers individuals of all

[16]Medical Expenditure Panel Survey, Agency for Healthcare Research and Quality, https://meps.ahrq.gov/.

ages, with age top-coded at 85; medical spending is reported at the individual level and cross-validated with medical providers and insurance companies. We use the 1999–2012 waves and keep household heads aged 20 and older, excluding observations with missing age, medical spending, health insurance coverage during working ages, or health status; the final sample has 120,731 individuals and 211,709 person-year observations (Table A3). The MEPS data are used to estimate the out-of-pocket medical spending profiles.

Table A3: MEPS Sample Selection

Selection	Individuals	Observations
Initial sample	259,263	491,795
Household heads	130,978	231,866
Age above 20	125,587	222,495
With insurance information at 25–65	121,385	213,005
Nonmissing health status	120,731	211,709

Notes: Number of individuals and of person-year observations remaining after each selection step. The last row is the sample used for the medical spending profiles. Source: MEPS 1999–2012.

A.3 Health and disability measures

In the PSID, health is constructed from three self-reported work-limitation questions asked in every wave: (1) *Do you have any physical or nervous condition that limits the type of work or the amount of work that you can do?* (2) *Does this condition keep you from doing some types of work?* (3) *For work you can do, how much does it limit the amount of work you can do—a lot, somewhat, or just a little?* The first question allows *Yes* or *No*. The second, available from 1986, allows *Yes*, *No*, or *Can do nothing*. The third has been coded *Not at all*, *Somewhat*, *Just a little*, or *A lot* since 1976, and *We can't work*, *It limits me a lot*, *Some, not much*, or *Limitation but not on work* before 1976. For observations before 1986, health is defined from the first and third questions only. Following Low and Pistaferri (2015), an individual is in good health ($h_t = 0$) with a *No* to the first question or a *Not at all* to the third; in poor health ($h_t = 1$) with a *Yes* to the first question and *Somewhat* or *Just a little* to the third; and has a disability ($h_t = 2$) with a *Yes* to the first question, *Can do nothing* to the second, and *A lot* to the third, which follows the Social Security Administration's criterion for disability insurance. The frailty index of Hosseini et al. (2022) uses PSID items available since 2003, and the PSID self-rated health question is available since 1984; neither covers the life cycle of the 1920–1940 cohort, which requires health information from 1968 on.

In the MEPS, health is based on self-rated health, following Pashchenko and Porapakkarm (2019). Each year has three interview waves, and in each the respondent rates his or her health as (1) excellent, (2) very good, (3) good, (4) fair, or (5) poor. An individual is in good health ($h_t = 0$) if the rating is in categories 1–3 in all three waves, has a disability ($h_t = 2$) if

the rating is poor in any wave, and is in poor health ($h_t = 1$) otherwise. Table A4 compares the shares of each health state by age band in the two samples. The shares with a disability are 9.3 percent (PSID) and 10 percent (MEPS) at ages 50–60, 12.9 and 11.7 percent at 60–70, and 19.7 and 15 percent at 70 and older; the MEPS classifies a larger share as being in poor health in every age band. The PSID column for ages 70 and older sums to 99.0 because of rounding.

Table A4: Health Measures by Age Band in the Two Surveys

	Ages 50–60		Ages 60–70		Ages 70 and older	
Health Status	MEPS	PSID	MEPS	PSID	MEPS	PSID
With a disability (%)	10	9.3	11.7	12.9	15	19.7
Poor health (%)	21	10.6	23.6	17.0	28.7	19.3
Good health (%)	69	80.1	64.7	70.1	56.3	60.0

Notes: Share of person-year observations in each health state, in percent, by age band and survey. PSID: work-limitation classification following Low and Pistaferri (2015); MEPS: self-rated health classification following Pashchenko and Porapakkarm (2019). The PSID column for ages 70 and older sums to 99.0 because of rounding. Sources: PSID 1968–2015; MEPS 1999–2012.

A.4 First-step procedures

Health transitions. The transition probabilities $\Pr(h_t = j \mid h_{t-1} = i)$, $i, j \in \{0, 1, 2\}$, of (3) are estimated by probit regressions of the indicator $\mathbb{1}\{h_t = j\}$ on a set of age dummies, on the subsample with $h_{t-1} = i$. The predicted values are the transition probabilities, which are then interpolated linearly across ages so that the transition matrices change smoothly over the life cycle. Panel (A) of Figure 1 plots the row for $i = 0$.

Survival probabilities. The survival function $S(h_t, t+1)$ of (4) is estimated with a logistic regression of an indicator of survival on a polynomial in age, previous health status, and their interactions whenever these are statistically significant, on PSID data. Table A5 reports the coefficients: age squared, dummies for poor health and disability in the previous period, their common interaction with age squared, and a constant, on 132,070 person-year observations. Panel (B) of Figure 1 plots the implied annual mortality rates.

Out-of-pocket medical spending. Out-of-pocket spending is total medical spending net of the amount covered by insurance, from the MEPS. Following Pashchenko and Porapakkarm (2019), total medical spending is regressed on age and year dummies separately for each health group, with the cross-sectional and longitudinal weights of the survey; the insurance coverage share is estimated in the same way. For working-age individuals, coverage is the amount paid by insurance programs; for individuals aged 65 and older, coverage is the amount paid by Medicare. Out-of-pocket spending at each age and health state is the product of the estimated coinsurance rate and total spending. Following De Nardi et al. (2025), the estimated spending

Table A5: Estimated Coefficients of the Survival Logit

	Coefficients	S.E.
Age^2	-.0006***	(0.0000)
$\text{Health}_{t-1} = 1$	-1.3198***	(0.2288)
$\text{Health}_{t-1} = 2$	-3.1419***	(0.4189)
$\text{Health}_{t-1} \times \text{Age}^2$	0.0109***	(0.0032)
Constant	7.8801***	(0.1373)
Observations	132,070	

Notes: Logistic regression; dependent variable: indicator of survival to the next period. $\text{Health}_{t-1} = 1$ and $\text{Health}_{t-1} = 2$ are indicators of poor health and of a disability in the previous period; $\text{Health}_{t-1} \times \text{Age}^2$ is the interaction of the previous health state with age squared. Robust standard errors clustered at the individual level in parentheses. $^{***}p < 0.01$. Source: PSID 1968–2015.

is multiplied by 1.60 for individuals younger than 65 and by 1.90 for those aged 65 and older, which makes it consistent with the aggregate spending reported in the National Health Expenditure Accounts, and the profiles are smoothed by regressing them on a quadratic in age. Panel (C) of Figure 1 plots the result; the dependence on participation in (6) comes from the coverage share, which differs between workers and nonworkers.

Hourly wages. Hourly wages are annual earnings divided by annual hours worked in the PSID, where respondents in survey year t report earnings and hours for year $t-1$; observations with hourly wages below \$6.50 or above \$250 are excluded. Because wages are observed only for participants, the age- and health-specific profiles $W(h,t)$ of (5) are estimated with the two-stage selection model of Heckman (1976), as applied by Guner et al. (2012). French (2005) and French and Jones (2011) instead estimate the wage equation inside the model by matching the profiles of participants; the selection model is used here because it fits the targeted moments and is computationally cheaper. In the first stage, a probit of participation on age, age squared, health, health interacted with age, family size, and year of birth, on all PSID observations, yields the inverse Mills ratio; in the second stage, the logarithm of the hourly wage is regressed on age, age squared, their interactions with health, and the inverse Mills ratio. Table A6 reports both equations; the coefficient on the inverse Mills ratio is 0.182 with a standard error of 0.0482, and the sample has 64,941 observations. The stochastic component of wages, $(\rho, \sigma_\varepsilon^2)$ in (5), is estimated from the residuals of the wage equation at ages 25 to 65, after dropping the top 0.5 percent of residuals to limit the influence of outliers on the variance, by maximum likelihood with the Kalman filter; the estimates are (0.981, 0.016). Panel (D) of Figure 1 plots $\exp W(h,t)$.

Initial distribution. The initial state at age 30 is drawn from the empirical joint distribution of wages, health, assets, and labor force participation of PSID sample members aged 28–32, with the mean of log wages adjusted to the estimated wage profile within each health

Table A6: Estimated Coefficients of the Wage and Selection Equations

	Coefficients	S.E.
Wage Equation		
Age	0.0582***	(0.00500)
Age^2	-0.00067***	(0.00006)
Health×Age	-0.0113***	(0.00127)
Health×Age^2	0.00013***	(0.00002)
Constant	1.447***	(0.104)
Selection Equation		
Age	0.0977***	(0.00453)
Age^2	-0.00162***	(0.00004)
Health	-1.371***	(0.0682)
Health×Age	0.0122***	(0.00119)
Family Size	0.0225***	(0.00369)
Birth Year	-0.0123***	(0.00080)
Constant	23.79***	(1.581)
Inverse Mill's Ratio	0.182***	(0.0482)
Observations	64,941	

Notes: Two-stage selection model. Wage equation: dependent variable, logarithm of the hourly wage in 2016 dollars. Selection equation: dependent variable, indicator of labor force participation; probit. Health is the health state (0, 1, 2). Robust standard errors clustered at the individual level in parentheses. $^{***}p < 0.01$. Source: PSID 1968–2015.

group. Initial average indexed earnings are imputed by assuming five years of work at 2,000 hours a year at the age-30 hourly wage, the procedure of French (2005). Table A7 summarizes the initial distribution. Mean assets are \$60,241 overall and \$61,388, \$42,022, and \$30,289 for individuals in good health, in poor health, and with a disability; mean hourly wages are \$19.19, \$15.13, and \$12.50; participation is 0.97, 0.89, and 0.45; and the three groups are 94.74, 4.17, and 1.09 percent of the 4,849 observations.

A.5 Target profiles

The target profiles are labor force participation by health, hours worked conditional on participation by health, mean and median assets, and the mean labor market reentry rate over the life cycle, estimated from the PSID with fixed-effects regressions following French (2005). Labor force participation is an indicator that annual hours exceed 800, which captures full- and part-time employment; hours are self-reported, with survey year t referring to year $t-1$. Assets are real estate, farm or business equity, vehicles, stocks, mutual funds, individual retirement accounts, Keogh plans, liquid assets, bonds, and investment trusts, net of mortgages and other debts, plus home equity net of debt; pension wealth and Social Security wealth are excluded. Following Borella et al. (2018), assets are divided equally between partners. Wealth

Table A7: Initial Conditions at Age 30

	Overall	Health Status		
		Good health	Poor health	With a disability
Assets (in 2016 dollars)				
Mean	60,241	61,388	42,022	30,289
Standard Deviation	146,975	149,670	91,894	19,310
Wages (in 2016 dollars)				
Mean	18.95	19.19	15.13	12.50
Standard Deviation	9.32	9.32	8.29	6.98
Participation				
Mean	0.96	0.97	0.89	0.45
Standard Deviation	0.20	0.18	0.31	0.50
Percentage		94.74	4.17	1.09
Observations	4,849	4,594	202	53

Notes: PSID sample members aged 28–32 born between 1920 and 1940. Assets are nonpension net worth and wages are hourly wages, in 2016 dollars; participation is the share with annual hours above 800; "Percentage" is the share of observations in each health state. Source: PSID 1968–2015.

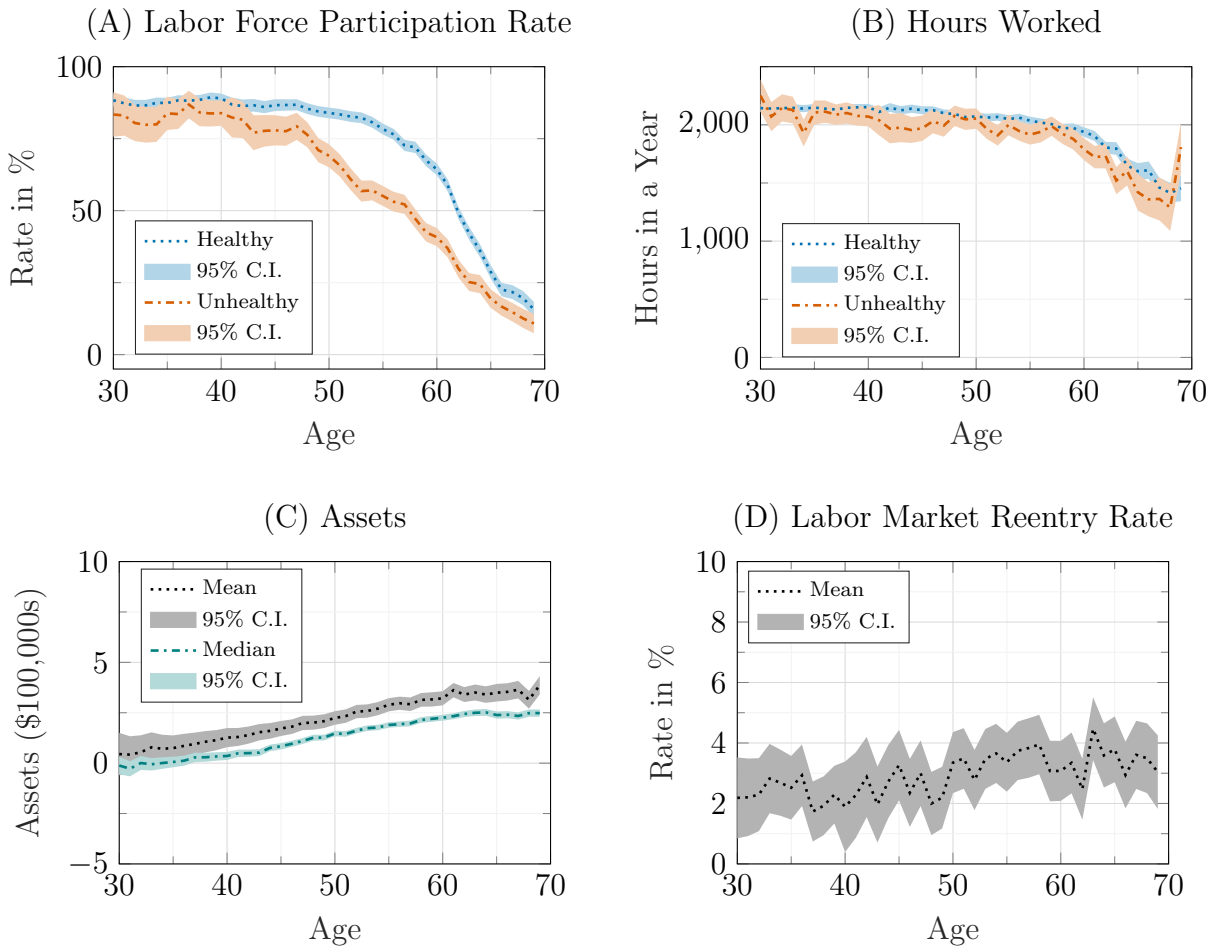
is observed only in selected survey years from 1984 on; for the other years assets are imputed from a fixed-effects regression of assets on age polynomials, log wages, family size, education, the unemployment rate, health indicators, and the interactions of the age polynomials with log wages and with health, and of health with education, family size, the unemployment rate, and wages; observed and imputed values are both used. Reentry is the indicator RE_t of (2): participation at t after nonparticipation at $t-1$.

The profiles are the age coefficients of (18), which includes an individual effect, age dummies interacted with good health and with poor health or disability, family-size dummies, and the unemployment rate. To remove family-size, business-cycle, and individual effects, the profiles are evaluated at a family size of 3, an unemployment rate of 6.5 percent, and the mean individual effect of individuals aged 50, in average health at 50, and born in 1935; the last normalization controls for cohort and for the correlation between the individual effect and health. The asset and reentry profiles are not conditioned on health. Figure A1 plots the profiles with their 95 percent confidence bands. Healthy individuals participate more than unhealthy individuals at every age, and the difference widens after age 50; hours conditional on participation decline with age in both groups; mean assets exceed median assets throughout; and the reentry rate stays below 5 percent at every age, with a maximum of 4.47 percent at age 63. The same data series appear as the black lines of Figure 2 and Figure B1.

A.6 Calibrated parameters

Table A8 lists the parameters that are fixed outside the method of simulated moments and are not in Table A1. The gross return is 4 percent, as in French (2005); the reduction from the

Figure A1: Target Profiles from the PSID



Notes: Profiles from (18), ages 30–69, with 95 percent confidence bands (shaded). (A) Labor force participation in percent, healthy (blue dotted) and unhealthy (red dash-dotted); (B) annual hours conditional on participation, same groups; (C) mean (black dotted) and median (teal dash-dotted) nonpension assets in hundreds of thousands of 2016 dollars; (D) mean labor market reentry rate in percent. "Unhealthy" pools poor health and disability. Source: PSID 1968–2015.

gross to the net return is 20 percent, as in Stantcheva (2017), so the net return is 3.2 percent; the consumption floor is \$3,500, as in French and Jones (2011); the borrowing limit is zero. The remaining rows restate the first-step inputs of Appendix A.4. The factor π^{db} in (11) is the probability of receiving disability benefits for older applicants with severe work limitations estimated by Low and Pistaferri (2015).

B Solution, Estimation, and Fit

B.1 Numerical solution

The problem (17) is solved by backward induction with value function iteration, starting from the terminal age $T = 85$, at which the continuation value is zero, and proceeding to

Table A8: Calibrated Parameters

Parameter	Description	Value	Source
Demographics			
t	Entry age; terminal age T	30; 85	
	Labor-market entry age; earnings history imputed at age 30	25; 5 years at 2,000 hours a year	French (2005)
Returns and budget			
r_f	Gross return on assets	4%	French (2005)
τ_k	Reduction from gross to net return	20%	Stantcheva (2017)
r	Net return, $(1-\tau_k)r_f$	3.2%	
$\underline{c}$	Consumption floor	\$3,500	French and Jones (2011)
$\underline{a}$	Borrowing limit	0	
Ability process			
ρ	Autoregressive coefficient of $\ln\theta_t$	0.981	PSID
σ_ε^2	Innovation variance	0.016	PSID
First-step data adjustments			
	Scaling of medical spending, ages below 65; 65 and older	1.60; 1.90	De Nardi et al. (2025)
	Hourly-wage trimming bounds	\$6.50; \$250	PSID
	Residual age window; top share dropped for $(\rho, \sigma_\varepsilon^2)$	25–65; 0.5%	PSID
Disability insurance			
y_{db}	Earnings threshold (1987 dollars)	\$3,600	SSA
π^{db}	Probability of receiving benefits	from Low and Pistaferri (2015)	Low and Pistaferri (2015)

Notes: Parameters fixed outside the method of simulated moments; the preference parameters are in Table 3 and the Social Security parameters in Table A1. Monetary values in 2016 dollars except the disability earnings threshold, which is in 1987 dollars. The scaling factors multiply the estimated medical spending profiles so that they match aggregate spending in the National Health Expenditure Accounts. Sources: PSID 1968–2015; SSA; the cited studies.

age 30. The continuous state variables are discretized. The grid has 30 points for assets, 10 for the wage, and 10 for average indexed earnings; together with the two claiming states, the three health states, and the two previous-participation states, this gives the finite state space on which the value function is computed directly. The expectation over next period's wage innovation ε_{t+1} in (5) is computed by five-node Gauss–Hermite quadrature (Judd 1998); the expectations over health and survival are finite sums over the three health states with the probabilities (3) and (4). The value function at points off the grid is evaluated by linear interpolation inside the grid and linear extrapolation outside it.

The solution consists of decision rules for consumption, hours, and claiming as functions of the state X_t and the parameters; the participation rule is the indicator that the hours rule is positive, and next period's assets follow from the budget constraint (15). Simulated histories are generated by forward induction: an initial state is drawn for each simulated individual, the age-30 decision rules are applied, the age-31 state is generated from the transition probabilities and the realized shocks, and the process is repeated through age T. Initial states are drawn

from the empirical joint distribution of wages, health, assets, and labor force participation of individuals aged 28–32 in the PSID sample, with the mean of log wages adjusted to the estimated age-30 profile within each health group; initial average indexed earnings are imputed from five years of work at 2,000 hours per year at the age-30 wage, following French (2005). Appendix A reports the initial distribution.

B.2 Moments and weighting

The second step estimates $\mathcal{P} = (\gamma, \nu, \phi_n(h{=}0), \phi_n(h{=}1,2), \phi_{RE}, L, \beta)$ by the method of simulated moments, given the first-step inputs χ: the net return r, the ability process $(\rho, \sigma_\varepsilon^2)$, the wage profiles $W(h,t)$, the health transition probabilities, the survival probabilities, the medical spending function, and the pension, Social Security, disability insurance, and tax rules. Poor health and disability share one participation cost because the PSID has few observations of individuals with a disability. The estimator (19) minimizes the weighted distance between the target profiles of Appendix A.5 and the corresponding profiles simulated from the model. With $\mathbb{T} = 40$ ages, the moment vector is

$$\hat{\varphi}(\mathcal{P};\chi) = \begin{bmatrix} \mathbb{E}[p_{iht} \mid h,t] - \int p_t(X,\mathcal{P},\chi)\, dF_{h,t}(X) \\ \mathbb{E}[n_{iht} \mid n_{iht} > 0,\, h,t] - \int n_t(X,\mathcal{P},\chi)\, dF_{h,t}(X \mid n_t > 0) \\ \mathbb{E}[a_{it} \mid t] - \int a_t(X,\mathcal{P},\chi)\, dF_t(X) \\ \operatorname{med}[a_{it} \mid t] - \operatorname{med}_{F_t}[a_t(X,\mathcal{P},\chi)] \\ \mathbb{E}[re_{it} \mid t] - \int re_t(X,\mathcal{P},\chi)\, dF_t(X) \end{bmatrix}, \tag{32}$$

for $t \in \{30,\ldots,69\}$ and $h \in \{\text{healthy, unhealthy}\}$, where p_t, n_t, a_t, and re_t are the participation, hours, asset, and reentry rules implied by the decision rules of Appendix B.1, F_t is the distribution of the state at age t in the simulated cohort, and $F_{h,t}$ is that distribution conditional on health h; the hours rows condition on positive hours in the data and in the simulation, since the target is hours conditional on participation. The blocks are: labor force participation by health (healthy, unhealthy) at ages 30–69, $2\mathbb{T}$ moments; hours worked conditional on participation by health, $2\mathbb{T}$ moments; mean and median assets, $2\mathbb{T}$ moments; and the mean labor market reentry rate, $\mathbb{T}$ moments—$7\mathbb{T} = 280$ moments in all, for seven parameters, which leaves 273 degrees of freedom for the χ^2 statistic of Table 3. The weighting matrix $\hat{\mathbf{W}}_I$ is the inverse of the $7\mathbb{T} \times 7\mathbb{T}$ diagonal matrix whose diagonal is that of the variance–covariance matrix of the data moments; I is the number of individuals in the data and ς the ratio of the number of data observations to the number of simulated observations. The identifying assumptions follow French (2005): work decisions do not affect health or wages, and age changes work and saving incentives but not preferences.

The estimation proceeds in five steps. First, the empirical life-cycle profiles of participation, hours, assets, and reentry are estimated from the PSID (Appendix A.5). Second, given the initial distribution of the state variables and χ, matrices of health and wage shocks are drawn. Third, for a value of $\mathcal{P}$ the model is solved by value function iteration and the decision rules are used to simulate profiles of participation, hours, assets, and reentry. Fourth, the moment conditions (32) and the criterion in (19) are computed. Fifth, a new $\mathcal{P}$ is chosen and the third and fourth steps are repeated until the criterion is minimized.

B.3 Fit of assets and reentry

Figure B1 completes Figure 2 with the three remaining moment blocks. The model's mean assets lie inside the 95 percent band through the 40s and below the data from the early 50s; the median is above the data from the 30s through the early 50s and below it from the mid-50s; at age 60 the model has 1.642 against 3.218 for the mean and 1.183 against 2.252 for the median, in hundreds of thousands of 2016 dollars, and at age 65 1.185 against 3.493 and 0.7582 against 2.385. The reentry rate is zero in the model at every age; in the data it lies between 1.73 and 4.47 percent, with the maximum at age 63. Section 4.5 relates the asset misfit to the absence of a bequest motive and to the saving-subsidy reforms.

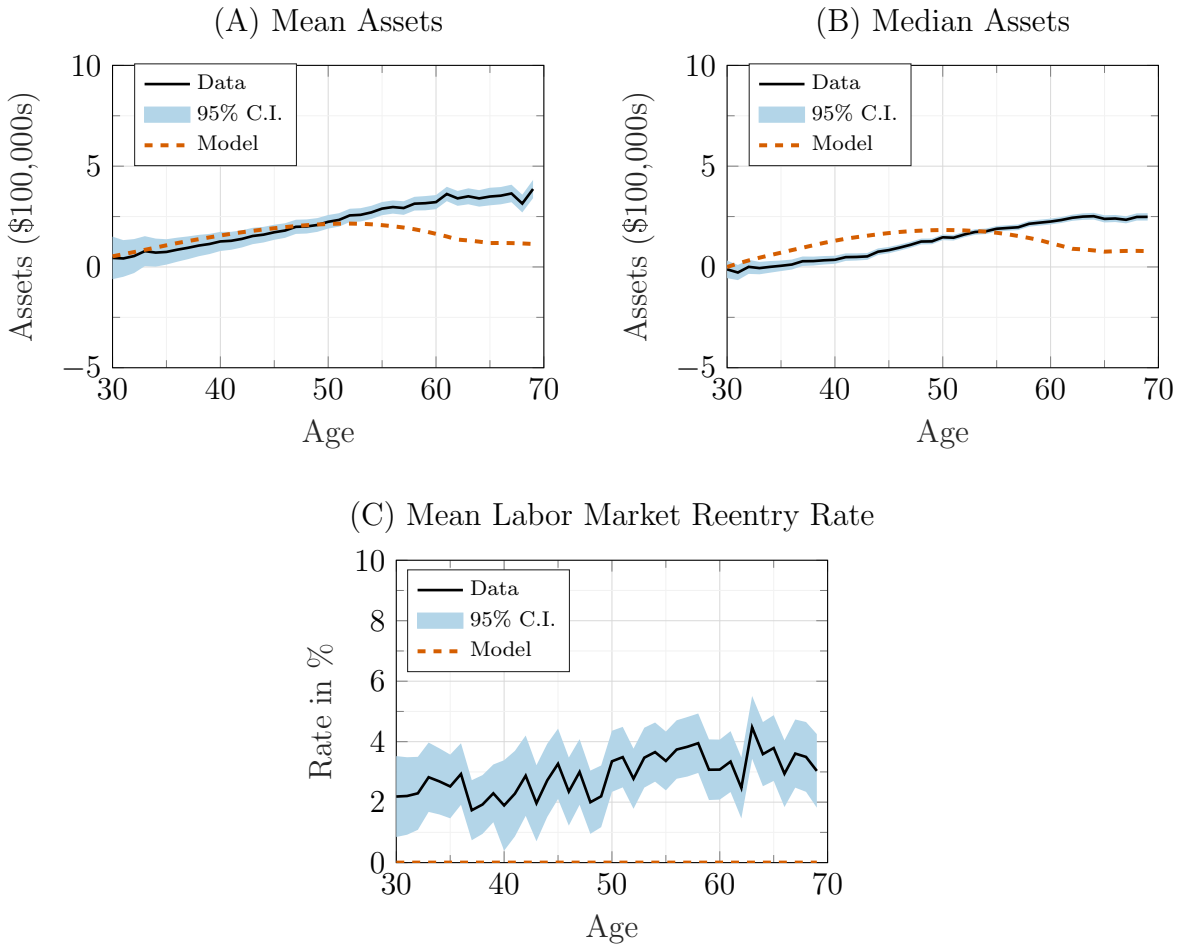
B.4 Re-estimation with logarithmic preferences

The model was also estimated with preferences that are separable in consumption and labor,

$$u(c_t, n_t; h_t) = \ln c_t - \varkappa \frac{n_t^{1+1/\epsilon}}{1+1/\epsilon} - \phi(h_t)\,\mathbb{1}\{P_t = 1\} - \phi_{RE}\,\mathbb{1}\{RE_t = 1\}, \tag{33}$$

where $\varkappa$ scales the disutility of hours, ϵ is the Frisch elasticity of labor supply, and $\phi(h_t)$ and ϕ_{RE} are fixed utility costs of participation and of reentry; the Frisch elasticity is set to $\epsilon = 0.5$ following Chetty (2012), and the remaining parameters are estimated. The health and wage processes, the policy rules, the targets, and the method of simulated moments are those of the baseline. Table B1 reports the estimates: a discount factor of 0.97, a reentry cost of 36.15, participation costs of 0.53 for healthy and 0.59 for unhealthy workers, and $\varkappa = 1.37 \times 10^{-10}$; the costs are in units of flow utility and are not comparable in level with the time costs of Table 3. As in the baseline, the utility cost of working is higher for unhealthy than for healthy workers. Figure B2 shows the fit. Mean assets are closer to the data at older ages than in the baseline (3.542 against 3.218 at age 60), while participation is matched less closely: healthy participation in the model is 96.86 percent against 83.94 in the data at age 50 and 68.44 against 64.34 at age 60. The reform experiments of Section 6 to Section 9 were not re-run under this specification; Section 11 states what this implies for the sensitivity of the results.

Figure B1: Model Fit: Assets and Labor Market Reentry



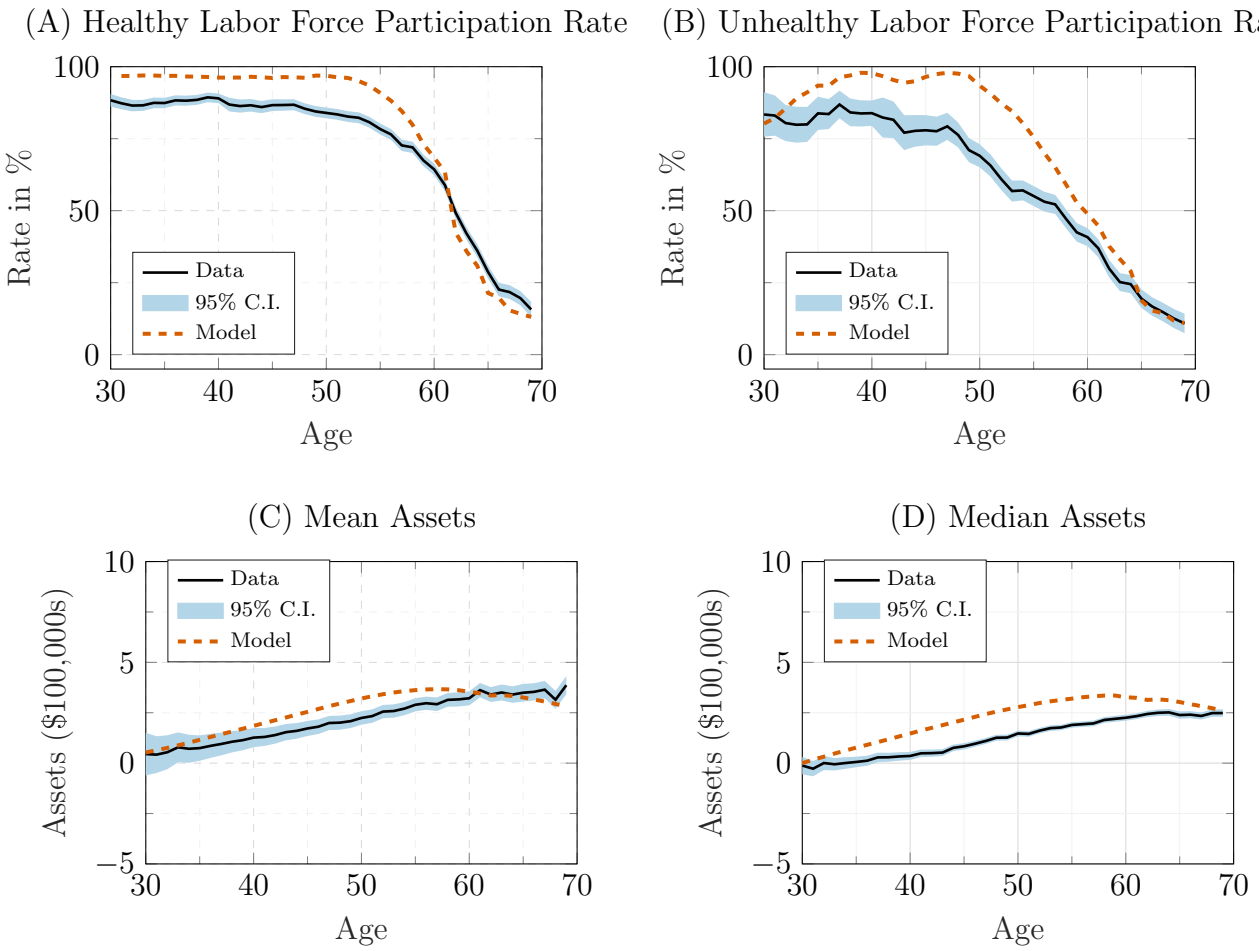
Notes: Black: PSID profiles from (18); shaded: 95 percent confidence bands; red dashed: model. Ages 30–69. (A) Mean and (B) median nonpension assets in hundreds of thousands of 2016 dollars; (C) mean labor market reentry rate in percent, the share of individuals participating at age t after not participating at $t-1$. Model series recomputed in 2026; data profiles and confidence bands are unchanged from the May 2025 version. Source: PSID 1968–2015.

Table B1: Estimated Parameters under Logarithmic Preferences

Parameter	Definition	Estimates	S.E.
β	Time discount factor	0.97	0.0015
ϕ_{RE}	Labor market reentry cost	36.15	1.2686
$\phi(h=0)$	Fixed cost of work, healthy	0.53	0.0082
$\phi(h=1,2)$	Fixed cost of work, unhealthy	0.59	0.0152
$\varkappa$	Scaling factor for work disutility	1.37×10^{-10}	0.14×10^{-10}

Notes: Method of simulated moments with the utility function (33), the Frisch elasticity fixed at $\epsilon = 0.5$, and the 280 moments of Table 3; standard errors in the last column. ϕ_{RE}, $\phi(h{=}0)$, and $\phi(h{=}1,2)$ are in units of flow utility. May 2025 estimates.

Figure B2: Model Fit under Logarithmic Preferences



Notes: Black: PSID profiles from (18); shaded: 95 percent confidence bands; red dashed: model estimated with the utility function (33). Ages 30–69. (A) and (B) labor force participation in percent by health; (C) mean and (D) median nonpension assets in hundreds of thousands of 2016 dollars. "Unhealthy" pools poor health and disability. May 2025 model series. Source: PSID 1968–2015.

C Reform Construction and Outcome Definitions

This appendix states the rule change behind each reform family of Table 2, restates the outcome definitions of Section 5 family by family, reports the full grid of benefit-formula reforms, plots the participation changes of the claiming-schedule reforms against their direct fiscal effect, and shows labor force participation by age under the elimination experiments of Section 6.

C.1 Construction of the reforms

Benefit levels and progressivity (P1, P2, P3, L). Reforms P1.1–P1.9 modify the first bracket of the primary insurance amount (8) by raising its rate from 0.90 by one percentage point per step, from 0.91 under P1.1 to 0.99 under P1.9. Reforms P2.0–P2.9 and P3.0–P3.9

target the second and third brackets: with the first rate at 0.99, P2.0–P2.9 raise the second rate from 0.32 by 0.32 percentage points per step to 0.352, and with the first two rates at 0.99 and 0.352, P3.0–P3.9 raise the third rate from 0.15 by 0.15 percentage points per step to 0.165; each step is one percent of the bracket's initial rate. Reforms L0–L9 raise the three rates proportionally by one percent per step. The full set of bracket triples is the second column of Table C1; L9 and P3.9 share the triple (0.99, 0.352, 0.165). The bend points, the cap, and every other rule are unchanged.

Claiming schedule (AN, AD, AL). Reforms AN1–AN3 raise the normal retirement age to 66, 67, and 68. The implementation shifts the status-quo schedule of Table 1 to the right by one year per year of increase and sets the factor at age 62 to $100 - 5\,(NRA + 1 - 62)$ percent of the primary insurance amount, that is, to 75, 70, and 65; the resulting schedules are rows AN1–AN3 of Table 6. Reforms AD1–AD5 raise the delayed retirement credit from the cohort's 4.5 percent per year of delay to 8, 8.5, 9, 9.5, and 10 percent and leave the factors before the normal retirement age unchanged. Reforms AL1–AL9 set the reduction per year of claiming before the normal retirement age and the credit per year of claiming after it to a common value of 7 to 15 percent, in steps of one percentage point. In every reform of this family the factor at the normal retirement age is 100, claiming remains possible from age 62 and mandatory by age 70, and the earnings test, the benefit formula, and disability insurance are unchanged.

Medicare Part A and Part B. Let $m^0(h, t, P)$ denote baseline out-of-pocket spending at ages 65 and older for health h, age t, and participation status P, and let ξ denote the average share of total medical spending at those ages paid by Medicare for the targeted group, so that total spending is $m^0/(1-\xi)$. A reform that raises the coverage share by $\Delta\xi$ sets

$$m^\rho(h, t, P) = m^0(h, t, P)\,\frac{1 - \xi - \Delta\xi}{1 - \xi}, \qquad t \geq 65, \tag{34}$$

for everyone aged 65 and older under Part A and for nonparticipants aged 65 and older ($P = 0$) under Part B, with $\Delta\xi = 0.20$. The experiments start from a re-simulated baseline: the coverage shares are computed from the MEPS for individuals aged 65 and older by health and participation status, the estimated model is simulated with these shares, and the two reforms are applied to that simulation.

Income tax after age 62 (I1–I5). After-tax income is $y_t - T_t(y_t) = \lambda_t y_t^{1-\tau_t}$ by (13), so a reduction in the tax is an increase in λ_t. The implementation note of the May 2025 appendix sets, for $t \geq 62$,

$$\tilde{\lambda}_t = \lambda_t + \Delta\lambda\,(t - 61), \qquad \Delta\lambda \in \{0.01,\, 0.02,\, 0.03,\, 0.04,\, 0.05\}, \tag{35}$$

with λ_t unchanged before age 62, and the table labels describe the same five reforms as increases of 1 to 5 percent in after-tax income after age 62. The direct effect is the change in the net present value from age 62 of income-tax collections, with the sign reversed so that revenue forgone is positive, as a share of baseline collections.

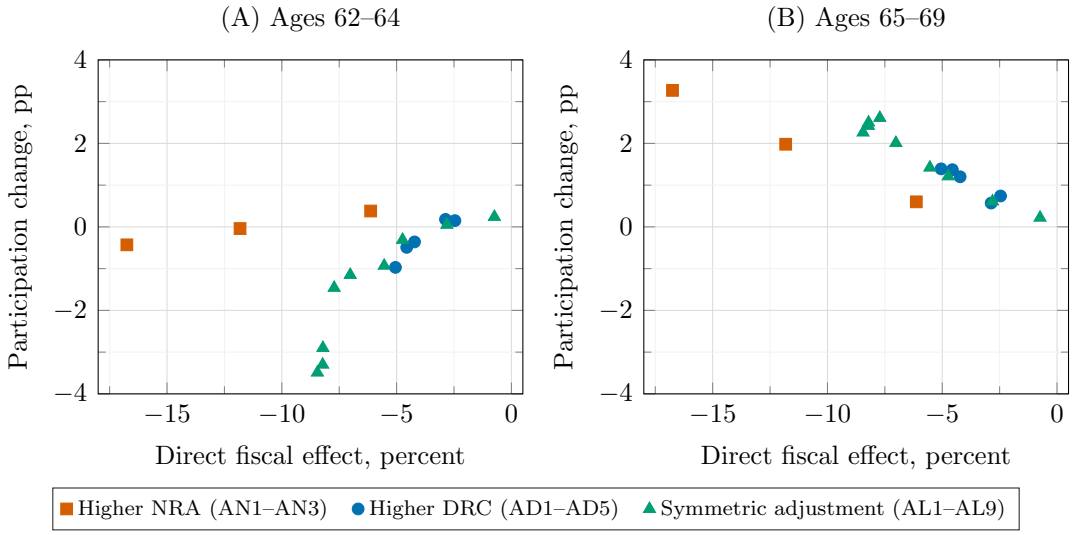
Saving subsidies (K1–K3). The net return in (15) is raised from $r = (1 - \tau_k)r_f$, equal to 3.2 percent, by 0.2, 0.4, and 0.8 percentage points at every age; the subsidy is the additional return $\Delta r\, a_t$ paid on assets held in each period. K3 sets the net return equal to the gross return of 4 percent and coincides with the elimination experiment D3 of Table 5.

C.2 Outcome definitions by family

The welfare columns are common to all families: WTP is $100\, x^{\mathrm{CE}}$ from (20), and the dollar value $\mathcal{W}$ in (22) uses baseline consumption from age 62 for the Social Security, Medicare, and income-tax families and from age 30 for the saving subsidies. The fiscal columns follow (23) with the objects listed below; the definitions used in computing the tables are quoted verbatim in the notes to Table 4. All net present values (NPVs) are per capita at age 62 at the net return r.

- *Social Security (P, L, AN, AD, AL).* B is the NPV of retirement benefits $b_t\, ss_t$ paid by the program; Direct is ΔB in percent of B^0; O is all other taxes and transfers, namely disability benefits and transfers net of income and payroll taxes; Indirect is ΔO in percent of O^0; Net is ΔG in percent of baseline government spending G^0.
- *Medicare (Part A, Part B).* B is the NPV of Medicare payments to the targeted group after 65 (everyone for Part A; nonparticipants for Part B); Direct is ΔB in percent of baseline payroll-tax collections $T^{p,0}$; O is all taxes and transfers, namely Social Security benefits and transfers net of income and payroll taxes; Indirect is ΔO in percent of O^0; Net is $(\Delta B + \Delta O)$ in percent of baseline government spending.
- *Income tax (I1–I5).* B is the NPV of revenue forgone after 62; Direct is ΔB in percent of baseline income-tax collections $|T^0|$; O is all other taxes and transfers, namely Social Security benefits and transfers net of payroll taxes; Indirect is ΔO in percent of O^0; Net is ΔG in percent of G^0.
- *Saving subsidies (K1–K3).* $B = Sub$ is the NPV of the subsidy; Direct is Sub in percent of baseline government spending G^0; Indirect is the change in government spending excluding

Figure C1: Participation Changes at Ages 62–64 and 65–69 against the Direct Fiscal Effect, Claiming-Schedule Reforms



Notes: Each marker is one row of Table 7. The horizontal axis is the Direct column, the change in the net present value of retirement benefits paid to the cohort as a percentage of its baseline value; the vertical axis is the change in labor force participation in percentage points (pp) at ages 62–64 (panel A) and 65–69 (panel B). Squares: a higher normal retirement age (AN1–AN3); circles: a higher delayed retirement credit (AD1–AD5); triangles: equal early-claiming reductions and delayed credits of ±7 to ±15 percent per year (AL1–AL9). Source: authors' simulations of the estimated model.

the subsidy, in percent; Net is the change in government spending including the subsidy, in percent; the MVPF is the interval $[\mathcal{W}/Sub,\ \mathcal{W}/\Delta G^{\text{full}}]$ of (28), whose upper bound uses the net cost under full realization and taxation of capital gains.

The MVPF of the first three families is $\mathcal{W}/\Delta G$ (25); when $\Delta G < 0 < \Delta B$ the tables report the lower bound $\mathcal{W}/\Delta B$ (27), and when $\mathcal{W} > 0$, $\Delta B < 0$, and $\Delta G < 0$ they report ∞. The participation columns are changes in labor force participation at ages 30–49, 62–64, and 65–69 in percentage points.

C.3 Benefit-formula reforms: the full grid

Table C1 reports all 39 rows of the grid of Section 8.1; Table 8 in the main text reproduces ten of them. The two sequences coincide at their endpoint, so the L9 row repeats the P3.9 row.

C.4 Participation and fiscal saving under the claiming-schedule reforms

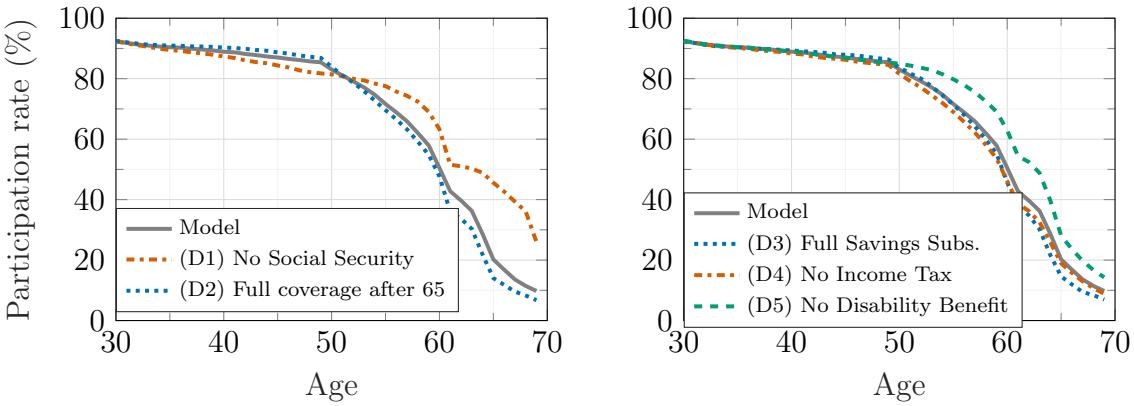
Figure C1 plots, for every row of Table 7, the change in participation at ages 62–64 and at ages 65–69 against the direct fiscal effect, with one marker per reform and the marker type by family.

Table C1: Social Security Progressivity and Level Reforms: Full Grid

Policy Reforms		%Δ Welfare		%Δ Budget			Δ Participation		
	PIA Brackets	MVPF	WTP	Net	Direct	Indir.	30–49	62–64	65–69
(P1.1)	(0.91, 0.32, 0.15)	$[0.52, +\infty)$	0.10	-0.03	0.57	-0.10	0.08	0.20	0.06
(P1.2)	(0.92, 0.32, 0.15)	0.97	0.22	0.09	1.23	-0.07	0.14	-0.07	0.00
(P1.3)	(0.93, 0.32, 0.15)	0.92	0.31	0.13	1.66	-0.09	0.14	0.15	0.01
(P1.4)	(0.94, 0.32, 0.15)	0.84	0.40	0.19	2.16	-0.10	0.14	-0.17	-0.23
(P1.5)	(0.95, 0.32, 0.15)	0.91	0.50	0.21	2.32	-0.09	0.16	-0.34	-0.31
(P1.6)	(0.96, 0.32, 0.15)	1.14	0.59	0.20	2.80	-0.16	0.19	-0.27	-0.15
(P1.7)	(0.97, 0.32, 0.15)	1.01	0.67	0.26	3.22	-0.16	0.21	-0.57	-0.50
(P1.8)	(0.98, 0.32, 0.15)	0.86	0.75	0.34	3.72	-0.15	0.24	-0.71	-0.49
(P1.9)	(0.99, 0.32, 0.15)	0.69	0.80	0.44	4.12	-0.11	0.25	-0.85	-0.60
(P2.0)	(0.99, 0.3232, 0.15)	0.56	0.83	0.57	4.75	-0.07	0.25	-1.10	-0.83
(P2.1)	(0.99, 0.3264, 0.15)	0.50	0.86	0.67	5.28	-0.05	0.24	-1.29	-1.03
(P2.2)	(0.99, 0.3296, 0.15)	0.49	0.88	0.70	5.99	-0.11	0.25	-1.05	-1.01
(P2.3)	(0.99, 0.3328, 0.15)	0.44	0.90	0.78	6.56	-0.10	0.24	-1.37	-1.15
(P2.4)	(0.99, 0.336, 0.15)	0.38	0.92	0.93	7.21	-0.05	0.24	-1.60	-1.63
(P2.5)	(0.99, 0.3392, 0.15)	0.37	0.93	0.98	7.95	-0.10	0.27	-1.63	-1.58
(P2.6)	(0.99, 0.3424, 0.15)	0.36	0.95	1.03	8.46	-0.11	0.28	-1.47	-1.68
(P2.7)	(0.99, 0.3456, 0.15)	0.33	0.94	1.11	8.99	-0.10	0.27	-1.74	-1.99
(P2.8)	(0.99, 0.3488, 0.15)	0.31	0.95	1.19	9.49	-0.10	0.28	-2.29	-2.1
(P2.9)	(0.99, 0.352, 0.15)	0.29	0.95	1.27	9.97	-0.09	0.28	-2.26	-2.05
(P3.0)	(0.99, 0.352, 0.1515)	0.29	0.95	1.26	10.04	-0.10	0.28	-2.22	-2.05
(P3.1)	(0.99, 0.352, 0.153)	0.29	0.96	1.27	10.13	-0.10	0.28	-2.30	-2.07
(P3.2)	(0.99, 0.352, 0.1545)	0.29	0.96	1.28	10.19	-0.10	0.28	-2.30	-2.07
(P3.3)	(0.99, 0.352, 0.156)	0.29	0.96	1.26	10.26	-0.13	0.28	-2.25	-2.03
(P3.4)	(0.99, 0.352, 0.1575)	0.29	0.96	1.27	10.34	-0.13	0.28	-2.23	-2.03
(P3.5)	(0.99, 0.352, 0.159)	0.29	0.97	1.29	10.41	-0.12	0.28	-2.41	-2.02
(P3.6)	(0.99, 0.352, 0.1605)	0.29	0.97	1.27	10.48	-0.14	0.28	-2.39	-2.02
(P3.7)	(0.99, 0.352, 0.162)	0.30	0.97	1.27	10.55	-0.15	0.28	-2.39	-2.02
(P3.8)	(0.99, 0.352, 0.1635)	0.30	0.97	1.25	10.63	-0.18	0.28	-2.37	-2.01
(P3.9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	-0.20	0.28	-2.37	-1.99
(L0)	(0.909, 0.3232, 0.1515)	0.51	0.16	0.12	1.34	-0.05	0.11	-0.05	0.01
(L1)	(0.918, 0.3264, 0.153)	0.59	0.32	0.21	2.47	-0.11	0.14	-0.15	-0.13
(L2)	(0.927, 0.3296, 0.1545)	0.49	0.46	0.36	3.48	-0.10	0.13	-0.49	-0.08
(L3)	(0.936, 0.3328, 0.156)	0.61	0.60	0.38	4.25	-0.18	0.18	-0.29	-0.3
(L4)	(0.945, 0.336, 0.1575)	0.59	0.74	0.49	5.28	-0.21	0.21	-0.48	-0.43
(L5)	(0.954, 0.3392, 0.159)	0.52	0.87	0.64	6.31	-0.19	0.23	-0.72	-0.68
(L6)	(0.963, 0.3424, 0.1605)	0.47	0.95	0.79	7.41	-0.20	0.24	-1.00	-1.22
(L7)	(0.972, 0.3456, 0.162)	0.42	1.00	0.92	8.66	-0.23	0.26	-1.40	-1.43
(L8)	(0.981, 0.3488, 0.1635)	0.35	1.02	1.14	9.80	-0.18	0.27	-1.76	-1.86
(L9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	-0.20	0.28	-2.37	-1.99

Notes: The second column gives the three bracket rates of the primary insurance amount (8) under the reform; the status quo is (0.9, 0.32, 0.15). P2.0–P2.9 are applied on top of P1.9 and P3.0–P3.9 on top of P2.9; L9 and P3.9 are the same policy. Columns as in Table 8 and Table 4: WTP in percent of consumption; Direct in percent of baseline retirement benefits; Indir. in percent of baseline other taxes and transfers; Net in percent of baseline net government spending; MVPF in dollars of welfare per dollar of net spending, with $[0.52, +\infty)$ the lower bound that counts only the direct cost in the one row in which net spending falls; participation changes in percentage points at ages 30–49, 62–64, and 65–69. All NPVs per capita at age 62 at the net return of 3.2 percent. Shading is blue for a welfare gain, a fiscal saving, a higher MVPF, or a participation increase and red for the opposite. Source: simulations of the estimated model.

Figure C2: Labor Force Participation by Age under the Elimination Experiments



Notes: Labor force participation rate in percent by age, 30–69, in the estimated model (gray) and in the elimination experiments of Table 5. D1 sets Social Security retirement benefits to zero; D2 sets out-of-pocket medical spending after age 65 to zero for participants and nonparticipants; D3 sets the net return on assets equal to the gross return (the K3 reform of Table 9); D4 sets the income tax to zero; D5 sets disability benefits to zero. Left panel: D1 and D2; right panel: D3–D5. Source: simulations of the estimated model.

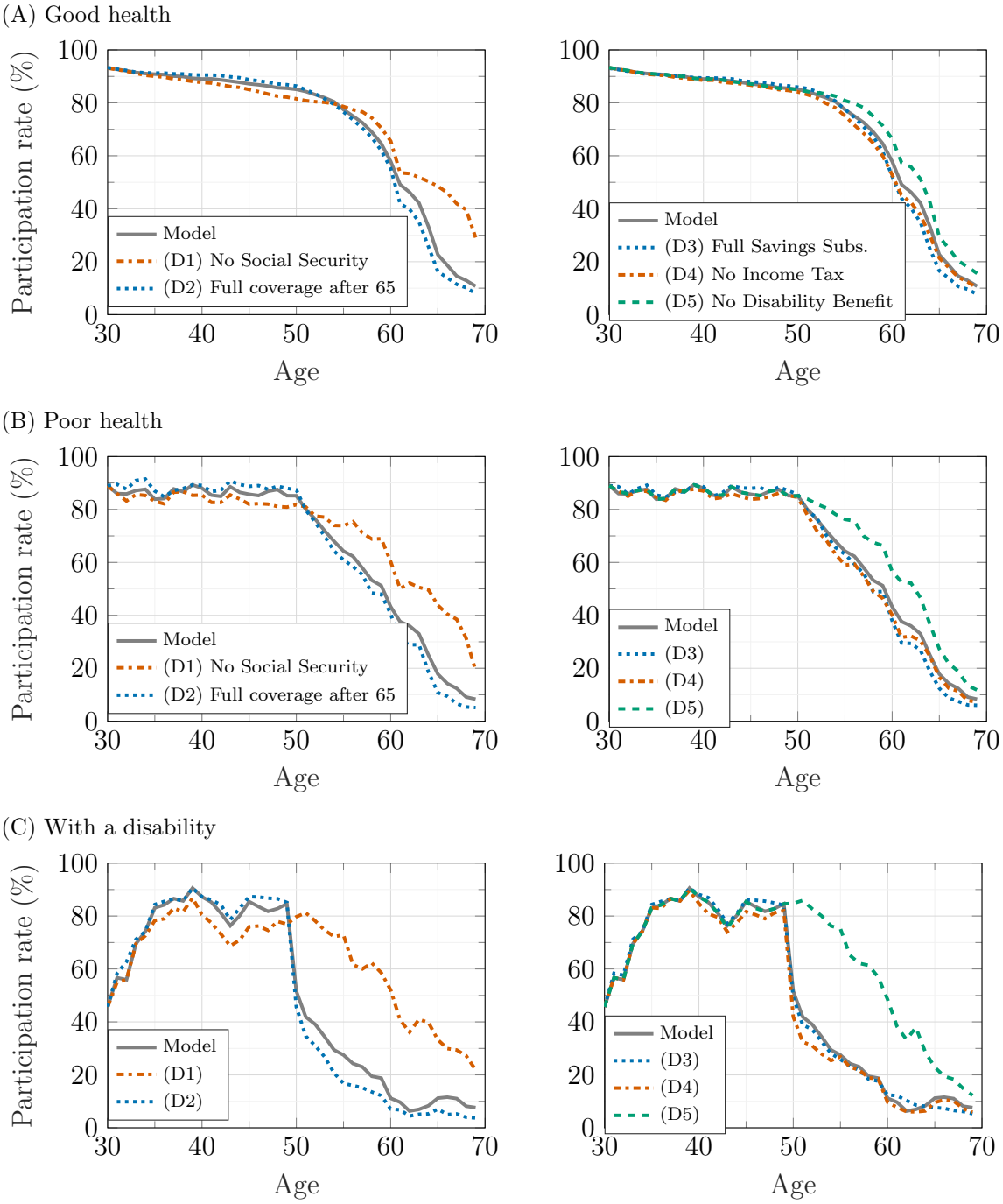
C.5 Participation under the elimination experiments

Figure C2 plots labor force participation by age in the estimated model and under the five elimination experiments of Section 6, and Figure C3 repeats the comparison by health state. These figures are the only exhibit of the participation responses to D1–D5; Table 5 reports welfare only. Removing Social Security benefits (D1) raises participation from age 63 by more than any other experiment; removing disability benefits (D5) raises it from age 50, by more than D1 at every age from 50 to 62 except age 60, where the two are within half a percentage point, and most of all among individuals with a disability, the group eligible for the benefit. Full coverage of medical spending after 65 (D2), the full saving subsidy (D3), and removal of the income tax (D4) lower participation at the older ages, D4 by less than the other two from the early 60s.

D Derivations for Sections 5 and 11

This appendix derives the welfare measures and column identities of Section 5 from the definitions of Section 3 (Appendix D.1), gives the algebra behind the sufficient-statistics mapping of Section 5.4 (Appendix D.2), derives the reduced-form factor-price adjustment discussed in Section 11 (Appendix D.3), and defines the cohort's net fiscal position on which the incidence stance of Section 2.4 rests (Appendix D.4). Notation is that of the main text: superscript 0 is the status quo, ρ a reform, Δ their difference, and $\mathbb{E}[\cdot]$ an average over the simulated cohort in which individuals who have died contribute zero.

Figure C3: Labor Force Participation by Age and Health under the Elimination Experiments



Notes: Labor force participation rate in percent by age, 30–69, by current health state, in the estimated model (gray) and in the elimination experiments of Table 5: D1 sets Social Security retirement benefits to zero; D2 sets out-of-pocket medical spending after age 65 to zero; D3 sets the net return on assets equal to the gross return; D4 sets the income tax to zero; D5 sets disability benefits to zero. In panels (B) and (C) the legend abbreviates the experiment labels to their identifiers. Source: simulations of the estimated model.

D.1 Welfare measures and the column identities

The consumption-equivalent change. Write the value function (17) of individual i at age 30 in sequence form,

$$V_{30}^{i} = \mathbb{E}_{30}\left[\sum_{t=30}^{85} \beta^{\,t-30}\, \mathbb{1}\{\text{alive}_{it}\}\, u(c_{it}, n_{it}; h_{it})\right], \tag{36}$$

where the expectation is over health, survival, wage, and medical-expense histories and the choices are the decision rules of Section 3.6. By (1), flow utility is homogeneous of degree $\gamma(1-\nu)$ in consumption: $u(e^{x}c, n; h) = e^{\gamma(1-\nu)x}\, u(c, n; h)$ for any x, because l_t in (2) does not depend on c_t. Multiplying status-quo consumption by e^{x} at every age and in every state, with hours, participation, and claiming held at their status-quo values, therefore multiplies every term of (36), and hence the cohort average V^0, by $e^{\gamma(1-\nu)x}$. The value of x at which the scaled status quo is as good as the reform solves $V^{\rho} = e^{\gamma(1-\nu)x}V^0$, which is (20); the factor $e^{x}-1$ is the proportional change in consumption, and $100\,x^{\text{CE}}$, the WTP column, is its percentage value to first order. With $\nu > 1$, both V^0 and V^{ρ} are negative and $\gamma(1-\nu) < 0$, so a reform that raises V (makes it less negative) has $V^{\rho}/V^0 \in (0,1)$, $\ln(V^{\rho}/V^0) < 0$, and $x^{\text{CE}} > 0$. The measure uses the consumption allocation only; it does not re-solve the individual's problem in the scaled status quo, and no re-solution is needed, because the scaling changes utility by the same factor in every state.

The leisure-equivalent change. Flow utility is homogeneous of degree $(1-\gamma)(1-\nu)$ in leisure. Multiplying l_t by e^{y} at every age and in every state multiplies V^0 by $e^{(1-\gamma)(1-\nu)y}$, so the leisure-equivalent change solves $V^{\rho} = e^{(1-\gamma)(1-\nu)y}V^0$ and

$$x^{\text{LE}} = \frac{\ln(V^{\rho}/V^0)}{(1-\gamma)(1-\nu)} = \frac{\gamma}{1-\gamma}\, x^{\text{CE}}, \tag{37}$$

which is (21). The ratio of the two measures is the preference parameter $\gamma/(1-\gamma)$ and carries no information about the reform; this is why the LE column appears only in Table 5.

From the columns to the MVPF. By (22) and (23), $\mathcal{W}_{\rho} = (\text{WTP}_{\rho}/100)\,\bar{C}_s$ and $\Delta G_{\rho} = (\text{Net}_{\rho}/100)\,G^0$, so (25) becomes

$$\text{MVPF}_{\rho} = \frac{\mathcal{W}_{\rho}}{\Delta G_{\rho}} = \frac{\bar{C}_s}{G^0}\,\frac{\text{WTP}_{\rho}}{\text{Net}_{\rho}} \equiv \kappa_f\,\frac{\text{WTP}_{\rho}}{\text{Net}_{\rho}}, \tag{38}$$

which is (26), and the same substitution with $\Delta B_{\rho} = (\text{Direct}_{\rho}/100)\,\text{base}_f$ gives the lower bound (27), $\mathcal{W}_{\rho}/\Delta B_{\rho} = (\bar{C}_s/\text{base}_f)\,\text{WTP}_{\rho}/\text{Direct}_{\rho}$. The constants κ_f and $\bar{C}_s/\text{base}_f$ are ratios of baseline aggregates that are common to every row of a family; within a family the

MVPF column therefore orders the rows exactly as the ratio WTP/Net does, and the lower bounds as WTP/Direct does. Across families the constants differ, through base_f and through the consumption window s (Table 4).

The identity (24) follows from (23) when $\Delta G = \Delta B + \Delta O$:

$$\begin{aligned}\text{Net}_\rho &= 100\,\frac{\Delta B + \Delta O}{G^0} = \frac{\text{base}_f}{G^0}\left(100\,\frac{\Delta B}{\text{base}_f}\right) + \frac{O^0}{G^0}\left(100\,\frac{\Delta O}{O^0}\right) \\ &= \omega_D\,\text{Direct}_\rho + \omega_O\,\text{Indirect}_\rho.\end{aligned} \tag{39}$$

The condition $\Delta G = \Delta B + \Delta O$ holds when every flow that the reform changes is counted either in B or in O. For the Social Security families $\text{base}_f = B^0$, so $\omega_D = B^0/G^0$ and $\omega_O = O^0/G^0$; the two weights sum to one only if $G^0 = B^0 + O^0$, which is not assumed, because the reported output does not state the items of G^0, so the identity is used only in the form (39). For Medicare the Direct base is baseline payroll-tax collections rather than baseline Medicare payments, so $\omega_D = T^{p,0}/G^0$ and the weights do not sum to one. For the income-tax family $\text{base}_f = |T^0|$ and $\omega_D = |T^0|/G^0$. For the saving subsidies the subsidy is zero at the baseline, so $O^0 = G^0$, and the Direct base is G^0 by definition; both weights equal one and Net is the sum of Direct and Indirect.

The four forms of the MVPF cell. Let $\mathcal{W} > 0$. If $\Delta G > 0$ the MVPF is a positive number. If $\Delta G < 0$ the reform raises welfare and lowers net spending and the ratio is not a price; the tables print ∞. When in addition $\Delta B > 0$, the behavioral saving outside the program exceeds the program's own cost, $\Delta O < -\Delta B < 0$. Crediting a fraction $\vartheta \in [0,1]$ of that saving gives the cost $C(\vartheta) \equiv \Delta B + \vartheta\,\Delta O$, which falls from $\Delta B > 0$ at $\vartheta = 0$ to $\Delta G < 0$ at $\vartheta = 1$ and vanishes at $\vartheta^* = \Delta B/(-\Delta O) \in (0,1)$. The MVPF computed with any crediting short of ϑ^* is

$$\frac{\mathcal{W}}{C(\vartheta)} \in \left[\frac{\mathcal{W}}{\Delta B},\,+\infty\right), \qquad \vartheta \in [0, \vartheta^*), \tag{40}$$

which is the interval $[\text{MVPF}^{\text{LB}}, +\infty)$ printed for P1.1: its lower end counts the program's own outlay and none of the saving outside it. If $\Delta B < 0$ as well as $\Delta G < 0$, as in every AD and AL row, $C(\vartheta) < 0$ for every ϑ and no finite positive ratio exists. If $\mathcal{W} < 0$ and $\Delta G < 0$, as in the AN rows, the ratio is positive and measures the welfare loss per dollar of net saving. Consider the reverse reform, which starts from the reformed economy and restores the status quo. Its CE change x' solves $V^0 = e^{\gamma(1-\nu)x'}V^\rho$, so $x' = -x^{\text{CE}}_\rho$; its change in net spending is $-\Delta G_\rho$; and its dollar WTP is $-x^{\text{CE}}_\rho\,\bar{C}^\rho_s$, where $\bar{C}^\rho_s$ is the consumption aggregate of (22) computed under the reform. Hence

$$\text{MVPF}_{\rho^{-1}} = \frac{-x^{\text{CE}}_\rho\,\bar{C}^\rho_s}{-\Delta G_\rho} = \text{MVPF}_\rho \times \frac{\bar{C}^\rho_s}{\bar{C}^0_s}, \tag{41}$$

so the printed AN entry is the MVPF of lowering the NRA back to 65 up to the ratio of the two consumption aggregates, which differ only through the reform's effect on consumption after age 62. For the saving subsidies, asset income ra_t is taxed as it accrues through y_t in (15), so the model's ΔG is the net cost under full realization and taxation of capital gains; the upper bound of (28) is $\mathcal{W}/\Delta G$ and the lower bound is $\mathcal{W}/Sub$, with $0 < \Delta G < Sub$ because $\Delta O < 0$ and Net is positive. By construction the ratio of the two bounds equals the ratio of the Direct to the Net column, $Sub/\Delta G$.

D.2 The MVPF and the sufficient-statistics objects

Identities. With $\Delta B = \Delta B^M + \Delta B^B$ as in (29) and FM, FM_B as in (30), $\Delta G = \text{FM}\ \Delta B^M$ and $\Delta B = \text{FM}_B\ \Delta B^M$ whenever $\Delta B^M \neq 0$. Defining $\hat{\pi}$ by $\mathcal{W} = (1+\hat{\pi})\,\Delta B^M$ gives

$$\text{MVPF} = \frac{\mathcal{W}}{\Delta G} = \frac{(1+\hat{\pi})\,\Delta B^M}{\text{FM}\ \Delta B^M} = \frac{1+\hat{\pi}}{\text{FM}}, \qquad \text{MVPF}^{\text{LB}} = \frac{\mathcal{W}}{\Delta B} = \frac{1+\hat{\pi}}{\text{FM}_B}, \tag{42}$$

which is (31). Nothing is approximated: $\hat{\pi}$ is defined so that the first equality holds for the discrete reforms the tables report. Three consequences follow. First, MVPF $= 1$ if and only if $1+\hat{\pi} = \text{FM}$, the condition under which the marginal dollar of program spending is worth exactly one dollar of public funds. Second, if $\mathcal{W} > 0 > \Delta G$ and $\Delta B^M > 0$, then $\text{FM} = \Delta G/\Delta B^M < 0$: the behavioral fiscal gains exceed the mechanical outlay. Third, a printed MVPF pins down only the ratio of the two objects: the pairs $(\hat{\pi}, \text{FM})$ consistent with it form the line $1+\hat{\pi} = \text{MVPF} \cdot \text{FM}$, and the lower bound adds the line $1+\hat{\pi} = \text{MVPF}^{\text{LB}} \cdot \text{FM}_B$; neither $\hat{\pi}$ nor FM is identified without ΔB^M.

The envelope approximation and the weights ω_t. Index a family of transfer rules by a scalar ξ and let $\delta_{it}^M(\xi)$ be the change in individual i's period-t resources in (15) produced by a unit change in ξ with behavior held fixed: the mechanical change in benefits, coverage, taxes, or asset returns. The shadow value of a dollar of resources at age t is $u_c(c_{it}, n_{it}; h_{it})$ whether or not the borrowing constraint binds, because a dollar received at t is consumed at t when $a_{t+1} = 0$ binds and is worth u_c at t by the Euler equation otherwise. Applying the envelope theorem to (36)—the first-order effect of re-optimizing consumption and hours is zero at an interior solution, and individuals who change a discrete participation or claiming decision in response to a marginal change in ξ are indifferent between the alternatives—gives

$$\frac{dV^0}{d\xi} = \mathbb{E}\left[\sum_{t=30}^{85} \beta^{\,t-30}\, u_c(c_t^0, n_t^0; h_t)\, \delta_t^M\right], \tag{43}$$

where the indicator of survival is absorbed in $\mathbb{E}[\cdot]$. To convert utils into the dollars of (22), differentiate (20): $dx^{\text{CE}} = dV/(\gamma(1-\nu)V^0)$, so $dW = \bar{C}_s\, dx^{\text{CE}} = dV/\bar{\lambda}$ with

$$\bar{\lambda} \equiv \frac{\gamma(1-\nu)\, V^0}{\bar{C}_s} = \frac{\mathbb{E}[\sum_t \beta^{\,t-30}\, u_c(c_t^0, n_t^0; h_t)\, c_t^0]}{\bar{C}_s} > 0, \tag{44}$$

where the second equality uses Euler's theorem for the homogeneous function u, $u_c\, c = \gamma(1-\nu)\, u$, summed over (36). Thus $\bar{\lambda}$ is the cohort's consumption-weighted lifetime marginal utility of consumption per dollar of the baseline consumption aggregate used in the dollar conversion. Define

$$\omega_t \equiv \frac{\beta^{\,t-30}\,(1+r)^{\,t-62}\, u_c(c_t^0, n_t^0; h_t)}{\bar{\lambda}}, \qquad \mathcal{W} \approx \sum_{t=30}^{85} (1+r)^{-(t-62)}\, \mathbb{E}\big[\omega_t\, \delta_t^M\big]\,, \tag{45}$$

which is the expression used in Section 5.4: ω_t is the marginal utility of a dollar at age t, discounted at β and compounded to age 62 at r, relative to $\bar{\lambda}$. By construction $\sum_t (1+r)^{-(t-62)}\mathbb{E}[\omega_t c_t^0] = \bar{C}_s$: with $s = 30$ the consumption-weighted lifetime average of ω_t is one, and with $s = 62$ it is $\bar{C}_{62}/\bar{C}_{30}$, so the convention of (22) for the Social Security, Medicare, and income-tax families values a util at the price that makes the CE change worth its application to consumption after age 62 only. The NPV of the mechanical resource changes is the mechanical outlay of the program plus the mechanical interactions outside it, $\sum_t (1+r)^{-(t-62)}\mathbb{E}[\delta_t^M] = \Delta B^M + \Delta O^M$, where ΔO^M collects the offsetting response of the floor transfer (16) for individuals at the floor and, for the saving subsidies, the income tax on the subsidized asset income through y_t in (15); retirement, disability, and transfer income are untaxed, so for the Social Security families ΔO^M is the floor interaction alone. Then

$$1 + \pi \approx \frac{\sum_t (1+r)^{-(t-62)}\, \mathbb{E}[\omega_t\, \delta_t^M]}{\sum_t (1+r)^{-(t-62)}\, \mathbb{E}[\delta_t^M]}, \tag{46}$$

the transfer-weighted average of ω_t, which coincides with $1 + \hat{\pi}$ of (42) to first order and up to ΔO^M. It exceeds one when the mechanical transfer accrues in states and ages with above-average marginal utility, and π is then the consumption-smoothing gain of Baily (1978) and Chetty (2006), extended to a dynamic setting with many states as in Chetty (2009), Landais and Spinnewijn (2021), and Kolsrud et al. (2024). From (1), $u_c = \gamma\, c^{\gamma(1-\nu)-1}\, l^{(1-\gamma)(1-\nu)}$, so with $\nu > 1$ the weight ω_t falls with leisure at given consumption; the insurance value of a dollar is lower after labor-force exit than during work at the same consumption, and higher in states of high mortality risk and low assets only through their effect on c_t^0 and on survival.

What is and is not identified. The expression (43) is first-order in the size of the reform. For discrete reforms of the size in the tables, individuals who switch participation or claim-

ing state are not indifferent between their status-quo and reformed choices, and their utility changes, second-order for a marginal change, are not negligible (Kleven 2021); the structural $\mathcal{W}$ includes them, and $\hat{\pi}$ is defined by (42) so that the identities hold exactly. On the cost side, the Direct column is $\Delta B = \Delta B^M + \Delta B^B$—for the Social Security families, ΔB^B is the change in outlays induced by changes in claiming ages, earnings histories, and the earnings test—and the Indirect column is ΔO. Computing ΔB^M requires applying the reformed rule to the status-quo histories with every decision rule held at its status-quo value, an accounting pass that is not part of the reported output; until it is run the tables identify $(1+\hat{\pi})/\,\mathrm{FM}$ and $(1+\hat{\pi})/\,\mathrm{FM}_B$ only, the fiscal multiplier cannot be read from the Direct column, and the relation to the estimates of Haller (2022) is one of objects, not of magnitudes.

The two valid sign inferences. Let ψ^0 and ψ^ρ be the status-quo and reformed claiming-age factors of (9) and hold every decision rule, hence every claiming age a^c and earnings history, at its status-quo value. Then $\Delta B^M = \sum_t (1+r)^{-(t-62)} \mathbb{E}\left[b_t^0 \left(ss_t^\rho - ss_t^0\right)\right]$ with ss_t from (10) evaluated at $\psi^\rho(a^{c,0})PIA_t^0$ and $\psi^0(a^{c,0})PIA_t^0$; because ss_t is nondecreasing in ssb_t, the sign of ΔB^M follows from the sign of $\psi^\rho - \psi^0$ at the claiming ages observed in the status quo (the crediting of benefits withheld after the NRA, which rises with the DRC, has the same sign). For AD1–AD5, $\psi^\rho(a) = \psi^0(a)$ for $a \leq 65$ and $\psi^\rho(a) > \psi^0(a)$ for $a > 65$, so $\Delta B^M \geq 0$, with strict inequality if anyone claims after 65 in the status quo; the printed Direct and Net are negative, so $\Delta B^B < -\Delta B^M \leq 0$, $\mathrm{FM}_B < 0$, and $\mathrm{FM} < 0$. For AN1–AN3, $\psi^\rho(a) < \psi^0(a)$ at every a, so $\Delta B^M < 0$; with $\mathcal{W} < 0$ and $\Delta G < 0$, both $1+\hat{\pi}$ and FM are positive and their ratio is the reversal value of (41); whether FM exceeds one is not determined, because the sign of ΔB^B is not. For AL1–AL9, $\psi^\rho - \psi^0$ is negative before the NRA and positive after it, so ΔB^M depends on the status-quo distribution of claiming ages and is not signed by the schedules. For the benefit-formula, Medicare, and income-tax families, $\Delta B^M > 0$ because the reformed rule pays more, or taxes less, at every status-quo history, and $\Delta B > 0$ in every printed row; but $\mathrm{FM} - 1 = (\Delta B^B + \Delta O)/\Delta B^M$ has the sign of $\Delta B^B + \Delta O$, in which ΔO is known from the Indirect column and ΔB^B is not, so whether FM exceeds one is undetermined for these families.

D.3 Factor prices: a reduced-form adjustment

Prices. Let aggregate output be $F(K, N)$ with constant returns to scale, K the capital stock and N labor in efficiency units, and let factor markets be competitive, so that the wage per efficiency unit and the gross return satisfy

$$w = F_N(K, N), \qquad r_f + \delta_K = F_K(K, N), \tag{47}$$

with δ_K the depreciation rate. Write $k \equiv K/N$ and $f(k) \equiv F(k,1)$, so that $F_K = f'(k)$ and $F_N = f(k) - kf'(k)$; let $s_K \equiv kf'(k)/f(k)$ be the capital share, $s_L \equiv 1 - s_K$ the labor share, and $\sigma \equiv d\ln k / d\ln(F_N/F_K)$ the elasticity of substitution. Differentiating $\ln(F_N/F_K)$ with respect to $\ln k$ gives $1/\sigma = -k^2 f''/(f - kf') - kf''/f' = -kf''f/\big[(f - kf')f'\big]$, hence $kf''/f' = -s_L/\sigma$ and $d\ln F_N / d\ln k = -k^2 f''/(f - kf') = s_K/\sigma$. Therefore

$$\begin{aligned} d\ln w &= \frac{s_K}{\sigma}\,(d\ln K - d\ln N), \qquad d\ln(r_f + \delta_K) = -\frac{s_L}{\sigma}\,(d\ln K - d\ln N), \\ dr_f &= -(r_f + \delta_K)\,\frac{s_L}{\sigma}\,(d\ln K - d\ln N). \end{aligned} \tag{48}$$

For the constant-elasticity-of-substitution technology $F = \big[\alpha K^{(\sigma-1)/\sigma} + (1-\alpha)N^{(\sigma-1)/\sigma}\big]^{\sigma/(\sigma-1)}$ the capital share is $s_K = \alpha k^{(\sigma-1)/\sigma}/\big[\alpha k^{(\sigma-1)/\sigma} + 1 - \alpha\big]$; the Cobb–Douglas case is $\sigma = 1$ and $s_K = \alpha$. The signs are those of a capital-deepening: a rise in K/N raises the wage and lowers the return, by amounts that fall with σ. In a small open economy r_f is given, k solves $f'(k) = r_f + \delta_K$ whatever the cohort does, and $w = f(k) - kf'(k)$ is pinned down with it; $d\ln K - d\ln N$ is then irrelevant and both terms in (48) are zero. This is the assumption of Section 3.7. In a closed economy, $d\ln K$ and $d\ln N$ are the reform-induced changes in the economy's aggregates. If the reform applies to every cohort in a stationary economy they equal the cohort's own proportional changes in assets and efficiency labor, which can be computed from the simulated panels behind the reform tables (assets, participation, and hours by age); if it applies to one cohort, they are the cohort's changes scaled by its shares of the aggregates.

Welfare. At fixed behavior, the price changes alter the resources of an individual at age t in (15) through earnings, net asset income, and benefits. Earnings $e_t = w_t n_t$ change by $e_t\, d\ln w$, of which the income tax takes $T_t'(y_t)$ and the payroll tax τ_t^p below the cap; net asset income $ra_t = (1-\tau_k) r_f a_t$ changes by $(1-\tau_k) a_t\, dr_f$, of which the income tax takes $T_t'(y_t)$; and retirement benefits change through (7)–(9), by $b_t\, ss_t\, \eta_{ss,w}\, d\ln w$, where $\eta_{ss,w}$ is the elasticity of the benefit to a proportional change in the wage profile. Valuing these at the weights (45),

$$\begin{aligned} \Delta\mathcal{W}^{fp} &= \sum_{t=30}^{85} (1+r)^{-(t-62)}\, \mathbb{E}\Big[\omega_t\, \delta_t^{fp}\Big]\,, \\ \delta_t^{fp} &\equiv \big(1 - T_t'(y_t) - \tau_t^p \mathbb{1}\{e_t < \bar{y}_t^p\}\big)\, e_t\, d\ln w + \big(1 - T_t'(y_t)\big)(1-\tau_k)\, a_t\, dr_f \\ &\quad + b_t\, ss_t\, \eta_{ss,w}\, d\ln w. \end{aligned} \tag{49}$$

The elasticity $\eta_{ss,w}$ lies in $[0,1]$: a proportional rise in the wage profile raises the AIME proportionally up to the cap, and the PIA formula (8) converts it at bracket rates that fall from 0.9 to 0.32 and 0.15, so the elasticity equals one only for an individual whose AIME lies in the first bracket, is below one above the first bend point, and is zero at the cap; averaged

over the cohort it is below one, and it is computable from (7)–(8) on the baseline histories. Disability benefits and pensions, which depend on the PIA, and floor transfers would add terms of the same form and are omitted.

Fiscal effect. The government's side of the same changes is

$$\begin{aligned} \Delta G^{fp} &= \sum_{t=30}^{85} (1+r)^{-(t-62)}\, \mathbb{E}\left[g_t^{fp}\right], \\ g_t^{fp} &\equiv -\left(T_t'(y_t) + \tau_t^p \mathbb{1}\{e_t < \bar{y}_t^p\}\right) e_t\, d\ln w - T_t'(y_t)(1-\tau_k)\, a_t\, dr_f + b_t\, ss_t\, \eta_{ss,w}\, d\ln w, \end{aligned} \tag{50}$$

so that $\delta_t^{fp} - g_t^{fp} = e_t\, d\ln w + (1-\tau_k) a_t\, dr_f$ at every age: the two terms partition the cohort's change in factor income between its members and the government, and the benefit term is a transfer between the two. The reduction τ_k of the gross return stands for capital taxes outside the explicit tax function and is not an item of G in Table 4; counting it would add $\tau_k a_t\, dr_f$ in revenue. The discount rate in the NPVs and in $\bar{C}_s$ is held at r; a change in r_f would also change $r = (1-\tau_k) r_f$ and hence the discount factors, but since $\mathcal{W}$ and ΔG are themselves first-order in the reform, the re-discounting is second-order and is not evaluated. The adjusted MVPF is

$$\text{MVPF}^{fp} = \frac{\mathcal{W} + \Delta \mathcal{W}^{fp}}{\Delta G + \Delta G^{fp}}. \tag{51}$$

Signs by family. In a closed economy the sign of the adjustment follows from the direction of $d\ln K - d\ln N$, which the participation columns of Tables 7–9 and the saving response indicate but do not measure. Holding K fixed, a reform that raises labor supply lowers the wage and raises the return: by (49) workers lose and asset holders gain, and by (50) labor-tax revenue falls while the benefit base and income taxes on asset income move in the opposite directions. A reform that raises saving raises K/N, raises the wage and lowers the return, with the opposite incidence. Table D1 records the direction of each term for each family, taking the sign of $d\ln N$ from the participation columns and the sign of $d\ln K$ from the saving response where the tables bear on it. The income-tax reductions raise participation in every age group, so $d\ln N > 0$ unambiguously; the steeper adjustments raise participation at ages 30–49 and 65–69 and, beyond the smallest steps, lower it at 62–64, so $d\ln N > 0$ unless the 62–64 loss dominates in efficiency units; a higher NRA raises participation at 65–69 by more than it lowers it at 30–49, but the sign of $d\ln N$ depends on the efficiency weights of the two groups; the benefit-formula and Medicare expansions lower participation at both older age groups beyond the smallest steps, so $d\ln N < 0$ if those groups dominate; and the saving subsidies raise K and lower old-age participation, so $d\ln K - d\ln N > 0$. The magnitude is bounded by the inputs: the participation changes in the tables are at most a few percentage points within age groups that are a minority of the cohort's efficiency labor, and (48) scales them

Table D1: Direction of the Factor-Price Adjustment by Reform Family in a Closed Economy

Family	$d\ln K - d\ln N$	$d\ln w$	dr_f	Wage term in $\Delta\mathcal{W}^{fp}$	Labor-tax term in ΔG^{fp}	Benefit term in ΔG^{fp}
Income tax (I1–I5)	−	−	+	−	+	−
Higher DRC, symmetric adjustment (AD, AL), if the gains at 30–49 and 65–69 dominate	−	−	+	−	+	−
Higher NRA (AN), if the gain at 65–69 dominates	−	−	+	−	+	−
Benefit formula (P, L) beyond the smallest steps; Medicare	+	+	−	+	−	+
Saving subsidy (K1–K3)	+	+	−	+	−	+

Notes: Signs from (48)–(50) with K held fixed except in the last row, where the subsidy raises saving. The sign of $d\ln N$ is read from the participation columns of Table 7, Table C1, and Table 9; where the columns move in opposite directions across age groups the row states the condition under which the sign holds. The asset-income terms have the sign of dr_f in $\Delta\mathcal{W}^{fp}$ and the opposite sign in ΔG^{fp}. A + in a ΔG^{fp} column is an increase in net government spending. In the small open economy of Section 3.7 every entry is zero. No magnitude is computed.

by a factor share and divides by σ. A number requires s_K, σ, and δ_K; $d\ln N$ and $d\ln K$ by reform from the simulated panels; the age profiles of the tax bases $\mathbb{E}[T_t' e_t]$, $\mathbb{E}[\tau_t^p e_t \mathbb{1}\{e_t < \bar{y}_t^p\}]$, $\mathbb{E}[T_t'(1-\tau_k)a_t]$, and $\mathbb{E}[b_t ss_t]$; and $\eta_{ss,w}$. None of these is reported here, and no adjusted MVPF is computed. The companion paper lists factor-price feedback among the questions it leaves open (Ndiaye and Yu 2026); the fixed-price environment is that of Farhi and Werning (2013) and Golosov et al. (2016).

D.4 The cohort's net position and the incidence stance

Let M_t denote the medical payments made by Medicare on behalf of the individual at age t, the part of total medical spending after age 65 not included in m_t in (6). The cohort's net fiscal position is the per-capita NPV at age 62 of what its members receive from the government less what they pay:

$$N_c \equiv \sum_{t=30}^{85}(1+r)^{-(t-62)}\,\mathbb{E}[b_t\,ss_t + db_t + M_t + tr_t + Sub_t] - \sum_{t=30}^{85}(1+r)^{-(t-62)}\,\mathbb{E}[T_t(y_t) + T_t^p]\,, \quad (52)$$

where Sub_t is the saving subsidy of Appendix C.1, $\Delta r\, a_t$ under K1–K3 and zero at the baseline. For each family, the aggregates B and O of Table 4 collect the items of N_c that the family's reforms can change: retirement benefits, disability benefits, transfers, and the two taxes for the Social Security families, to which the Medicare family adds M_t and the saving-subsidy family the subsidy. An item that a reform leaves unchanged drops out of ΔN_c, so

$$\Delta N_c = \Delta G \quad (53)$$

under the same condition as (39), that every flow the reform changes is counted in B or in O; the Net column of every reform table is therefore the change in the cohort's net claim on the government in percent of baseline government spending on it, G^0.

The incidence stance of Section 2.4 can be stated with these objects. Let μ be the social value, in dollars of the cohort's consumption, of a dollar of government resources outside the cohort: to taxpayers or beneficiaries of other cohorts, or to a transition that the model does not describe. The social value of a reform is $\mathcal{W} - \mu\,\Delta G$, so a reform with $\Delta G > 0$ is worth undertaking if and only if MVPF $> \mu$, and a reform with $\mathcal{W} > 0 > \Delta G$ is worth undertaking for every $\mu \geq 0$. The paper's statements take $\mu = 1$—fiscal savings at face value, a unit welfare weight on their recipients outside the cohort—and report the MVPF so that a reader with another μ can re-rank the rows with $\Delta G > 0$ by comparing each MVPF with μ (Hendren and Sprung-Keyser 2020). Within the cohort, incidence is valued by the ex ante average at age 30 of (20) and not by the distribution of gains and losses across realized histories. Across cohorts, nothing is said about the welfare of others: the exercise is for one cohort facing permanent rules from age 30, with no transition. A reform contributes to closing a financing gap of the kind measured by the Trustees' actuarial deficit if $\Delta N_c < 0$; expressing ΔN_c in percent of the cohort's taxable payroll would require the NPV at age 62 of $\min\{e_t, \bar{y}_t^p\}$ over the cohort, which the reported baseline aggregates do not include, so the comparison with that benchmark is one of sign (Section 7.5).